Chapter 1

# COMMUNITY HEALTH CARE CONTEXT

”

We enter the new millennium, the opportunities and challenges in nursing are boundless and ever-changing.

New biotechnologies offer opportunities not experienced before: women can give birth to eight living babies in one delivery; multiple defective organs can be replaced; and through genomic typing we soon will be able to anticipate the anomalies that lead to illness.

However, challenges abound: millions of people die each year from conflicts resulting in wars, drought, and starvation; from preventable ancient, new, and reemerging infectious diseases; and from unhealthy lifestyle choices.

”

# COMMUNITY HEALTH

The communities in which we live and work have a profound influence on our collective health and well-being (WHO, 2006a).

And, since the beginning, people have attempted to create healthier communities.

A community that achieves a high level of wellness is composed of healthy citizens, functioning in an environment that protects and promotes health.

# Concept of Community

- ***Community:*** **is a group of people in defined geographical area with common goals and objectives and the potential for interacting with one another.**
- ***Community:*** **refers to a collection of people who interact with one another and whose common interests or characteristics form the basis for a sense of unity or belonging.**

## Cont. ..... Concept of Community

❖ ***List some communities:***

- **It can be a society of people holding common rights and privileges (e.g. a citizens of a town).**
- **Sharing common interests (e.g. a community of farmers).**
- **Living under the same laws and regulations (e.g. a prison community).**

# Critical Attributes of Communities

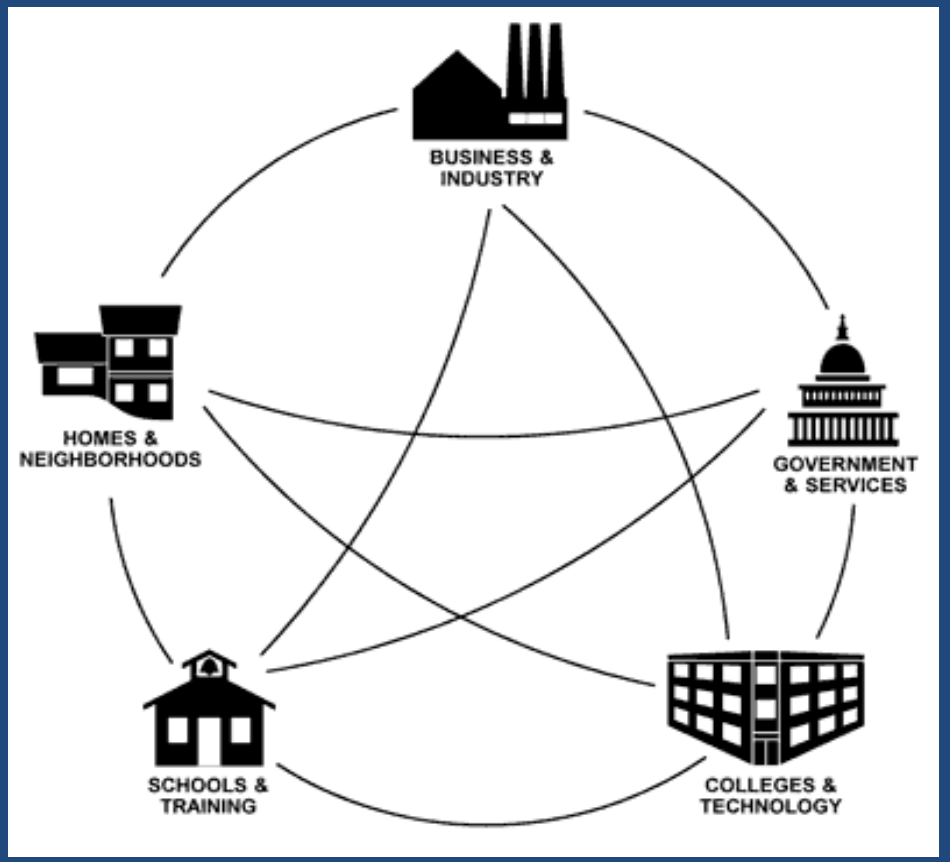


1. Group Orientation (people).
2. Bounded Between Individuals (ethnicity, culture, or geographic).
3. Social Interaction Between Individuals



# Components of Community

- People
- Goals and needs
- Environment
- Services system
- Boundaries



# Types of Communities

- Three types of communities that have relevance to community health practice:

## 1. Geographic Community:

- City, town, neighborhood.
- ***In community health, it is useful to identify a geographic area as a community:***

- A community demarcated by geographic boundaries, such as city or county, become clear target for analysis of health needs.

- **Available data, such as morbidity and mortality figures, can augment assessment studies to form the basis for planning health programs.**

- **Media campaigns and other health education efforts can readily reach intended audiences.**

# 2. Common-Interest Community

- **A community also can be identified by a common interest or goal.**

✓**The members of a national professional organization.**

✓ **Women who have mastectomies.**

**Are all common-interest communities**

# Example of Common-Interest Community

- **Disabled individuals scattered throughout a large city may emerge as a community through a common interest in promoting adherence to guidelines for wheel-chairs access, parking spaces, toilet facilities, elevators, or other facilities for the disabled.**

# 3. Community of Solution

- **Group of people who come together to solve a problem that affects all of them.**
- ***The shape of this community varies with:***

1. **The nature of the problem.**
2. **The size of the geographic area affected.**
3. **The number of resources needed to address the problem.**

# Examples: Community of Solution

- A water pollution problem may involve several counties whose agencies and personal must work together to control upstream water supply, industrial waste disposal, and city water treatment.
- Several schools may collaborate with law enforcement and health agencies, as well as legislators and policy makers, to study patterns of substance abuse among students and design possible preventive approaches.

# Community Function

- Creation and distribution of goods and services.
- Provision for socialization.
- Control of social behavior.
- Provision of a sense of identity and mutual support.
- Coordination, control and direction of activities to attain other community goals.

( Katz & Kahn, 1966; Warren, 1987).

# Community Health

- Human beings are social creatures. All of us, with rare exception, live out our lives in the company of other people.
- Communities are an essential and permanent feature of human experience.
- The communities in which we live and work have a profound influence on our collective health and well-being.

# Example:

- **Research has established that both smoking and passive exposure to tobacco smoke are directly associated with serious negative health effects and premature mortality among more than 1 billion smokers worldwide.**
  **(Institute of Medicine, 2001)**

- **Many communities and organization including (hospitals, schools, restaurants and airlines) have developed regulatory approaches to smoking as:**

1. **They have removed cigarette vending machines.**
2. **Conducted antismoking advertising campaigns.**
3. **Prohibited televised commercials.**
4. **Increased cigarette taxes.**
5. **Restricted smoking in public places.**

- **Such community rules and educational efforts:**
  - **Protect nonsmokers.**
  - **Promote quitting smoking.**
  - **Decrease the potential for heart and lung disease.**

- **Community health, as field of practice, seeks to provide organizational structure, a broad set of resources, and collaborative activities needed to accomplish the goal of an optimally healthy community.**
- **Community health is concerned with the interchange between population groups and their total environment, and with the impact of that interchange on collective health.**

## Other community and national concerns include the rising incidence and prevalence of:

- **Sexually transmitted disease.**
- **Substance abuse.**
- **Tuberculosis.**
- **Teen pregnancy.**
- **Family and teen violence.**
- **Terrorism.**
- **Pollution-driven environmental hazards.**

## Many different professionals work in community health to form complex team as follows:

- City planner designing an urban renewal project.
- The social worker providing counseling about child abuse or the use of chemical substance among adolescents.
- A physician treating clients affected by a sudden outbreak of hepatitis.
- Prenatal clinics.
- Meals for elderly.
- Genetic counseling centers.
- Educational programs for early detection of cancer.

**All are part of the community health efforts**

# Definition of Public Health

- ***Winslow's classic(1920) definition of Public Health***
  **Public health is the science and art of preventing disease, prolonging life, and promoting physical health and efficiency through organized community efforts for the following purposes:**

1. **Sanitation of the environment.**
2. **Control of communicable infections.**
3. **Education of the individual in principles of personal hygiene.**
4. **Organization of medical and nursing services for the early diagnosis and preventive treatment of disease.**
5. **Development of the social machinery that will ensure a standard of living adequate for the maintenance of health for every individual in the community.**

# Definition of Community Health

- ***The concept of Community Health can be defined as:***

**"The identification of needs and the protection and improvement of collective health within a geographically defined area".**

# Populations and Aggregates

**Populations:**

- **All of the people occupying an area, or all of those who share one or more characteristics.**
- ***A population may be defined:***

1. **Geographically:** **Such as the population of Palestine or a city's population.**

***This designation of population is useful in community health:***

- **Epidemiologic study.**
- **Collecting demographic data for purposes such as health planning.**

## 2. Common Qualities or Characteristics

- **Might be any thing thought to relate to health, such as:**

✓ **age, race, sex, social class, medical diagnosis, level of disability, exposure to toxins, or participation in a health seeking behavior, such as smoking cessation.**

***This meaning become useful in C.H.***

- **When a specific group of people (e.g., homeless individuals) is targeted for intervention.**

## Aggregate:

- **A mass or grouping of distinct individuals who are considered as a whole and who are loosely associated with one another.**

**For Example:**

***Aggregate can be identified by:***

- **Virtue of setting (those enrolled in a well-baby clinic).**
- **Demographic characteristics (women).**
- **Health status (smokers, or those with hypertension).**

## *Community health nurse, needs to define the community targeted for study and intervention:*

- **Who are the people who compose the community?**
- **Where are located, and what are their characteristics?**
- **A clear delineation of the community or population must be established.**
- **The complex nature of communities.**
- **The characteristics of the people in terms of: age, sex, race, socioeconomic level, and health status.**
- **What are the resources?**
- **Is the community under rapid change? What are the changes?**

# Definition of Community Health Nursing

***The American Nurses Association(ANA), defines Community Health Nursing as:***

**"Community Health Nursing is a synthesis of nursing practice and public health practice applied to promoting and preserving the health of populations. The practice is general and comprehensive, it is not limited to a particular age group or diagnosis, and it is continuing, not episodic.**

**The dominant responsibility is to population as a whole; nursing directed to individuals, families, or groups contributes to the total population. The focus of community health nursing is on the prevention of illness and the promotion and maintenance of health".**

# THE CONCEPT OF HEALTH

- **Health refers to a person's physical, mental, social and spiritual state.**
- **It can be positive (as being in good health), or negative (as being in poor health).**
- ***The World Health Organization(WHO)* defines health positively as:**

"*a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity*".

## THE CONCEPT OF HEALTH Cont. …….

- **Wellness: Is the process of moving toward integrating human functioning and maximizing potential.**

- **Wellness, includes the definition of health incorporates the capacity to develop one's potential to lead fulfilling and productive life; can be measured in terms of quality of life.**

## THE CONCEPT OF HEALTH Cont. …….

- **Wellness** is the ability to adapt, to relate effectively, and to function at near-maximum capacity and **includes:**

- Self-responsibility.
- Nutritional awareness.
- Physical fitness.
- Stress management.

## THE CONCEPT OF HEALTH Cont. …….

- Environmental sensitivity.
- Productivity.
- Expression of emotion.
- Creativity.
- Personal care.
- Home safety.

## THE CONCEPT OF HEALTH Cont. …….

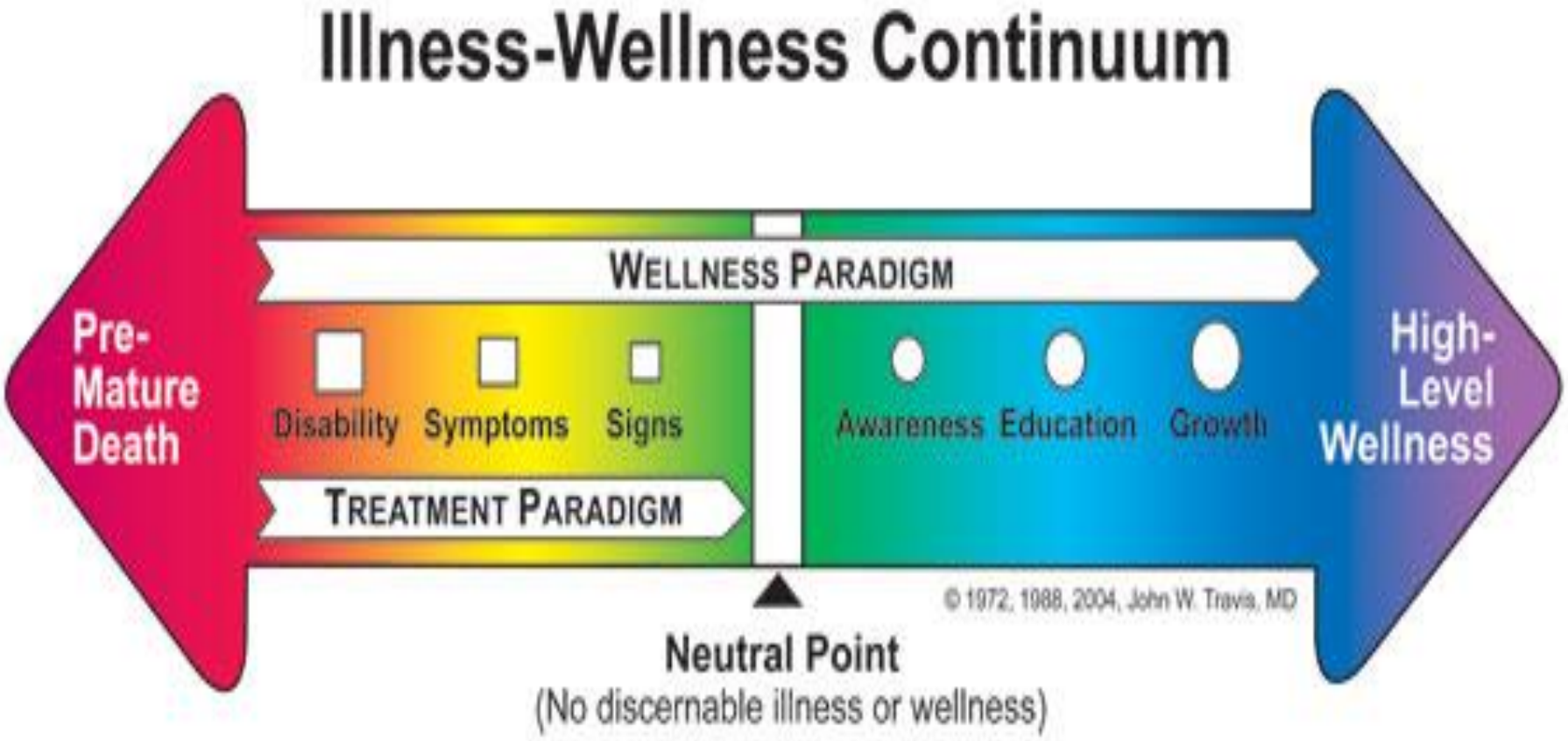
- **Well-being:** a state of positive health or a person's perception concerning positive health.
- **Influences on Health**

Many factors, influence the health including:

- Lifestyle - Genetics - The environment

- ***Florence Nightingale,*** explored the health and illness connection with the environment.
  She believed that a person's health was greatly influenced by ventilation, noise, light, cleanliness, diet, and restful bed.

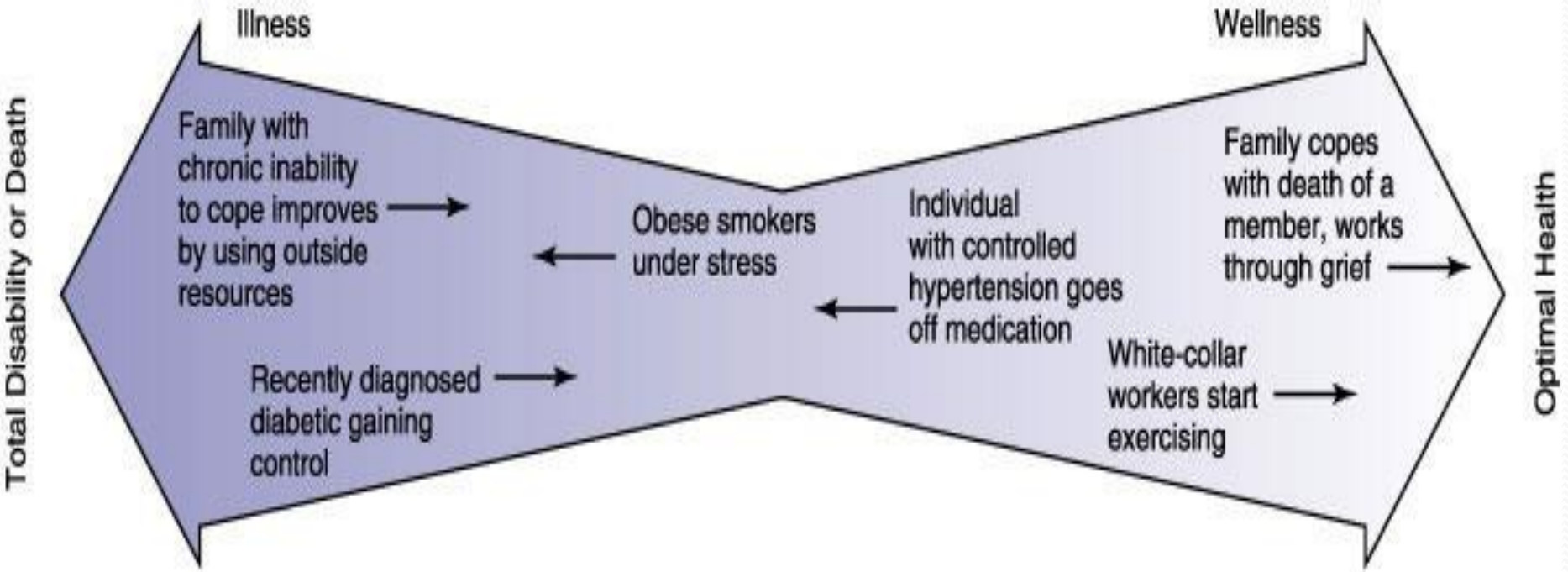
# The Health Continuum: Wellness–Illness



# The Health Continuum: Wellness–Illness

- Society suggests a polarized or "either/or" way of thinking about health: people either are well or they are ill. Yet wellness is a relative concept, not an absolute, and **illness is** a state of being *relatively unhealthy. There are many levels and* degrees of wellness and illness, Because health involves a range of degrees from optimal health at one end to total disability or death at other, it often describes as a continuum.

# The Health Continuum: Wellness–Illness



## The Health Continuum: Wellness–Illness Cont. ……

**It ranges from:**

- High level wellness (Optimal Health)
- Good health
- Normal health
- Illness
- Critical illness
- Total disability or Death

**A person's state of health is dynamic, varying from day to day and even hour to hour.**

## The Health Continuum: Wellness–Illness Cont. ……

This **health continuum** applies not only to individuals but also to families and communities. A nurse might speak of a dysfunctional family, meaning one that is experiencing a relative degree of illness; or a healthy family might be described as one that exhibits many wellness characteristics, such as effective communication and conflict resolution, as well as the ability to work together and use resources appropriately.

## The Health Continuum: Wellness–Illness Cont. ……

Likewise, a community, as a collection of people, may be described in terms of degrees of wellness or illness. The health of an individual, family, group, or community moves back and forth along this continuum throughout life.

## The Health Continuum: Wellness–Illness Cont. ……

A healthy community, first described by Cottrell (1976) as a competent community, is one in which the various organizations, groups, and aggregates of people making up the community do at least four things:

1. They collaborate effectively in identifying the problems and needs of the community.
2. They achieve a working consensus on goals and priorities.
3. They agree on ways and means to implement the agreed on goals.
4. They collaborate effectively in the required action.

## The Health Continuum: Wellness–Illness Cont. ……

**Community health practice ranges over the entire continuum; it always works to improve the degree of health in individuals, families, groups, and communities.**

**In particular, community health practice emphasizes the promotion and preservation of wellness and the prevention of illness or disability.**

- **Health as a state of being:**

➢ **That includes all of many characteristics of a person, family, or community, whether physical, psychological, social, or spiritual.**

➢ **These characteristics often indicate the degree of wellness or illness of an individual or community and suggest the presence or absence of vitality and well-being.**

- **Subjective and Objective Dimensions of Health:**

➢**Health involves both subjective and objectives dimensions:**

**- Subjective involves how well people feel.**

**- Objective refers to how well they can function in their environment.**

➢**Most often, functional performance diminishes dramatically toward illness end of health continuum.**

- **Continuous and Episodic Health Care Needs**

➢ **Community health practice encompasses populations in all age groups with birth-to-death development health care needs.**

➢ **Continuous needs may include, for example:**

✓ **Assistance with providing a toddler-proof home.**

✓ **Establishing a positive toilet-training techniques.**

✓ **Help in effectively dealing with the progressive emancipation of preteens and teenagers.**

- **Population may have one-time, specific, negative health events, such as an illness or injury, that are not an expected part of life.**

**These episodic needs might derive from:**

- ✓ **The birth of an infant with Down syndrome.**
- ✓ **A head injury incurred from an automobile crash.**
- ✓ **Diagnosis of HIV/AIDS.**

- **Community health nurse may interact with clients having either continuous or episodic health care needs, or both.**

- **For example:**

**When can parents expect a child with Down syndrome to begin toilet training.**

# HEALTH FOR ALL

- **A Primary Health Care Initiative**

**"Health for All" was formally expressed in the *Declaration of Alma-Ata* in 1978.**

- **Functions:**

**The Declaration of Alma-Ata lists the eight basic functions of PHC:**

1. **Education concerning prevailing health problems and the methods of preventing and controlling them.**

2. **Promotion of food supply and proper nutrition.**
3. **An adequate supply of safe water and basic sanitation.**
4. **Maternal and child health, including family planning.**
5. **Immunization against major infectious diseases.**
6. **Prevention and control of locally endemic diseases.**
7. **Appropriate treatment of common diseases and injuries.**
8. **Provision of essential drugs.**

# Major Goals for Community Health Nursing

1. Care of ill, disabled, and suffering in non-hospital setting.
2. Support of development and well-being throughout the lifecycle.
3. Promotion of human relatedness and mutual caring.
4. Promotion of self-responsibility regarding health and well-being.
5. Promotion of relative safety in the environment while conserving resources.

# COMPONENTS OF COMMUNITY HEALTH PRACTICE "Mission of the CHN "

**These components are:**

**(1) Promotion of health,**
**(2) prevention of health problems,**
**(3) treatment of disorders,**
**(4) rehabilitation,**
**(5) evaluation, and**
**(6) research**

# 1) Promotion of health

## Definition:

1. "Actions related to lifestyles and choices that maintain or enhance a populations health". (WHO)
2. "Any combination of educational, organizational, economic, and environmental supports for behavior and conditions of living conductive to health".

- Promotion of health is recognized as one of the most important components of public health and community health practice.
- Health promotion includes all efforts that seek to move people closer to optimal well-being or higher levels of wellness. Nursing, in particular, has a social mandate for engaging in health promotion.
- Community health promotion, then, encompasses the development and management of health care services that are responsive to community health needs. Wellness programs in schools and industry are examples, they are useful when they are accompanied by desire, opportunity, and resources that encourage more healthful practices.

- Health promotion programs and activities includes many forms of **health education**.

- For example:

- Physical activity and fitness.
- Nutrition.
- Tobacco use.
- Teaching the dangers of alcohol and drug use.
- Family planning.
- Mental health and mental disorders.
- Violent and abusive behavior.
- Educational and community-based programs.

- The goal of health promotion is to raise levels of wellness for individuals, families, populations, and communities.
- Community health efforts accomplish this goal through a three-pronged effort to:

1. Increase the span of healthy life for all citizens.
2. Reduce health disparities among population groups.
3. Achieve access to preventive services for everyone.

## 2) Prevention of health problems

- Prevention of health problems constitutes a major part of community health practice.
- Prevention means anticipating and averting problems or discovering them as early as possible to minimize potential disability and impairment.
- It is practiced on three levels in community health:

(1) primary prevention,
(2) secondary prevention, and
(3) tertiary prevention (Neuman, 2001).

# I. Primary Prevention:

- Primary prevention obviates the occurrence of a health problem; it includes measures taken to keep illness or injuries from occurring.
  It is applied to a generally healthy population and precedes disease or dysfunction.

- Examples:

- Genetic counseling

- Immunization

- Healthy diet and healthy lifestyle

- Other examples of primary prevention activities by a community health nurse include:

✓ Encouraging elderly people to install and use safety devices (e.g. grab bars by bathtubs, hand rails on steps) to prevent injuries from falls.

✓ Teaching young adults healthy lifestyle behaviors so that they can adopt changes for a lifetime, for themselves and their children.

✓ Working through a local health department to help control and prevent communicable diseases such as rubeola, poliomyelitis, or varicella by providing regular immunization programs.

# II. Secondary Prevention

- Involves efforts to detect and treat existing health problems at the earliest possible stage when disease or impairment already exist.
- Examples:

✓ Hypertension and cholesterol screening programs in many communities help to identify high-risk individuals and encourage early treatment to prevent heart attacks or stroke.

✓ Teaching breast and testicular self-examination, encouraging regular mammograms and pap smears for early detection of possible cancer.

✓ Providing skin testing for tuberculosis for high-risk groups.

- Secondary prevention attempts to discover a health problem at a point when intervention may lead to its control or eradication.
- This is the goal behind testing of water and soil samples for contaminants and hazardous chemicals in the field of community environmental health.
- It also prompts community health nurses to watch for early signs of child abuse in a family, emotional disturbances in a group of widows, or alcohol and drug abuse among adolescents.

## III. Tertiary Prevention

- Tertiary prevention attempts to reduce the extent and severity of a health problem to its lowest possible level, so as to minimize disability and restore or preserve function.
- Examples:

✓ Treatment and rehabilitation of persons after a stroke to reduce impairment.

✓ Postmastectomy exercise programs to restore functioning.

✓ Early treatment and management of diabetes to reduce problems or slow their progress.

## 3) Treatment of disorders

- It focuses on the illness end of the continuum and is the remedial aspect of community health practice.

This occurs by three methods:

1) Direct service to people with health problems, e.g. home visits, provides an educational program and support group for people wanting to stop smoking or lose weight.

2) Indirect service that helps people with health problems to obtain treatment.

- e.g. a young woman with postpartum bleeding, assisted by the community health nurse, can obtain an immediate appointment with a physician at the local clinic.

3) Development of programs to correct unhealthy conditions.

- e.g. one community with a high incidence of alcoholism and drug abuse initiated a chemical dependency counseling and treatment center.

## 4) Rehabilitation

Involves efforts to reduce disability and, as much as possible, restore function.

## 5) Evaluation

- Is the process by which that practice is analyzed, judged, and improved according to established goals and standards.
- Evaluation of health and health care should be an integral part of every kind of health service, from individual practice to national and international programs.

## 6) Research

Is a systematic investigation to discover facts affecting community health and community health practice, solve problems, and explore improved methods of health service.

**Epidemiology** (the study of health and disease determinants and distribution in populations), and **Biostatistics** (the science of statistically measuring population health conditions) are the primary public health measurement and analytic sciences underlying community health practice.

# CHARACTERISTICS OF CHN

Eight characteristics of community health nursing are particularly salient to the practice of this specialty:

(1) it is a field of nursing;
(2) it combines public health with nursing;
(3) it is population focused;
(4) it emphasizes prevention, health promotion, and wellness;
(5) it promotes client responsibility and self-care;
(6) it uses aggregate measurement and analysis;
(7) it uses principles of organizational theory; and
(8) it involves interprofessional collaboration.

## (1) it is a field of nursing;

The two characteristics of any specialized nursing practice are:

(1) specialized knowledge and skills, and

(2) focus on a particular set of people receiving the service.

- These two characteristics are true for community health nursing.
- As a specialty field of nursing, community health nursing adds public health knowledge and skills that address the needs and problems of communities and aggregates and focuses care on communities and vulnerable populations.

## (2) it combines public health with nursing;

**Community health nursing, then, as a field of nursing;**

- It "Combines nursing science with public health science to formulate a community-based and population-focused practice" (Williams, 2000).
- It "synthesizes the knowledge from the public health sciences and professional nursing theories to improve the health of communities and vulnerable populations" (ANA, 1999, 2000).

Knowledge of the following elements of public health is essential to community health nursing (ANA, 1999):

1. History and philosophy of public health, including emphasis on the greatest good for the greatest number.
2. Concept of aggregates—assessing needs, planning and providing services, and evaluating services' impact on population groups—including aggregate-level decision-making.
3. Priority of preventive, protective, and health-promoting strategies over curative strategies.

4. Means for measurement and analysis of community health problems, including epidemiologic concepts and Biostatistics.
5. Influence of environmental factors on aggregate health.
6. Principles underlying management and organization for community health, because the goal of public health is accomplished through organized community efforts.
7. Public policy analysis and development.
8. Health advocacy and the political process.

## (3) it is population focused;

The central mission of public health practice is to improve the health of population groups, community health nursing shares this essential feature, it is **population focused; meaning** that it is concerned for the health status of population groups and their environment.

## (4) it emphasizes prevention, health promotion, and wellness;

- In community health nursing, the promotion of health and prevention of illness are of first-order priority. There is less emphasis on curative care.
- These include services to mothers and infants, prevention of environmental pollution, school health programs, senior citizens' fitness classes, and "workers' right-to-know" legislation that warns against hazards in the workplace.

## (5) it promotes client responsibility and self-care;

- The goal of public health, “to increase quality and years of healthy life and eliminate health disparities” (UDHHS, 2000), requires a partnership effort.
- Clients’ health status and health behavior will not change unless people accept and apply the proposals (developed in collaboration with clients) presented by the community health nurse.

## (6) it uses aggregate measurement and analysis;

- Community health nursing uses aggregate measurement and analysis. The need to collect and examine data on the entire population under study before making intervention decisions is fundamental to community health nursing in a public health practice.
- Health status, environmental factors, health related services, economic patterns, and social policy are among the many foci of community health research and evaluation.

## (7) it uses principles of organizational theory;

- Community health nursing uses principles from organizational theory to provide effective administration of health care services.
  Public health has long been defined as the protection and improvement of community health through organized community efforts.
- As community health nurses carefully assess group and community needs, establish priorities, and plan, implement, and evaluate services, they are using public health management and organizational principles.

## (8) it involves interprofessional collaboration;

- Community health nurses must work in cooperation with other team members, coordinating services and addressing the needs of population groups.
- This interprofessional **collaboration** among health care workers, other professionals, and clients is essential for establishing effective services and programs.

# Standards of Community Health Practice

## ☒Standards 1: Assessment

- The public health nurse systemically collects data that are comprehensive and accurate.

## ☒Standards 2: Population Diagnosis and Priorities

- The public health nurse analyzes the assessment data to determine the population diagnosis and priorities.

## Standards of Community Health Practice Cont. …….

### ☒Standards 3: Outcomes identification

- The public health nurse identifies expected outcomes for a plan that is based on population diagnosis and priorities.

### ☒Standards 4: Planning

- The public health nurse develops a plan that reflects best practices by identifying action plans and alternatives to attain expected outcomes.

## Standards of Community Health Practice Cont. …….

### ☒Standards 5: Implementation

The public health nurse implements the identified plan by partnering with others.

### ☒Standards 5A: Coordination

The public health nurse coordinates programs, services and other activities to implement the identified plan.

## Standards of Community Health Practice Cont. …….

☒**Standards 5B: Health Education & H. Promotion**

The public health nurse employs multi strategies to promote health, prevent disease, and ensure a safe environment for population.

☒**Standards 5C: Consultation**

The public health nurse provides consultation to various community groups and officials to facilitate the implementation of programs and services.

## Standards of Community Health Practice Cont. …….

☒**Standards 5D: Regulatory Activities**

The public health nurse identifies, interprets and implements public health laws, regulations and policies.

☒**Standards 6: Evaluation**

The public health nurse evaluates the health status of the population. (ANA 2007, Newly edition in early 2013)

# Chapter 2

# HISTORY AND EVOLUTION OF COMMUNITY HEALTH NURSING

# Historical Development of Community Health Nursing

- Community health nursing is the product of centuries of responsiveness and growth. Its practice has adapted to accommodate the needs of a changing society, yet it has always maintained its initial goal of improved community health.

- Community health nursing’s development, which has been influenced by changes in nursing, public health, and society, can be traced through several stages.

- The history of public health nursing, since its recognized inception in Europe, and more recently in America, encompasses continuing change and adaptation. (Hein, 2001)

- The historical record reveals a professional nursing specialty that has been on the cutting edge of innovations in public health practice and has provided leadership to public health efforts.

- In tracing the development of public health nursing and, later, community health nursing, the leadership role has been clearly evident throughout its history.

- Nurses in this specialty have provided leadership in planning and developing programs, in shaping policy, in administration, and in the application of research to community health.

## Four general stages mark the development of community health / public health nursing:

1) The early home-care stage (before the mid-1800s) emphasized care to the sick poor in their homes by various lay and religious orders.
2) The district nursing stage (mid-1800s) included voluntary home nursing care for the poor by specialists or “health nurses” who treated the sick and taught wholesome living to patients.

3) The public health nursing stage (1900 to 1970) was characterized by an increased concern for the health of the general public.
4) The community health nursing stage (1970 to the present) includes increased recognition of community health nursing as a specialty field, with focus on communities and populations.

## TABLE 2.1 Development of Community Health Nursing

| Stages | Focus | Nursing Orientation | Service Emphasis | Institutional Base (Agencies) |
|---|---|---|---|---|
| Early home care (Before mid-1800s) | Sick poor | Individuals | Curative | Lay and religious orders |
| District nursing (1860–1900) | Sick poor | Individuals | Curative; beginning of preventive | Voluntary; some government |
| Public health nursing (1900–1970) | Needy public | Families | Curative; preventive | Government; some voluntary |
| Emergence of community health nursing (1970–present) | Total community | Populations | Health promotion; illness prevention | Many kinds; some independent practice |

# Societal Influences On The Development Of CHN

Many factors have influenced the growth of CHN:

**Six are particularly significant:**

1) Advanced technology,
2) Progress in causal thinking,
3) Changes in education,
4) Demographic changes and the role of women,
5) The consumer movement,
6) Economic factors.

# 1- Advanced Technology

- Advanced technology has contributed in many ways in shaping the practice of community health nursing.
- For example, technologic innovation has greatly improved health care, nutrition, and lifestyle and has caused a concomitant increase in life expectancy.
- Better meeting the needs of the elderly population and addressing chronic conditions.
- Advanced technology has been a strong force behind industrialization, large-scale employment, and urbanization. We are now primarily an urban society, with approximately 75% of the world's population living in urban or suburban areas.

# However:

- Population density leads to many health-related problems, particularly the spread of diseases and increased stress.
- In addition, changes in transportation and energy sources in industry also have increased environmental pollution and industrial hazards.
- Community health nurses are learning how to combat these urban health problems.
- Community health nurses now are in the business of information distribution, and they use computer technologies to enhance the efficiency and effectiveness of their services.
- Geographic Information Systems (GIS) is an example of emerging computer technology that the community health nurse can use to design and evaluate population-focused programs.

# 2- Progress in Causal Thinking

- Relating disease or illness to its cause is known as causal thinking in the health sciences.
- Progress in the study of causality, particularly in epidemiology, has significantly affected the nature of community health nursing.
- *The germ theory of disease causation;* established in the late 1800s, was the first real breakthrough in control of communicable disease.
- At that time, it was established that disease could be spread or transmitted from patient to patient or from nurse to patient by contaminated hands or equipment.
- Nurses incorporated the teaching of cleanliness and personal hygiene into basic nursing care.

- A second advance in causal thinking was the model interactions among:
  - a causative agent,
  - a susceptible host, and
  - the environment.
- This information offered community health nurses new ways to control and prevent health disorders.
  - For example, nurses could decrease the vulnerability of individuals (hosts) by teaching them healthier lifestyles.
  - They could instigate measles vaccination programs as a means of preventing the organism (agent) from infecting children.
  - They could promote proper disinfection of a neighborhood swimming pool (environment) to prevent disease.

Further progress in causal thinking led to the recognition that not just one single agent, but many factors a multiple causation approach contribute to a disease or health disorder.

For example

A food poisoning outbreak that is associated with a restaurant might be caused not only by the *Salmonella* organism but also by improper food handling and storage, lack of adherence to minimum food preparation standards, and lack of adequate health department supervision and enforcement.

- Nursing’s contribution to public health adds a further application to causal thinking.
- That is, nursing seeks to identify and implement the causes, or contributing factors, of wellness.
- Community health nurses do more than prevent illness; they seek to promote health.
- **Nurses promote healthier lifestyle practices as:**
  - Eating low-fat diets,
  - Exercising,
  - Managing stress, and
  - Maintaining social support systems

# 3- Changes in Education

- Changes in nursing education, had an important influence on community health nursing practice.
- Health teaching had greater importance in practice. For the learner, education has led to more responsibility.
- Scientific inquiry (health research) considered basic to progress, has created a dramatic increase in knowledge.

- Now, community health nurses need skills in such areas as: population assessment, policy making, research, management, collaborative functioning, and critical thinking.
- As the result of expanding education, community health nurses have to improve their skills, sharpen their knowledge, and clarify their roles.

# 4- Demographic Changes and the Role of Women

- Changing demographics, such as changing role of women, varying numbers of births and deaths, immigration and rapidly increasing population of elderly persons, affect community health nursing. Monitoring these changes is essential for relevant and effective nursing services.
- Equally important is a diverse and representative workforce in nursing; salaries for nurses compare favorably with those for other workers who have four years of education in fields other than health care, such as education, human services (social work), and business.

- Although the diversity of career options and employment opportunities for women has been a positive social factor, these gains have decreased the number of women entering nursing.
- As a profession, nursing's contributions and status have improved, but its ability to compete with careers offering higher pay and status remains problematic.
- Nurses still struggle for equality—equality of recognition, respect, and autonomy, as well as job selection, equal pay for equal work, and equal opportunity for advancement in the health field.
- The women's movement has contributed to community health nursing's gains in assuming leadership roles, but a need for greater influence and involvement remains for all nurses.

# 5- Consumer Movement

- Health care consumers are viewed as active members of the health team rather than as passive recipients of care.
- They may contract with the community health nurse for family care or group services, represent the community on the local health board, or act as ombudsmen by serving as representatives or advocates for their community constituents.
- The consumer movement has also contributed to increased concern for the quality of health services, including a demand for more humane, personalized health care.
- Community health nurses seek to provide holistic care by collaborating with others to offer more coordinated, comprehensive, and personalized services—a case-management approach.

# 6- Economic Forces

- Unemployment and the rising cost of living, combined with increasing health care costs, have resulted in:

- Numerous people with no health insurance.
- With limited or no access to health services.
- Poor people become more vulnerable to health problems.

## Other economic forces affecting C.H.N. are:

- Changing health care financing patterns (including prospective payment and diagnosis-related groups);
- Decreased federal, state, and local subsidies of public health programs;
- Pressures for health care cost containment through managed care;
- Increased competition and managed competition among providers of health services.

# PREPARATION FOR C.H.N.

- This day may not represent the bigger picture of the community health nurse's role in community advisory panels, grant writing for new programs, or participation in or presentation of in-service programs.
- Academic preparation for this role is necessary, as is continuous professional development, and this training must meet the requirements of both employers and, in many instances, state regulations.

# Academic Preparation:

- The minimum preparation for community health nurses in many states has been graduation from a baccalaureate-level nursing program, only those who have completed a baccalaureate nursing program can apply for this certificate, and only people with the certificate can take jobs as public health nurses.
- A comprehensive nursing education that is rich in leadership, management, research, health maintenance and promotion, disease prevention, and community health experience is needed to meet the demands of this specialty.

# Professional Development

- Completion of a baccalaureate education may not be sufficient educational preparation for the more demanding community health nursing settings.
- It is mandated that nurses participate in continuing education programs and receive continuing education units.
- A master's degree program in nursing is a longer commitment and gives the nurse greater marketability.
- A master's degree in business, public health, education, or epidemiology can lead to management positions, agency teaching, or research positions.
- A doctoral program may be the next educational step for those wanting university teaching, research, or upper-level administrative positions.

Chapter 3

# Roles of Community Health Nurse

Historically, community health nurses have engaged in many roles.

From the beginning, nurses:

- Provided care to the sick,
- Taught positive health habits and self-care,
- Advocated on behalf of needy populations,
- Developed and managed health programs,
- Collaborated with others to implement changes in health services.

# CORE PUBLIC HEALTH FUNCTIONS

Community health nurses work as partners within a team of professionals (in public health and other disciplines), nonprofessionals, and consumers to improve the health of populations.

The various roles and settings for practice based on three primary functions of public health:

- assessment,
- policy development, and

They are foundational to all roles and are applied at three levels of service:

- to individuals,
- to families, and
- to communities.

Regardless of the role or setting of choice, these three essential responsibilities direct the work of all community health nurses.

# Assessment

Assessment, means that the community health nurse must gather and analyze information that will affect the health of the people.

Assessment is the systematic collection, assembly, analysis, and dissemination of information about the health of a community.

The nurse needs to determine health needs, health risks, environmental conditions, resources, depending on the persons, community, or population targeted for intervention.



# Policy Development

Policy development is enhanced by the synthesis and analysis of information obtained during assessment.

Typically, the nurse recommends specific training and programs to meet identified health needs of target populations.

That by raising the awareness of key policy makers about factors such as health regulations and budget decisions that negatively affect the health of the community.

Additional data may be needed to identify trends in groups or clusters of families, so that

# Assurance

Assurance– activities that make certain that services are provided based on standards of care.

Community health nurses perform the assurance function by:

- provide service to target populations,
- improve quality assurance activities,
- maintain control of communicable disease,
- participate in research,
- provide expert consultation,
- and provide services based on standards

# TEN ESSENTIAL SERVICES OF PUBLIC HEALTH

1. Monitor health status to identify community health problems.
2. Diagnose and investigate health problems and health hazards in the community.
3. Inform, educate, and empower people about health issues.
4. Mobilize community partnerships to identify and solve health problems.
5. Develop policies and plans that support individual and

# TEN ESSENTIAL SERVICES OF PUBLIC HEALTH

6. Enforce laws and regulations that protect health and ensure safety.
7. Link people to needed personal health services and assure the provision of health care when otherwise unavailable.
8. Assure a competent public health and personal health care workforce.
9. Evaluate effectiveness, accessibility, and quality of personal and population-based health services.
10. Research for new insights and innovative solutions to health problems.

# Basic Requirements to be C.H.N.

1. Studying epidemiology, statistics, research.
2. Orientation to health care system.
3. Identification of high risk population.
4. Intervention with high risk population.
5. Application of public health concepts to the care.
6. Orientation to the regulation affecting public health nursing practice.

# Roles of Community Health Nurses

Over time, the role of the community health nurse has broadened.

Community health nurses wear many hats while conducting day-to-day practice.

This chapter examines seven major roles:

(1) clinician, (2) educator, (3) advocate, (4) manager, (5) collaborator, (6) leader, and (7) researcher

# 1. Care Provider "Clinician Role"

- The most familiar role of the community health nurse is that of clinician or care provider.
- The Clinician role in community health means that the nurse ensure that health services are provided not just to individuals and families, but also to groups and populations.
- In community health the clinician views clients in the context of larger systems. The family or group must be considered in totality.

- The community health nurse provides care along the entire range of the wellness-illness continuum; however, promotion of health and prevention of illness are emphasized.
- Skills in observation, listening, communication, counseling, and physical care are important for the community health nurse.
- Recent concerns for environment, sociocultural, psychological, and economic factors in community health have created a need for stronger skills in assessing the needs of

- For community health nurses, the clinician role involves certain emphases that are different from those basic nursing.

  Three clinician emphases, in particular, are useful to consider here:

- Holism.
- Health promotion.
- Skill expansion.

# Holistic Practice

- Holistic nursing care encompasses the comprehensive and total care of the client in all areas, such as physical, emotional, social, spiritual, and economic.

# Focus on Wellness

- The clinician role in community health also is characterized by its focus on promoting wellness. The community health nurse provides services along the entire range of the health continuum but especially emphasize promotion of health and prevention of illness.

- Nursing service includes seeking out clients who are at risk for poor health and offering preventive and health promotion service, rather than waiting for them to come for help after problems arise.
- The community health nurse may help employees of a business learns how to live healthier lives or work with the expected changed behavior, or work with a group of people who want to quit smoking

- Groups and populations are identified that may be vulnerable to certain health threats, and preventive and health promoting programs can be designed. Examples include:

- Immunization of preschoolers.
- Family planning programs.
- Cholesterol screening.
- Prevention of behavioral problems in adolescents.
- Protecting and promoting the health of vulnerable population is an important component of the clinician role.

# Expanded Skills

- Many different skills are used in the role of the C.H. clinician.
- In the early years of C.H.N. , emphasis was placed on physical care skills.
- With time, skills in observation, listening, communication, and counseling become integral to the clinician role as it grew to encompass an increased emphasis on psychological and sociocultural

- Recently, environmental and community-wide consideration such as:

- Problems caused by pollution.
- Violence and crime.
- Drug abuse.
- Unemployment.
- Poverty.
- Homelessness.

Have created a need for stronger skills in assessing the needs of groups and populations and intervening at community level.

- The clinician role in population-based nursing also requires skills in:
  - Collaboration with consumers and other professionals.
  - Use of epidemiology and biostatistics.
  - Community organization and development.
  - Research.
  - Program evaluation.
  - Administration and leadership.

# 2. Educator Role

- A second important role of the community health nurse is that of Educator or health teacher. Health teaching is one of the major function of the community health nurse.
- The educator role is especially usefulness in promoting the public's health for at least two reasons:

1. Community clients usually are not actually ill and can absorb and act on health information.

For example:

A class of expectant parents, unhampered by significant health problems, can grasp the relationship of diet to fetal development.

They understand the value of specific exercises to childbirth process, are motivated to learn, and are more likely to perform those exercises. Thus, the educator has the potential for finding

2. The educator role in community health nursing is significant because a wider audience can be reached.

With an emphasis on population and aggregates, the educational efforts of community health nursing care appropriately targeted to reach many people.

# 3. Advocate Role

- The issue of clients' rights is important in health care. Every patient or client has the right to receive just, equal, and human treatment.
- Our current health care system often is characterized by fragmented and depersonalized services, and many clients- especially the poor, the disadvantaged, those without health insurance, are denied their rights. They become frustrated, confused, degraded, and unable to cope

- The community health nurse often acts as an advocate for clients, pleading their cause or acting on their behalf, Clients may need some one:

- To explain which services to expect, which services they ought to receive.
- To make referrals as needed.
- To write letters to agencies or health care providers for them.
- To assure the satisfaction of their needs.

# Advocacy Goals:

- There are two underlying goals in client advocacy:

1. To help clients gain greater independence or self-administration, until they can reach the needed information and access health and social services for themselves, the community health nurse acts as an advocate for the clients by:

- Showing them what services are available.
- The ones to which they are entitled, and

# Advocacy Goals:

2. To make the system more responsive and relevant to the needs of clients, by calling attention to inadequate, inaccessible, or unjust care, community health nurses can influence change.

# Advocacy Actions

- The advocate role incorporate four characteristics actions:

1. Being assertive.
2. Taking risks.
3. Communicating and negating well.
4. Identifying resources and obtaining results.

# 4. Manager Role

- Community health nurses, like all nurses, engage in the role of managing health services.
- The nurse serves as a manger when:
  - Overseeing client care as a case manager.
  - Supervising ancillary staff.
  - Running clinics.
  - Conducting community health needs assessment projects.

- In each instance, the nurse engages in four basic functions that make up the management process:

1. Planning.
2. Organizing.
3. Leading.
4. Controlling and evaluating.

These activities are sequential and yet also occur simultaneously for managing service objectives.

# 1. Nurse as Planner

- The first function in the management process is planning. A planner sets the goals and direction for the organization or project and determines the means to achieve them.
- Specifically, planning includes:
  - Defining goals and objectives.
  - Determining the strategy for reaching them.
  - Designing a coordinated set of activities

# 2. Nurse as Organizer

- The second function of the manager role is that of Organizer.
- This involves:

Designing a structure within which people and tasks function to reach the desired objectives.

# 3. Nurse as Leader

- In the manager role, the community health nurse also must act as A Leader.
- As a leader the nurse directs, influences, or persuade others to effect change so as to positively affect people's health and move them toward a goal.

- The leading functions include:

- Persuading and motivating people.
- Directing activities.
- Ensuring effective two-way communication.
- Resolving conflicts.
- Coordinating the plan.

# 4. Nurse as Controller and Evaluator

- The fourth management function is to control and evaluate projects or programs.
- A Controller monitors the plan and ensure that it stays on course.
- In this function, the CHN must realize that plans may not proceed as intended and may need adjustments or corrections to reach the desired results or goals.
- At the same time, the nurse must compare and judge performance and outcomes against previously set goals and standards; a process that forms the Evaluator aspect of management.

# 5. Collaborator Role

- Community health nurses seldom practice in isolation.
- They must work with many people, including:

| | |
|---|---|
| Clients | Other nurses |
| Physicians | Health educators |
| Social workers | Physical therapists |
| Nutritionists | Occupational therapists |
| Psychologists | Epidemiologists |
| Biostatisticians | Attorneys |
| Secretaries | Environmentalists |
| City planners | Teachers |

- As members of health team, community health nurses assume the role of Collaborator.
- Which means to work jointly with others in a common endeavor, to cooperate as partners.
- Successful community health practice depends on his multidisciplinary collegiality and leadership.

- The community health nurse's collaborator role requires:
  - Skills in communicating.
  - Interpreting the nurse's unique contribution to the team.
  - Acting assertively as an equal partner.
  - The collaborator role also may involve functioning as a consultant.

## The following example shows a C.H.N functioning as collaborator:

- A school nurse noticed a rise in the incidence of drug use in her schools. She initiated a counseling program after joint planning with students, parents, teachers, the school psychologist, and a local drug rehabilitation center.

# 6. Leadership Role

- CHNs are becoming increasingly active in the leadership role, the leadership role focuses on affecting change, thus the nurse becomes an agent of change.
- As leaders, CHNs seek to initiate changers that positively affect people's health.
- They also seek to influence people to think and behave differently about their health and the factors contributing to it.

## 6. Leadership Role Cont. ……

- At the community level, the leadership role may involve working with a team of professionals to direct and coordinate such projects as a campaign to eliminate smoking in public areas.

# 7. Researcher Role

- In the Researcher role, community health nurses engage in systematic investigation, collection, and analysis of data for solving problems and enhancing community health practice.

# The Research Process

- CHNs practice the researcher role at several levels. In addition to everyday inquiries, community health nurses often participate in agency and organizational studies to determine such matters as job satisfaction among CHNs and risk associated with home visiting.
- The researcher role, at all levels, helps to determine needs, evaluate effectiveness of care, and develop theoretic bases for

- Research literacy means to search again – to investigate, to discover, and interpret facts.
- All research in community health, from the simplest inquiry to the most complex epidemiologic study, use the same fundamental process.

# The research process involves the following steps:

1. Identify an area of interest.
2. Specify the research question and statement.
3. Review of literature.
4. Identify the conceptual framework.
5. Select research design.
6. Collect and analyze data.
7. Interpret the results.
8. Communicate the findings.

- Research has a significant impact on community health and nursing practice in three ways:

1. It provides a new knowledge that help to shape the health policy.
2. Improves the service delivery.
3. Promotes the public's health.

- Nurses must become responsible users of research, keeping abreast of new knowledge and applying it in practice.
- Nurses must learn to evaluate nursing research articles critically, assessing their validity and applicability to their own practice.
- Nurses should subscribe to and read nursing research journals and discuss research studies with colleagues and supervisors.

- More community health nurses must also conduct research studies of their own or in collaboration with other community health professionals.
- A commitment to use and conduct of research will move the nursing profession forward and enhance its influence on the health of at-risk populations.

# SETTING FOR CHN PRACTICE

- The setting in which the CHN are practiced their different roles are grouped into seven categories:

1. Homes.
2. Ambulatory service settings.
3. Schools.
4. Occupational health settings.
5. Residential institutions.
6. Faith communities
7. The community at large.

# 1. Homes

- For long time, the most frequently used setting for community health nursing practice was the home.
- In the home, all of the community health nursing role's, to varying degrees, are performed.

- Clients who are discharged from acute care institutions, such as hospitals or mental health facilities, are regularly referred to community health nurses for continued care and follow-up. Here, the community health nurse can see clients in a family and environment context, the service can be tailored to the client's unique needs.

## For example:

- Mr. White, 67 years of age, was discharged from the hospital with a colostomy. The community health nurse immediately started home visits. She met with Mr. White and his wife to discuss their needs as a family and plan for Mr. Whit's care and adjustment to living with a colostomy.

- Practicing the clinician and educator roles, she reinforced and expanded on the teaching started in the hospital for colostomy care; including:

- Bowel training.
- Diet.
- Exercises.
- Proper use of equipments.

- As a part of total family care plan, CHN provided some forms of physical care for Mr. White as well as counseling, teaching, and emotional support for both Mr. White and his wife.
- In addition to consulting with the physician and social service worker, she arranged and supervised visits from the home health aide, who gave personal care and homemaker services. The CHN thus performed the manager, leader, and collaborator roles.

- The home also is a setting for health promotion. Many community health nursing visits focus on assisting families to understand and practice healthier living behaviors.
- Nurses may, for example, instruct clients on parenting infant care, child discipline, diet, exercise, coping with stress.

- In the home, unlike most other health care setting, clients are on their own turf, they feel comfortable and secure in familiarity surroundings and often are better able to understand and apply health information. Client self-respect can be promoted, because the client is host and the nurse is a gust.

- Sometimes, the thought of visiting in client's home can cause anxiety for the nurse. This may be the nurse's first experience outside the acute care, long-term care, or clinic setting.
- Visiting clients in their own environment can make the nurse feel uncomfortable. The nurse may be asked to visit families in unfamiliar neighborhoods and must walk through those neighborhoods to visit the client.

- Frequently, fear of unknown is the real fear, and often it has been enhanced by stories from previous stories from previous nurses. This may be the same feeling as that experienced when caring for your first client, first having a client in the ICCU.
- The community health nurse can collaborate with various types of home care providers, including hospitals, other nurses, physicians, rehabilitation therapist, and durable medical equipment companies to ensure

# 2. Ambulatory Service Settings

- Ambulatory Service Settings include a variety of venues for CHN practice in which the clients come for day or evening services that do not include overnight stays.

Example of an ambulatory service setting:

- Multiple clinics offering comprehensive services are community based or are located in outpatient departments of hospitals.
- A single clinic, such as, family planning

- Some kinds of day care centers, such as those for physically disabled or emotionally disturbed adults.
- Health offices are another type of ambulatory care setting, for example: a community health nurse associate with a health maintenance organization sees clients in the office and undertakes:

- Screening.
- Referral.

- Another type of ambulatory service setting includes places where services are offered to selected groups.

For example:

- CHN practice in migrant camps.
- CHN practice in remote mountain and coal-mining communities.

In each ambulatory setting, all of the community health nursing roles are used to varying degrees.

# 3. Schools

- Schools of all levels make up a major group of settings for CHN practice.
- The CHN may work with groups of students in preschool settings, as well as vocational or technical schools, colleges and university settings.
  Specialized schools, such as those for the developmentally disabled, are another setting for CHN practice.

# Responsibilities of School Nurse

- School nurse: a specialty branch of professional nursing that serves the school-age population.
- The primary responsibilities of the school nurse are to prevent illness, to promote and maintain the health of the school community.
- The National Association of School Nurses (NASN) created seven roles to describe their practice; the school nurse:

1. Provides direct healthcare to students and staff

2. Provides leadership for the provision of health services.
3. Provides screening and referral for health conditions.
4. Promotes a healthy school environment.
5. Promotes health.
6. Serves in a leadership role for health policies and programs.
7. Serves as a liaison between school personnel, family, community and health care providers.

- The school nurse is responsible for providing a broad range of services, including the following eight components of a coordinated school health program:

1. Health services: Serves as the coordinator of the health services program, provides nursing care, and advocates for health rights.

Health services include programs such as:

- Vision and hearing screening.
- Scoliosis screening.

- Oral health.
- TB screening.
- Immunization assessment, and monitoring medication administration.
- Care of children with specialized health care needs.
- First aid.
- Assessment of acute health problems.
- Health examinations (especially for athletic participation or school entry).

## 2. Health education:

Provides appropriate health information that promotes informed healthcare decisions, promotes health, prevents disease, and enhances school performance.

## 3. Environment:

Identifies health and safety concerns in the school community, promotes a safe and nurturing school environment.

- The function of the school nursing practice is the promotion of healthful school living.
- Emphasis on healthful physical environment includes:

  Proper selection, design, organization, and maintenance of the physical plant

Consideration should be shown for areas such as adaptability to students needs:

1. Safety, safety of the school bus.
2. Visual, thermal, and acoustic factors.
3. Aesthetic values.
4. Sanitation.

## 4. Nutrition:

Supports school food service programs and promotes the benefits of healthy eating patterns. Safety of food service, school breakfast and lunches.

## 5. Physical education/activity:

Promotes healthy activities, physical education, and sports policies/ practices that promote safety and good sportsmanship.

## 6. Counseling/mental health:

Provides health counseling, assesses mental health needs, provides interventions, refers students to appropriate school staff or community agencies, and provides follow-up once treatment is prescribed.

## 7. Parent/community involvement:

Promotes community participation in assuring a healthy school and serves as school liaison to a health advisory committee.

## 8. Staff wellness:

Provides health education and counseling, promotes healthy activities and environment for school staff.

# 4. Occupational Health Setting

- Business and industry provide another group of settings for community health nursing practice.
- Employee health has long been recognized as making a vital contribution to individual lives, productivity of business, and well being of the entire nation.
- Organizations are expected to provide a safe and healthy work environment in addition to offering insurance for health care.

# Occupational Health Nurses (OHN)s:

- Are registered nurses who independently observe and assess the worker's health status with respect to job tasks and hazards.
- Using their specialized experience and education, these registered nurses recognize and prevent health effects from hazardous exposures and treat workers' injuries/illnesses.

# Occupational Health Nurses (OHN)s:

1. Have special knowledge of workplace hazards and the relationship to the employee health status.
2. Understand industrial hygiene principles of engineering controls, administrative controls, and personal protective equipments.
3. Have knowledge of toxicology and epidemiology as related to the employee and the work site.

# The role of the occupational health nurse

- Occupational health nurses work in a variety of settings mainly industry, health services, commerce, and education.
- They can employed as independent practitioners or as part of a larger occupational health service team, often attached to a personnel department.
- Occupational health nurses are considered to be leaders in public health in the workplace setting.

# The O.H.N. role include:

1. The prevention of health problems, promotion of healthy living and working conditions.
2. Understanding the effects of work on health and health at work.
3. Basic first aid and health screening.
4. Workforce and workplace monitoring and health need assessment.
5. Health promotion.

6. Education and training.
7. Counseling and support.
8. Risk assessment and risk management.
9. Act as employee advocate, assuring appropriate job assignments for workers, and adequate treatment for job-related illness or injuries.

# 5. Residential Institutions

- Any facility where clients reside can be a setting in which a community health nursing is practiced.

- Residential institutions can include:

1. A halfway house in which clients live temporary while recovering from drug addiction.
2. Inpatient hospice program in which terminally ill clients live.

3. A continuity care center; in this setting, residents usually are elderly, increasingly more dependent and have many chronic health problems.

- The CHN functions as advocate and collaborator to improve the service.

- Residential institutions provide unique settings for the CHN to practice health promotion.

# 6- Faith Communities

- Faith community nursing finds its beginnings in an ancient tradition.
- The beginnings of CHN can be traced to religious orders, and for centuries, religious and spiritual communities were important sources of health care.
- In faith community nursing today, the practice focal point remains the faith community and the religious belief system provided by the philosophical framework.

Faith communities, where care is based on the philosophy of the religious organization.

# 7. Community at Large

- The community at large for community health nursing practice is not confined to  a specific philosophy, location, or building.
- When working with groups, population, or total community, the nurse may practice in many different places.

For example:

A community health nurse, as clinician and health educator, may work with a parenting group in town hall.

Another nurse, as client advocate, leader, and researcher, may study the health needs of a neighborhood's elderly population by collecting data throughout the area and meeting with resources people in many places.



# Environmental Health and Safety

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**Environmental Health** is concerned with assessing, controlling, and improving the impact people make on their environment and the impact of the environment on them.

The field of environmental health is concerned with all elements of the environment that influence people's health and well-being.

The conditions of workplaces, homes, or communities, including the many chemical, physical, and psychological forces present in the environment that affect human health, are important considerations.



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# Definition of Environmental Health

**Institute Of Medicine(IOM);** defines the **Environmental Health** as "freedom from illness or injury related to exposure to toxic agents and other environmental conditions that potentially detrimental human health" (IOM,1995, P.15).

# Importance of environmental health

- Preventive: a sanitary clean environment is free of vectors of diseases, rodents which may be reservoirs of infection, pathogenic agents of communicable diseases (infectious and parasitic)
- General welfare and health promotion of the population
- Comfort and increased productivity and quality of work
- Ethics: a clean environment is essential for human dignity

# Concepts and Theories Central to Environmental Health

- Assessing environmental health means more than looking for illness or disease-causing agents; it also means examining the quality of the environment.

- The nurse needs to consider the conceptual and theoretic approaches that are essential to assessing and controlling environmental health.



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# Preventive Approach

- The concept of prevention is vital to environmental health; those concerned with human and environmental health must monitor new developments and intervene to prevent problems from occurring.

- Health practitioners need to determine causal links between people and their environment to improve the health and well-being of both.



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## Preventive Approach…….

- Nurses must be particularly aware of environmental factors that have the potential to either promote or adversely affect the health of communities.
- All three levels of prevention must be employed, but most important is primary prevention. Prevention of a disease or hazard from occurring at all has the greatest benefit to the community.



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# Ecologic Perspective

An **ecosystem** is a community of living organisms and their interrelated physical and chemical environment. Within an ecosystem, any manipulation of one element or organism may have hazardous effects on the rest of the system. Therefore, no one factor, whether organism or substance, can be viewed in isolation.



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## Ecologic Perspective……

- For example, in produce processing, there are several points at which contamination can occur:

1- at production and harvest (growing, picking, and bundling);

2- during initial processing (washing, waxing, sorting, and boxing);

3- during distribution (trucking); and

4- during final processing (slicing, squeezing, shredding, and peeling).



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**Environmental justice** is a movement that has sought to ensure that no particular part of the population is disproportionately burdened by the negative effects of pollution.

- The U.S. Environmental Protection Agency (EPA) uses the following definition: “Environmental Justice is the fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies”.



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## Environmental justice Cont. …….

- It will be achieved when everyone enjoys the same degree of protection from environmental and health hazards and equal access to the decision-making process to have a healthy environment in which to live, learn, and work.



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# Long-range Environmental Impact

- When studying environmental health, it is important to consider the effect of positive or negative changes on the environment and on the people, animals, and plants living in it.
- This environmental impact must be viewed not only in terms of its consequences for people living now but also in terms of long-range impact on the human species.



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## Long-range Environmental Impact……..

- One must consider the health of future generations as well as present ones. Considerations should include food and fuel limitations of the natural environment, attendance to conservation through balancing of present and future needs, and prevention of the consequences of environmental abuse.



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# Evolution of Environmental Health

- Environmental influences on health have been present throughout human history. Interactions with the environment, and the conditions of that environment, have shaped humans' mental, emotional, and physical health since the beginning of time.

- From ancient tribal practices of burial of excreta to modern-day sewage treatment, humans have been concerned with how the environment would provide for their needs and affect their well-being



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# Evolution of Environmental Health

- **Florence Nightingale** identified the environment as one of the concepts in the nursing profession, she viewed the environment as a factor external to the patient that might influence health and illness. Then Florence Nightingale identified the need for *a clean environment*; she outlined *five essential points* in securing the health of houses; **"pure air, pure water, efficient drainage, cleanliness and light" (Nightingale, 1860)**
- **The Hippocratic** professed by modern physician concludes that the well-being of man is influenced by all *environmental factors*; the quality of the air, water, food, the wind and the topography of the land.

# Nursing Involvement in Environment Issues

- **In the 21st century** the relationship between the environment and human health will play a significant role in how we assess and address public health and the delivery of health care services.
- **In 2010 the American Nursing Association (ANA)** In Scope and Standards of Practice (ANA, 2010); in this document ANA clearly pronounced that environmental health is an essential component for all professional nursing practice.
- **Today** all nurses are challenged to address environmental problems. The healthy people 2020 objectives address many occupational and environmental health issues.

# Assessment of Environmental Hazards

- The primary environment for most individuals can be divided into three broad areas:
  The home, the worksite, and the community.
- Physical hazards in the community include biological and chemical hazards that exist in the environment.
- Psychological hazards in the community such as violence.

# Sources of Environmental Hazards

1) Environmental Hazards at Home
2) Environmental Hazards in the Occupational Setting (work)
3) Environmental Hazards in the Community

# Major Global Environmental Concerns

- The global perspective, and specifically the national perspective of environmental health provides a picture of humankind's attempts to protect populations. However, for much of our world's history; protecting people, other living beings, or the Earth itself has not been a priority or even a concern.
- Because of this history, major global environmental concerns now face the world, including overpopulation, ozone depletion, global warming, deforestation, wetlands destruction, desertification, energy depletion, inadequate housing, aesthetics, and environmental justice issues.



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# Overpopulation

- Globally, nearly one-third of the population is younger than 15 years. Until most recently, the elderly population (those 65 years and over) made up a relatively small percentage of the world's population. In 2002, this age group accounted for approximately 7% of the overall population.
- Compare that to projections for 2050, when the absolute number is expected to be tripled, bringing the percentage up to nearly 17% .
- The growth in the population over 65 will increasingly strain existing health care, pension, and social service systems.



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## Overpopulation…….

- In some nations, the population is projected to shrink. If low fertility rates continue in Germany, Italy, Russia, and Spain for instance, their populations will decrease by anywhere from 2% to 10 % by the year 2025.
  In contrast, countries such as Nigeria, Zambia, and Jordan have high fertility rates, and it is likely that their populations will increase by 30% to 60% over the same period.



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## Overpopulation…….

- When a population exceeds the ability of its ecosystem to either support it or acquire the support needed, or when it exceeds its ability to migrate to other ecosystems in a manner that preserves its standard of living, the population is said to be experiencing demographic entrapment.



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# Government's Role

- The government has a responsibility to prevent a population from exceeding the limits of the nation's resources and boundaries.
- The possible methods of preventing overpopulation, or solutions to it, are controversial, depending on one's culture, religious beliefs, and personal values and convictions. Ideally, the political system governing a country has a responsibility to provide a well-formed infrastructure of health and safety services for its population; economic development that provides employment, housing, and services; and political strength to provide stability to the nation.
- Population stability clearly has an impact on developing and developed societies alike.



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# ***Nurse's Role***

- Public health professionals, including community health nurses, have a responsibility in the area of overpopulation, both globally and locally.
- Interventions include the following:

(a) teaching families that birth spacing improves child and maternal survival and that a planned family is the best environment for a child's development;

(b) preventing high-risk pregnancies, such as those among teens and adult women who are infected with (HIV) or (AIDS);



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***Nurse's Role ……***

(c) preventing the growing epidemic of HIV/AIDS;

(d) Providing family planning education to prevent worldwide deaths from unsafe abortions; and

(e) providing prenatal care—because healthy mothers equal healthy children.

- These are key areas in which public health efforts can reap major rewards for families.



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# Town planning

Town planning is the policy of putting a scheme for the establishment of cities, towns and districts, taken into consideration future development and extension.

**The principles of town planning:**

- Division of town districts: industrial, commercial, and residential.
- Sufficient wide streets, parks, and play grounds.
- Sufficient spacing and open areas between building for good ventilation.
- General public services.

# A healthful housing environment

A healthful housing environment should meet the following criteria:

- Sanitary site and specifications of building
- Suitable number of rooms according to family size
- Adequate ventilation and lighting
- Heating and cooling system is necessary
- Sanitary water supply and waste disposal
- Cleanliness and insect control
- Safety measures for prevention of home accidents
- Damp-proofing of the foundations and plumbing system to prevent dampness of the building walls

# Hazards of bad/poor housing

- A higher incidence of general morbidity and mortality specially of infectious disease e.g. TB, diarrhea, common cold, influenza

- Increase risk of human accidents

- Social problems

# Ventilation

**Ventilation:** implies replacement of vitiated air and supply of fresh out door air, and control of quality of air with regard to temperature, humidity, and purity with a view to provide thermal environment that is comfortable and free from infection.

**Function**: gas exchange, cooling power, feeling of comfort and better performance of work.

**Ventilation can be achieved by:**

- Natural Ventilation: is brought about through doors and windows facing each other to allow cross ventilation.
- Mechanical Ventilation: include exhaust ventilation and air conditioning.

# Ventilation

## Community laws and regulation for ventilation:

- All building should have cross ventilation: 10 % for windows from the floor area or residential building, 20% for schools and public places.

# Lighting

**Types of lighting: is natural or sunlight, or artificial e.g. electricity, oil lamps and candles.**

## Effect of poor lighting

a. Strong and concentrated light produce eye fatigue and strain.
b. Poor light leads to accidents.
c. In factories strong shadows over a job can be dangerous.

## Community laws and regulations for good lighting:

The lighting level must be uniform and not patchy, cause no dazzle, and avoid dark spots.

# Environmental Hazards at Home

## a) Hygiene and sanitation

- Hygiene and household cleanliness contribute to the maintenance of family health.
- Maintaining a clean home and using closed containers for food sources can significantly reduce the risk of insects such as roaches and ants, as well as rodents (mice / rats), thus avoiding the need for pests control measures.

## b) Radon:

- Radon has been of great concern in recent years. radon is an odorless, colorless and tasteless radioactive gas that is a byproduct of the decay of uranium, it occurs naturally in the soil.

## c) Lead:

- The reduction of childhood lead poising is one of the most significant public health achievements in the recent years. Lead –based paints were both banned in the 1970s. lead - based paints, including dust and chips, airborne from steel structure, contaminated water from lead pipes or tanks, food lead containers, stained glass and glazed pottery.

# Table 1: Lead Poisoning; Symptoms and Effects

| |
|---|
| **≤ 9 µg/dl** |
| Tranplacental transfer, minimal observable effects |
| **10-20 µg/dl** |
| Causes toxicity, effects on growth& development, hearing, IQ. |
| **21-44 µg/dl** |
| Affects vitamin D metabolism, Hb synthesis, mild fatigue |

# Table 1: Lead Poisoning; Symptoms and Effects

| |
|---|
| **45 - 69 µg/dl** |
| Myalgia, irritable, lethargy, gastrointestinal complaints |
| **≥70 µg/dl** |
| CNS Symptoms (e.g., paralysis, encephalopathy, coma, seizure), and renal effects. |

## d) Garbage:

- A potential source of disease that often taken for granted is the method of garbage disposable.
- American produce almost 4.5 Lb. solid wastes per day.
- Garbage leads to contamination of water supplies, and contributes to breeding of bacteria and viruses that can be transmitted to humans via insects and rodents and possibly serves as a reservoir for diseases “e.g., gastroenteritis, hepatitis, polio".

## e) Formaldehyde:

- Is a colorless, flammable gas tat has a distinct pungent odor. Formaldehyde is a common building block for the synthesis of more complex compounds and materials. Worldwide, over 50 million tons of Formaldehyde are used annually in a wide range of products including; household products (such as carpets, draperies, shampoos, and cosmetics). The most common acute symptoms associate with exposure are: “eye and nose irritation, respiratory symptoms, nausea, headache and fatigue”.

## f) Carbon Monoxide:

- Is a colorless and odorless gas that is a byproduct of combustion. Sources are improperly vented furnaces, blocked flues or chimneys and automobile exhaust. Low level of CO can cause dizziness, headache, drowsiness, nausea, and flu- like symptoms, a higher level can result in unconsciousness and death.

## g) Environmental Tobacco Smoke:

- A major source in door air pollution is environmental tobacco smoke, tobacco has been recognized as a cause of lung cancer and other respiratory diseases in smokers.

# Implication for Nurses

- Primary prevention involves educating people to eliminate hazards before exposure occurs.
- Secondary prevention or screening for environment hazardous.

➢ The nurse can determine hazard that may threaten the health or safety of person in their home.

➢ Family history should include the physical condition of residency, school, worksite, home remodeling, eating, source and quality of drinking water.

# Home Safety

1. **Poisoning:** is the major cause of accidental death (e.g., lead Poisoning, pesticide Poisoning, mercury Poisoning, carbon monoxide Poisoning).
   29.500 deaths in 2007 (CDC, NCH, 2011).
2. **Falls:** falls is the other major cause of accidental death, mostly such accidents happen in the older (poor light, loose electric cords and rugs).
   For children, windows that do not have guards are hazardous.
   22.500 deaths in 2007 (CDC, NCH, 2011).
3. **Burns:** fire and burns are the third highest cause of home deaths.

**4. Suffocation and Drowning:** are a serious causes of accidental death, infants 0- 12 months of age rate 22.5 per 100.000, older than 65 years rate 8 per 10.0000 (CDC,2011).

**5. Firearms:** it occur annually while people are playing with or cleaning firearms, the number highest for the 15-24 years old group and it increases to double in suicidal and more in homicide, suicide rate (10.2 per 100.000) was higher than the homicide (6.1 per 100.000), approximately 85% of these deaths are in males. (CDC, NCH, 2010).

**6. Psychosocial hazard**: The last category of hazard is probably the most difficult to describe because it is more difficult to measure, environmentally induced is a natural byproduct of our fast paced society as well as a result of natural or man-made disasters.

# The Nurse's Responsibilities

## Primary prevention

- Advocate safer environment design of products, automobiles, equipments and buildings.
- Teach home safety related to falls and fire.
- Counsel women of child bearing.
- Develop work site health and safety program.

- Advocate vehicle protection system such as seat belt.
- Immunization for occupationally exposed workers for hepatitis B.
- education and support programs for waste management and recycling.

## Secondary prevention

- Assess homes, schools, worksites and communities for environmental hazard.
- Screen children from 6 months to 5 years age for blood lead level.
- Monitor workers for levels of chemical exposure.
- Screening at risk workers for lung diseases, cancer and hearing loss.

# Tertiary prevention

- Encourage limitation of activity when air pollution is high.
- Support cleanup of toxic waste sites and removal of other hazards.
- Refer homeowners to approved lead abatement resources.

# Environmental Hazards in the Community

## Types of Hazards:

- **Chemical:**

1. **Insulation (formaldehyde)**:
   expected harmful effect (respiratory allergies)
2. **Automobile exhaust (lead):**
   expected harmful effect (behavior disorders, neurological symptoms)
3. **Pesticides (polychlorinated biphenyls {PCB}):**
   expected harmful effect (Chloracne, liver diseases, headache, birth defects)

- **Biological and Infectious:**

1. **Water supply (Giardia lamblia, Cryptosporidium):**
   expected harmful effect (diarrhea, bloating, malabsorption)
2. **Food (Salmonella, E. Coli):**
   expected harmful effect (fever, nausea, diarrhea)
3. **Mosquito (Malaria, West Nile Virus):**
   expected harmful effect (fever, chills)

- **Physical**

**1. Physical hazard (faulty construction):**
expected harmful effect (death or injuries)

**2. Noise (motor vehicles):**
(e.g. airplanes, cars, motorcycle) expected harmful effect (hearing problem, stress, fatigue)

**3. Radiation (radon gas):**
expected harmful effect (infertility, birth defects, leukemia)

- **Psychological**
  1. **Natural disaster (e.g. flooding, forest fire):** expected harmful effect (death or injuries)
  2. **Low economic status (unemployment):** expected harmful effect (heart disease, ulcer)
  3. **Multiple role demand (working parents):** expected harmful effect (depression, anxiety)

# Air Pollution

- **Pollution** refers to the act of contaminating or defiling the environment to the extent that it negatively affects people's health. Air pollution is now recognized as one of the most hazardous sources of chemical contamination.
- It is especially prevalent in highly industrialized and urbanized areas where concentrations of motor vehicles and industry produce large volumes of gaseous pollutants.
- Air pollution is a global problem. Recognizing the threat, the WHO published the first global standards for air quality in 2005.



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## Air Pollution …….

Air pollution comes from many different sources:

- Natural processes that affect air quality include volcanoes, which produce sulfur, chlorine, and ash. Wildfires produce smoke and carbon monoxide. Cattle and other animals emit methane as part of their digestive process.
- Many forms of air pollution are human-made. Industrial plants, power plants and vehicles with internal combustion engines produce nitrogen oxides, carbon monoxide, carbon dioxide, sulfur dioxide. Example: cars in megacities and farmers burning their crop waste.



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## Air Pollution …….

- Outdoor air pollution contributes to cardiovascular and respiratory diseases and is believed to be responsible for nearly 1 million lung cancer deaths yearly.
  With respect to children, infant mortality in the first year of life, bronchitis, asthma, and reduced lung development are additional health threats.



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## Air Pollution ……

- The EPA has developed a tool to provide the public with timely and easy-to-understand information on local air quality and whether air pollution levels pose a health concern.
- The Air Quality Index (AQI) lets the public know how clean the air is and whether they should be concerned for their health.
- The AQI is focused on health effects that can happen within a few hours or days after breathing polluted air (Table 9.1).



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# Dusts, Gases, and Naturally Occurring Elements

- Dusts can contain numerous types of chemical irritants and poisons. Many hazardous dusts are associated with the workplace; for example, coal miners have developed black lung disease from inhaling coal dust, and a respiratory disease called silicosis is caused by exposure to silica dust (common in mining, sandblasting, and tunnel work).



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## Dusts, Gases, and Naturally Occurring Elements

- Dusts are also associated with farming and grain elevator work, as well as highway construction.
- Asbestos fibers, which are found in insulation and fireproofing materials, textiles, and many other products, have been associated with lung cancer.



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## Dusts, Gases, and Naturally Occurring Elements

- Although people who smoke are at 30 times greater risk of developing lung cancer than those who do not smoke, environmental tobacco smoke (ETS) is estimated to cause approximately 3,000 lung cancer deaths in nonsmokers each year, and 150,000 to 300,000 infants and children younger than 18 months of age experience lower respiratory tract infections annually because of exposure to second-hand smoke. In addition, asthma and other respiratory conditions are triggered or worsened by tobacco smoke (USDHHS, 2000).



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- Radon is a risk to all of us since it can seep into houses and public buildings through cracks in basement walls or through sewer openings, exposing people who inhabit, attend school, or work in the affected buildings. Furnaces and exhaust fans can help pull radon into a house, although the highest levels tend to be found in basements, where the gas enters.
- Home testing for radon was recommended by the EPA and the U.S.



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# Acid Precipitation (Acid Rain)

- The emission of hazardous chemicals into the Earth's atmosphere has a serious effect on the environment. Air pollutants such as sulfur dioxide from power plant emissions or nitrogen oxides from motor vehicle exhaust combine with rainwater, snow, and other forms of precipitation to produce sulfuric and nitric acid, commonly called "acid rain" (EPA, 2007a; Friis, 2007).



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## Acid Precipitation (Acid Rain)

- Although acid rain does not seem to pose any direct danger to humans, it kills small forms of life and endangers forest and freshwater ecologies.



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# Ozone Depletion and Global Warming

- Two new and very threatening environmental hazards, stratospheric ozone layer depletion and global warming, are closely related. Global warming is the trapping of heat radiation from the Earth's surface that increases the overall temperature of the world, causing a "greenhouse effect."
- This warming is caused by carbon dioxide and other gases that enter the atmosphere through a depleted ozone layer and become trapped.



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## Ozone Depletion and Global Warming

- The direct health effects of ozone depletion include increased risks for skin cancer and cataracts.
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- In the past, changes in the Earth's climate resulted from natural causes and enabled the evolution of species over hundreds of thousands of years.
  The accumulation of stratospheric ozone shields the Earth from damaging ultraviolet light, filtering and reducing the extent of radiation that reaches the Earth.
- Today, human activities are affecting our climate in serious and immediate ways. The ozone layer is being destroyed as a result of chemical interactions among air pollutants, primarily chlorofluorocarbons (CFCs).



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## Ozone Depletion and Global Warming

- Most of these ozone-destroying chemicals and greenhouse gases are produced in the wealthier countries.
- Continued worldwide efforts to phase out these chemicals are maintained.



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# ***Government's Role***

- Government regulation of air pollution has been relatively slow in developing.
  In 1963, the U.S. federal government passed the first series of Clean Air Acts.
- These set standards for air quality and industrial emissions and delegated funds to assist in pollution control programs.
  Although progress has been made, further public health efforts are needed to help identify pollution sources and related health hazards.



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- Containment: Control at the every point of its origins in industries.
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- Legislation : Air pollution can be controlled in many countries by suitable legislation as (no smoking).



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# ***Nurse's Role***

- Community health nurses can influence air quality through detection, community education, and lobbying for appropriate legislation. People are exposed to numerous impurities in the air in their homes and workplaces.
- Nurses can promote health by helping to detect indoor pollutants and informing people of existing or potential dangers. Many household products and building materials emit vapors that can cause problems.



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## ***Nurse's Role***

- Nurses can assist with the prevention or elimination of these health hazards by ensuring that the indoor environment is well ventilated (oxygenated) and heating equipment properly maintained, and by looking for possible sources of pollution.



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# Water Pollution

- Water is such an essential element to human survival that the available quantity and quality of water within a community has become a prime environmental health issue. Water has many uses other than consumption by humans.
- Water also acts as a medium for sustaining other living organisms, as a home to plant and animal life, and as a means of carrying and distributing necessary nutrients in the environment.



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## Water Pollution

- Although nursing's environmental health role concerns the safe consumption of water by humans, it is important, taking an ecologic perspective, to keep in mind water's other uses and users.
- Drinking water comes from two main sources: surface water (such as lakes and streams) and underground sources (called groundwater), which collects in areas known as aquifers and comes to the surface through wells and springs.



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## Water Pollution……

- In general, underground sources are thought to be less subject to contamination than surface sources, which are open to runoff from agricultural pesticides or industrial wastes. However, groundwater may be contaminated if seepage occurs.



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## Water Pollution…….

As much as 20% of the world's population may not have access to safe sources of drinking water. Water can be contaminated and made unsafe for drinking in many different ways.

Three are discussed here:

1. Water may be infected with bacteria or parasites that cause disease.



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## Water Pollution…….

2. Toxic substances, such as pesticides, are introduced by humans into water systems and constitute another form of water pollution.

3. Power plants or other industries dissipate excess heat into lakes and streams and cause water temperatures to rise. This thermal pollution kills off beneficial organisms in the water.



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## ***Government's Role***

- Globally, because of the enormous health problems in developing nations caused by unclean water, the WHO declared the 1980s as the International Clean Water Decade and established a goal to have safe drinking water for all by the year 1990.
- This goal was not met; only 50% of the world's population had a safe water supply in 1980, 55% in 1985, and 66% in 1990 . By 2002, the percentage had increased to 83%, yet slow progress in areas such as Sub-Saharan Africa tempered the achievement.



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- In 2005, a new global initiative, the International Decade for Action: Water for Life 2005–2015 was launched with the goal of increasing awareness of the basic need for safe drinking water and finding solutions to achieve global access (WHO, 2005).

- Efforts to address water purity globally continue to assume high priority, however the focus is shifting from drinking water quality alone toward overall improvement of the environment.



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# ***Nurse's Role***

- As nurses work in a community, they can help by examining household or city drinking water.

- Asking clients to observe and report changes in water quality further assists the nurse in the monitoring process. If such changes occur, the proper authorities, such as health department officials, should be notified and water samples tested.



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## ***Nurse's Role***

- Although water quality monitoring is ultimately the responsibility of environmental health authorities, it is incumbent on the nurse, as a collaborating member of the health team, to observe and report any information that would further the goal of safe and healthy water for communities.



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# Characteristics of potable water:

- Physical: odourless, colourless, clear and agreeable taste.
- Chemical: neutral or slightly alkaline.
- Bacteriological: allowable total bacterial count is 100 organism per millilitre.



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# Activities to provide safe water supply

- The main water sources in every place must be sampled to study the bacteriological contamination, once in every three months
- Identify the private wells which are being used to supply through tankers in houses must be inspected and disinfected frequently
- Construct good drainage system around the well
- Community must be educated to practice household method of purification of water for drinking purposes
- Educate the community about; the hygienic methods of collection, storage and use of water to prevent water borne diseases

# Dusts, Gases, and Naturally Occurring Elements

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# ***Nurse's Role***

- Community health nurses can influence air quality through detection, community education, and lobbying for appropriate legislation. People are exposed to numerous impurities in the air in their homes and workplaces.
- Nurses can promote health by helping to detect indoor pollutants and informing people of existing or potential dangers. Many household products and building materials emit vapors that can cause problems.



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## ***Nurse's Role***

- Nurses can assist with the prevention or elimination of these health hazards by ensuring that the indoor environment is well ventilated (oxygenated) and heating equipment properly maintained, and by looking for possible sources of pollution.



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# Water Pollution

- Water is such an essential element to human survival that the available quantity and quality of water within a community has become a prime environmental health issue. Water has many uses other than consumption by humans.
- Water also acts as a medium for sustaining other living organisms, as a home to plant and animal life, and as a means of carrying and distributing necessary nutrients in the environment.



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## Water Pollution

- Although nursing's environmental health role concerns the safe consumption of water by humans, it is important, taking an ecologic perspective, to keep in mind water's other uses and users.
- Drinking water comes from two main sources: surface water (such as lakes and streams) and underground sources (called groundwater), which collects in areas known as aquifers and comes to the surface through wells and springs.



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## Water Pollution……

- In general, underground sources are thought to be less subject to contamination than surface sources, which are open to runoff from agricultural pesticides or industrial wastes. However, groundwater may be contaminated if seepage occurs.



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## Water Pollution…….

As much as 20% of the world's population may not have access to safe sources of drinking water. Water can be contaminated and made unsafe for drinking in many different ways.

Three are discussed here:

1. Water may be infected with bacteria or parasites that cause disease.



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## Water Pollution…….

2. Toxic substances, such as pesticides, are introduced by humans into water systems and constitute another form of water pollution.

3. Power plants or other industries dissipate excess heat into lakes and streams and cause water temperatures to rise. This thermal pollution kills off beneficial organisms in the water.



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## ***Government's Role***

- Globally, because of the enormous health problems in developing nations caused by unclean water, the WHO declared the 1980s as the International Clean Water Decade and established a goal to have safe drinking water for all by the year 1990.
- This goal was not met; only 50% of the world's population had a safe water supply in 1980, 55% in 1985, and 66% in 1990 . By 2002, the percentage had increased to 83%, yet slow progress in areas such as Sub-Saharan Africa tempered the achievement.



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- In 2005, a new global initiative, the International Decade for Action: Water for Life 2005–2015 was launched with the goal of increasing awareness of the basic need for safe drinking water and finding solutions to achieve global access (WHO, 2005).

- Efforts to address water purity globally continue to assume high priority, however the focus is shifting from drinking water quality alone toward overall improvement of the environment.



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# ***Nurse's Role***

- As nurses work in a community, they can help by examining household or city drinking water.

- Asking clients to observe and report changes in water quality further assists the nurse in the monitoring process. If such changes occur, the proper authorities, such as health department officials, should be notified and water samples tested.



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## ***Nurse's Role***

- Although water quality monitoring is ultimately the responsibility of environmental health authorities, it is incumbent on the nurse, as a collaborating member of the health team, to observe and report any information that would further the goal of safe and healthy water for communities.



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# Characteristics of potable water:

- Physical: odourless, colourless, clear and agreeable taste.
- Chemical: neutral or slightly alkaline.
- Bacteriological: allowable total bacterial count is 100 organism per millilitre.



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# Activities to provide safe water supply

- The main water sources in every place must be sampled to study the bacteriological contamination, once in every three months
- Identify the private wells which are being used to supply through tankers in houses must be inspected and disinfected frequently
- Construct good drainage system around the well
- Community must be educated to practice household method of purification of water for drinking purposes
- Educate the community about; the hygienic methods of collection, storage and use of water to prevent water borne diseases

# Deforestation, Wetlands Destruction, and Desertification

- Deforestation is the clearing of tropical and temperate forests for cropland, cattle grazing, or urbanization. Elimination of these natural habitats is dooming some species of insects and animals to extinction, the loss of a species from the Earth forever.



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## Deforestation, Wetlands Destruction, and Desertification ……..

- Wetlands are natural inland bodies of shallow water, such as marshes, ponds, river bottoms, and flood plains, which filter contaminated surface waters and support wildlife reproduction and growth.

- The destruction of the wetlands has caused numerous species of wildlife to disappear from the area or become extinct.



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## Deforestation, Wetlands Destruction, and Desertification

- Desertification refers to the conversion of fertile land into desert, which is unable to support crop growth or wildlife.
- The destruction of forests and the upturning of Earth for urban sprawl uncovers organisms hidden for eons, to which humans and animals are then exposed.
- In addition, gases that were once absorbed by lost trees remain in the atmosphere and contribute to ozone depletion, which increases global warming.



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## Deforestation, Wetlands Destruction, and Desertification ……..

- Deforestation, in turn, contributes to desertification, because forests provide protection for the surrounding topsoil by way of their roots, fallen leaves, and undergrowth. When this protection is lost, landslides and other geographic changes occur. Global temperature increases cause riverbeds to dry up and create desert areas that are unable to support the people who once inhabited them.



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## ***Government's Role***

- The government has the power to make decisions that save the wetlands and forests in any country.

- Often, developers of housing and industrial facilities are more influential, and their interests prevail. If the importance of the wetlands and forests is not recognized where decisions are made, then they will be lost.



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## ***Nurse's Role***

- CHN can makes a difference in this area.
- Perhaps no other person knows a community more intimately than the CHN. This role gives a valid voice of concern at the local level.
- By using leadership and collaborative skills, the nurse can initiate grassroots efforts to save wetlands and forests in the community.



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# Energy Depletion

- Most of the energy sources we use today are not renewable.

1- Wood has been used for thousands of years and was our first fuel. It is still a primary source of home heating for most of the world's population.

2- Natural gas for heat and fuel can be a highly efficient energy source.



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## Energy Depletion

3- Coal is energy source, but  not all coal extraction is environmentally or economically feasible.

Some countries do not have coal as a natural resource and have similar problems with natural gas.



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4- Nuclear energy has been used for at least 50 years. This source of energy has proved to be an effective power product. Nevertheless, there have been some near-disasters and real disasters caused by human error.

The building of nuclear power plants is only one community concern; the disposal of the nuclear waste products also raises environmental, safety, and ethical issues.



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## Energy Depletion

The permanent disposal of all high-level radioactive waste falls under the regulations of the Nuclear Waste Policy Act of 1982. The act provides that all high-level waste will be disposed of underground and in a deep geologic repository, and that the Yucca Mountain, Nevada site is the current preferred site.



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## ***Government's Role***

- Other renewable sources of energy need to be discovered, rediscovered, or tapped. Newer and more "environmentally friendly" energy sources are used experimentally in limited areas. They include landfill gas recovery, solar power, and wind power.
- These sources can make a difference. so It will take both a global effort to increase awareness and accompanying technology to use these energy sources in enough areas to make an environmental difference.



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# ***Nurse's Role***

- CHN nurse may not have a direct role in the creation of new energy sources or the use of a particular source.

- However, the nurse can educate people about energy conservation, discuss alternative energy sources presently available in the community, and encourage people to become interested in and knowledgeable about the importance of the potential for energy depletion in the future.



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# Unhealthy or Contaminated Food

- The food supply, and particularly the quality of that food, is affected by the environment, and numerous health hazards are associated with food.
- The CHN needs to ask how the environment influences the safety of food for human consumption.
- Three types of hazardous foods must be considered when examining food as a possible health problem: inherently harmful foods, contaminated foods, and foods with toxic additives.



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# Food Safety

- Food is a potential source of infection.
- It can be contaminated by bacteria and other microorganisms and parasites.
- Food safety implies hygiene in production, handling, distribution and serving of all types of foods.

Many communicable diseases can be caused by poor personal hygiene and contamination of food.

- Any untidy household, dirty hands and poor environment create suitable conditions for growth of microorganism.

# Foodborne infections and diseases

- Viral e.g. viral hepatitis, poliomyelitis
- Bacterial e.g. typhoid, paratyphoid, and diarrhea
- Protozoa e.g. amoebiasis
- Parasites e.g. ascariasis, tape worm
- Other e.g. food poisoning

# Food Handlers

Food handlers, cooking, serving and distributing food should be periodically examined for any infection especially those who remain as carriers.

- **Types of examinations:**
  - Clinical examination: for skin diseases, TB
  - Lab examination: blood for typhoid and paratyphoid, stool for bacillary dysentery and parasites and amoebic dysentery
  - Swabbing: throat swab, nose swab to be examined for diphtheria or staphylococci
  - Chest x-ray

## Food Handlers Cont. …….

- Validity of the food certificates
- Sanitary precautions during work

-Cleanliness of the body clothing

-Responsibility of manager to exclude from work until cured any case of common cold, other infectious diseases, wounds, ulcers of the skin

- The health authority has the right to call any food handlers at any time for examination according to the necessity

# Food Utensils (Containers)

Containers or articles used for storage and serving food should be:

- Made of a safe materials not including toxic metals especially lead and arsenic
- Kept always clean
- Copper utensils need be soldered with tin periodically

# Community laws and regulations for food sanitation

## A. General measures at home:

- Keep food refrigerated
- Cover food to protect it from flies
- Wash vegetables with clean and safe water
- Wash hands with clean and soap before preparing and eating food
- Cook any only sufficient food for one time meal

## B. Sanitary public:

Restaurants, Cafeteria, Canteens, groceries, hotels and food store:

- Should fulfil sanitary requirements to be licensed by the local municipal or health authorities

# Waste Disposal

With the vast amounts of waste produced in the form of household garbage, human excreta, and agricultural and industrial byproducts, including hazardous chemical and radioactive substances, it is no wonder that waste management and disposal has become an important and pressing topic in recent decades.



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## Waste Disposal

New technology has effectively addressed some of the problems, but there is still much need for improvement. Solid and hazardous wastes pose a wide range of public health concerns. Therefore, it is imperative that health officials, including community health nurses, become aware of the possible health hazards that these wastes present to individuals and to communities.



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# Types of refuse

**Refuse** is the solid waste matter of the community.

- Building refuse: garbage (food waste) and rubbish (tins, paper, glass).
- Street refuse: refuse of public places .
- Hospital refuse: building refuse and dressing refuse.
- Industrial refuse: building refuse and toxic refuse.
- Agriculture: operation refuse.

# Disposal of Human Waste

- One of the oldest environmental health hazards comes from improper disposal of human excreta.
- Although industrialized nations have successfully addressed the problem, it continues to be widespread in developing nations and in rural, poverty-stricken communities.
- Human wastes, particularly feces, provide a perfect environment in which bacteria and disease-causing parasites can live and reproduce.

# Methods of waste (refuse) disposal

## A. Incineration / burning:

- Refuse can be disposal of hygienically by burning or incineration.
- Hospital wastes which is dangerous with large quantities and variety of pathogens must be disposed by incineration.
- Refuse containing sand dust and other inorganic materials like the glass and tins etc, which create problem in burning must be removed.
- It is a good method for cities where suitable land is not adequate
- Burning, although it reduces the volume of garbage, produced noxious odors and pollutes the air.

# Methods of waste (refuse) disposal

## B. Dumping:

- Bumping is a simple method. Low lying areas are selected to dump the wastes
- Dumping is problematic, because garbage dumps provide perfect conditions for the breeding of rats, flies, and other disease-carrying organisms and may potentially be a source of water contamination from runoff.

## C. Fermentation/ composting:

- It is a method of combined disposal of refuse and excreta.
- It is a process whereby organic matter breaks down under bacterial action resulting in the formation of a relatively stable humus-like material called compost which is a good manure.

# Disposal of Hazardous Waste

- Disposal of toxic chemical and radioactive wastes produced by industry is another grave concern.

- The threat is serious. Many of these wastes escape containment or accidentally leak into water systems and into the soil to contaminate drinking water and food.

**Community laws and regulations for refuse:**

- Collection of refuse adequately

- Adequate refuse disposal to prevent health hazards

# ***Government's Role***

- The government's role is to establish standards for safe waste disposal and to monitor and enforce compliance.
- More emphasis must be placed on transforming waste into usable products, increasing the amounts and kinds of recycling done, and reducing the amount of refuse produced in the first place.



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# ***Nurse's Role***

- Community health nurses can encourage the positive actions described by educating the public and lobbying for enabling legislation.
- Nurses can promote greater sensitivity among citizens to the problems of accumulating waste with its potential health hazards.
- Encourage clients to buy products that can be recycled.
- Discourage use of aerosol spray containers, plastics, and other nonrecyclable items.



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# Sewage disposal

- One of the oldest environmental health hazards comes from improper disposal of human excreta.
- Human wastes, particularly feces, provide a perfect environment in which bacteria and disease-causing parasites can live and reproduce.

## Component of sewage:

**Sewage** is the collected liquid wastes of the community

- Domestics waste: excreta of man and waste of kitchen and baths.
- Municipal waste: washing streets or rain.
- Industrial wastes.

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# Health hazards from sewage

- Spread of infection from flies
- Contamination of soil
- Pollution of river and shallow wells
- Contamination of vegetables by human fertilizers
- Pollution of sea water

**Diseases associated with improper excreta disposal:**

- Typhoid, paratyphoid, cholera, diarrhoea, and dysentery
- Helminth infection: hookworm, ascariasis
- Viral hepatitis, poliomyelitis

# Sewage disposal

## There are several methods of sewage disposal:

- Some are applicable to rural areas (Latrines and septic tanks).

- Some to cities and towns water carriage system and sewage treatment; sewage posses through a number of steps in special tanks where it is submitted to a series of physical and biological processes of finally separates into liquid part.

# Community laws and regulations for sewage disposal

- Water-closets in schools, industry public spaces should fulfil sanitary requirements, cleanness, ventilation, no insects, and periodically maintenance.

- Every building should have a sewage drainage system.

- Disposal at sea: proper selection of the site of disposal, pipes go sufficiently far from the shore.

# Insect and Rodent Control (Vector Control)

- All human communities are affected by the insects and rodents living in their environment.
- On the least dangerous level, they can cause irritation and discomfort.
- They can also pose a direct threat to health through such things as attacks by diseased rats or squirrels.



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## Insect and Rodent Control (Vector Control)

- Insects and rodents can consume and, in turn, contaminate food.
- However, by far the most serious health hazard they impose is through their role as vectors—nonhuman carriers of disease organisms that can transmit these organisms directly to humans.



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## Group of insects:

- Insect e.g. mosquitoes, flies, fleas, bed bugs, cockroaches
- Arachnid e.g. ticks, mites

## Health hazards from insects:

- Directly invade the body e.g. scabies
- Irritation, discomfort and hypersensitivity by bites

## Control measures of insects:

- Sanitation and cleanliness of the environment
- Control of breeding places
- Application of insecticides
- Sanitary disposal of wastes

# Nurse's Role Insect and Rodent Control

- Ensure that screens exist on all open windows, and use screen doors.
- Wash dishes, pots, and pans after meals, and clean counter surfaces.
- Keep foodstuffs that insects may infest, such as cereals, corn meal, and flour, in closed plastic containers.
- Keep floors swept and vacuumed in rooms where people eat, to eliminate food supply for rodents and insects; preferably, eat only in the kitchen or dining room.
- Remove trash bags that include food scraps and food packaging from the home daily and place in garbage containers that are kept outside and have tight-fitting lids.

# Industrial sanitation

**Sanitation of the work environment:**

a. Proper design and general cleanliness of industrial plants.
b. Adequate natural and mechanical ventilation.
c. Lighting: optimum power and distribution.
d. Safe water supply for drinking and personal cleanliness.
e. Toilet and washing facilities e.g. W.C. baths.
f. Waste disposal for plants and industrial wastes.

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# STRATEGIES FOR NURSING ACTION IN E. H.

The following strategies for collaboration and participation provide a summary of the nurse's role and can assist the nurse in addressing this two-parts goal:

1. Learn about possible environmental health threats. The nurse has a responsibility to keep abreast of current environmental issues and to know the proper authorities to whom problems should be reported.



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2. Assess clients' environment and detect health hazards. Careful observation and an environmental checklist can assist in this assessment.

3. Plan collaboratively with citizens and other professionals to devise protective and preventive strategies. Remember that environmental health work is generally a team effort.

4. Assist with the implementation of programs to prevent health threats to clients and the environment.



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5. Take action to correct situations in which health hazards exist. Nurses can use direct intervention (e.g., in an unsafe home situation), notify proper authorities, or publicly protest if corrective measures are beyond their sphere.

6. Educate consumers and assist them to practice preventive measures. Examples of preventive measures include; radon testing in homes and well water testing in rural communities.



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7. Take action to promote the development of policies and legislation that enhance consumer protection and promote  a healthier environment.

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Community as Client

Community health nurses work with clients at several levels as *individuals, Families, Groups, Subpopulation, Populations, and Communities.*

*The American Public Health Association (APHA) (1996) defines public health nursing as "the practice of promoting and protecting the health of populations using knowledge from nursing, social, and public health sciences"*

Although community health nurses work at all six levels of practice, working with communities is a primary mission for two important reasons:

1- Community directly influences the health of individuals, families, groups, subpopulation, and populations who are a part of it.

*For example, if a city fails to take aggressive action to stop air pollution, the health of all its citizens will be adversely affected.*

2. Provision of most health services occurs at the community level.

*Community agencies help develop specific health programs and disseminate health information to many types of groups and populations.*

The community health nurse, then, must work with community as the client.

The community as client refers to the concepts of a community- wide group of people as the focus of nursing service.

*Understanding the concept of the community as client is a prerequisite of effective service at every level of community nursing practice.*

## DIMENSIONS OF THE COMMUNITY AS CLIENT

The health of a community can be characterized through a number of perspectives. One such view is by examining three dimensions:

- status,
- structure,
- process

- Status is the most common measure of the health of a community. It typically comprises morbidity and mortality data identifying the physical, emotional, and social determinants of health.
- Structure of a community refers to its services and resources.
- Process reflects the community's ability to function effectively.

- Another perspective identifies the community as having three features: a location, a population, and a social system.
- Figure 15.1 presents a visual interpretation of this perspective. Each of these features has several components that need to be addressed and represents further information that must be collected and analyzed when assessing the health of a community.

- It is useful to think of these three dimensions of every community as providing a rough map that one can follow in assessing needs or planning for services provision.
- Further guidance in assessing the health of a community is provided in the community Profile Inventory found Tables 15-1, 15-2, and 15-3.
- In considering each dimension, one should pay particular attention to the questions that must be asked to assess the health of a community.

## *Location*

Every physical community carries out its daily existence in a specific geographic location.
The health of a community is affected by location, because placement of health services, geographic features, climate, plants, animals and the human made environment are intrinsic to geographic location. The location of a community places it in an environment that offers resources and also poses threats.

- The healthy community is one that makes wise use of its resources and is prepared to meet threats and dangers. In assessing the health of any community, it is necessary to collect information not only about variables specific to location but also *about relationships between the community and its location*.
- Do groups cooperate to identify threats? Does the community make certain that its members are given available information about resources and dangers?

Table 15-1 describes the location perspective of the community profile inventory, including the six location variables: (community boundaries, location of health services, geographic features, climate, flora and fauna, and the human-made environment).

## Location variables

**(1)Community boundaries**
**Community boundaries serve as basis for measuring incidence of wellness and illness, and for determining spread of disease.**

Where is the community located?
What is its boundary?
Is it a part of a larger community?
What smaller communities does it include?

**(2) location of health services**
**Use of health services depends on availability and accessibility.**

Where are the major health institutions located?
What necessary health institutions are outside the community?
Where are they?

**(3) geographic features**
**Injury, death, and destruction may be caused by floods, earthquakes, volcanoes, tornadoes, or hurricanes.**

What major landforms are in or near the community?
What geographic features pose possible threats?
What geographic features offer opportunities for healthful activities?

**(4) climate,**
**Extremes of heat and cold affect health and illness. Extremes of temperature and precipitation may tax community's coping ability.**

What are the average temperature and precipitation?
What are the extremes?
What climatic features affect health and fitness ?
Is the community prepared to cope with emergencies?

**(5)flora and fauna**
**Poisonous plants and disease- carrying animals can affect community health**
**Plants and animals offer resources as well as dangers.**

What plants and animals pose possible threats to health?

**(6) The human-made environment**
**All human influences on environment (housing, dams, farming, type of industry, chemical waste, air pollution, and so forth) can influence levels of community wellness.**

What are the major industries?
How have air, land, and water been affected by humans?
What is the quality of housing?
Do highways allow access to health institutions?

# Population

When one considers the community as the client, the second dimension to examine is the total population of the community. Population consists not of a specialized aggregate but of all the diverse people who live within the boundaries of the community.

The health of any community is greatly influenced by the attributes of its population. Various features of the population suggest health needs and provide a basis for health planning.

- Community health nurses can better understand any community by knowing about its population variables: size, density, composition, rate of growth or decline, cultural characteristics, social class structure, and mobility.
- Table 15-2 presents the population perspective section of the community profile inventory.

## Population variables:

**(1)Population Size**

- **The number of people influences number and size of health institutions.**
- **Size affects homogeneity of the population and its needs.**

What is the population of the community?
Is it an urban, suburban, rural community?

## (2) Density

- Increased density may increase stress
- High and low density often affect the availability of health Services.

What is the density of the population per square mile?

## (3) Composition

- **Composition of the population often determines types of health needs.**

What is the age composition of the community?
What is the sex composition of the community?
What is the marital status of community members?

**(4) Rate of growth or decline**

- **Rapidly growing communities may place excessive demands on health services.**
- **Marked decline in population may signal a poorly functioning community.**

How has population size changed over the past two decades?

What are the health implications of this change?

## (5) Cultural differences

- Health needs vary among sub-cultural and ethnic population.
- Utilization of health services varies with culture. Health practices and extent of knowledge are affected by culture.

# DIMENSIONS OF THE COMMUNITY AS CLIENT: population

- ✓ What is the ethnic breakdown of population?
- ✓ What racial groups are represented?
- ✓ What subcultural populations exist in the community?
- ✓ Do any of the subcultural groups have unique health needs and practices?
- ✓ Are different ethnic and cultural groups included in health planning?

## (6) Social class

- **Class differences influence the utilization of health services.**
- **Class composition influences cost of public health services.**

- ✓ What percentage of the population falls into each social class?
- ✓ What do class differences suggest for health needs and services?

## (7) Mobility

**Mobility of the population affects continuity of care. Mobility affects availability of service to highly mobile populations**

- ✓ How frequently do members move into and out of the community?
- ✓ How frequently do members move within the community?

# DIMENSIONS OF THE COMMUNITY AS CLIENT: population

- ✓ Are there any specific populations, such as migrant workers, that are highly mobile?
- ✓ How does the pattern of mobility affect the health of the community?
- ✓ Is the community organized to meet the health needs of mobile groups?

# Social system

- In addition to location and population, every community has a third dimension- a social system.
- The various parts of a community's social system that interact and influence the system are called social system variables. These variables include the following systems: health, family, economic, educational, religious, welfare, political, recreational, legal and communication.

- The community health nurse needs to understand the community as a social system. A community health nurse working in a tiny village in Alaska needs to grasp the social system of that village no less than a nurse working in New York City.
- Table 15-3 guides the nurse in assessing a community's social system variables.

## Social System variables include:

- Health system
- Family system
- Economic system
- Educational system
- Religious system
- Welfare system
- Political system
- Recreational system
- Legal system
- Communication system

# The concept of a social system

A social system is an abstract concept and can be more readily understood by first considering the people who make up the community's population. Each person enacts multiple roles, such as parent, spouse, employee, citizen, and political volunteer. People in certain roles tend to interact more closely with others in related roles. The patterns and interactions that emerge from these interactions among roles form the basis of organizations.

# The concept of a social system

Some organizations are informal (e.g., an extended family group).

Other organizations, such as a city police department or a software business, are more formal. However, all organizations are constructed from roles that are enacted by individual citizens. Organizations, in turn, interact with one another, forming linkages.

# The concept of a social system

For example, a medical equipment company and a laboratory establish contracts (linkages) with a home care agency.

When a group of organizations are linked and have similar functions, such as all those providing social services, they form a community system or subsystem.

# The concept of a social system

The various community systems have a profound influence on one another. Because this interaction among parts determines the health of the whole, it is the total social system that concerns community health nurses.

## The health care delivery system as part of the social system

Although community health nurse must examine all the systems in a community and must understand how they interact, the health system is of particular importance. Studying the health system in a community can be compared with assessing an individual client.

## The health care delivery system as part of the social system

The latter involves a head-to-toe examination looking for indications of wellness and illness in the respiratory, musculoskeletal, glandular, skin, and circulatory systems, among others, Initial assessment of a community also begins with a survey of its ten major social systems, before asking how well  the specific parts are functioning.

## The health care delivery system as part of the social system

The major function of the health system is to promote the health of the community. Community assessment asks not merely whether, but also how well, the system is functioning. What is the level of health promotion carried out by the health system of a community?
To answer this question, one needs a clear notion about the subsystems, organizations, and roles that make up the system.

# Nursing Process Characteristics Applied To Community As Client

Three characteristics support the use of the nursing process in community health nursing.

First, the nursing process is a problem-solving process that addresses community health problems at every aggregate level with the goals of preventing illness and promoting public health.

Second, it is a management process that requires situational analysis, decision making, planning, organization, direction, and control of services, as well as outcome evaluation.

Third, it is a process for implementing changes that improve the function of various health-related systems and the ways that people behave within those systems.

## Deliberative

The nursing process is deliberative purposefully, rationally, and carefully thought out.

It requires the use of sound judgment that is based on adequate information.

## Deliberative

Deliberative problem solving is a necessary skill for working with the community health team to address the needs and problems of aggregates in the community.

The nursing process is a decision-making tool to facilitate these determinations.

## Adaptable

The nursing process is adaptable. Its dynamic nature enables the community health nurse to adjust appropriately to each situation and to be flexible in applying the process to aggregate health needs. Furthermore, its flexibility is a reminder to the nurse that each client group and each community situation is unique.

## Cyclic

The cyclic nature of the nursing process enables the nurse to engage in a constant information feedback loop: The information gathered and lessons learned at each step of the process promote greater understanding of the group being served, the most effective way to provide quality services, and the best methods of raising this group’s level of health.

## Client-focused

The nursing process is client-focused; it is used for and with clients.

Community health nurses use the nursing process for the express purpose of addressing the health of populations.

## Interactive

The nursing process is interactive, in that nurse and clients are engaged in a process of ongoing interpersonal communication. Giving and receiving accurate information is necessary to promote understanding between nurse and clients and to foster effective use of the nursing process. Furthermore, because of the movement toward informed use of medical care, demands for clients' rights and the concept of self-care have gained emphasis.

## Need-oriented

The nursing process is need-oriented. A long association with problem solving has tended to limit the focus of the nursing process to the correction of existing problems. This focus is needed if the goals of community health—to protect, promote, and restore the people's health—are to be realized.

# Forming Partnerships and Building Coalitions

Another important consideration in community-level nursing practice requires teamwork. The job of planning for the health of an entire community or a community subsystem requires that the nurse collaborate with other professionals.

Each group member brings expertise and a particular view of the problem. These interprofessional work groups are often formed as either partnerships or coalitions.

## Partnerships

Partnerships are agreements between people that support a joint purpose. A partnership can be large or it can be a more modest endeavor.

Community-wide partnerships require more planning and coordination than do small partnerships.

## Coalition

A coalition is an alliance of individuals or groups that work together to influence the outcomes of a specific problem. Coalitions are an effective means to achieve a collaborative and coordinated approach to solving community problems.

Steps to coalition building include defining goals and objectives, conducting a community assessment, identifying key players or leaders, and identifying potential coalition members.

# Community Assessment

Assessment is the first step of the nursing process. Assessment for nurses means collecting and evaluating information about a community's health status to discover existing or potential needs as a basis for planning future action.

Assessment involves two major activities:

- The first is collection of data.
- The second is analysis and interpretation of data.

**Community needs assessment:**

- Is the process of determining the real or perceived needs of a defined community.
- The type of assessment depends on variables such as the needs that exist, the goals to be achieved, and the resources available for carrying out the study.

# Types of Community Assessment

## 1- Familiarization or "Windshield Survey":

Familiarization assessment involves studying data already available on a community, and gathering a certain amount of firsthand data, to gain a working knowledge of the community. Such an approach, sometimes called “windshield survey” is used by new staff members in community health agencies.

# Types of Community Assessment:

**Familiarization or "Windshield Survey":**

Nurses drive (or walk) around the community; find health, social, and governmental services; obtain literature; introduce themselves and explain that they are working in the area; and generally become familiar with the community.

## 2- Problem-Oriented Assessment:

A second type of community assessment, problem-oriented assessment, begins with a single problem and assesses the community in terms of that problem.

The problem-oriented assessment is commonly used when familiarization is not sufficient and a comprehensive assessment is too expensive. This type of assessment is responsive to a particular need. The data collected will be useful in any kind of planning for a community response to the problem.

## 3- Community Subsystem Assessment:

- In Community subsystem assessment, the community health nurse focuses on a single dimension of community life.

For example; the nurse might decide to survey youth organizations to discover their roles in the community.

## Community Subsystem Assessment:

- Community subsystem assessment can be a useful way for a team to conduct a more thorough community assessment.

For example; if five members of a nursing agency divide up the ten systems in the community and each person does an assessment of two systems, they could then share their findings to create a more comprehensive picture of the community and its needs.

## 4- Comprehensive Assessment:

- Comprehensive assessment is used to discover all relevant community health information. A survey compiles all the demographic information on the population, such as its size, density, and composition.
- Key informants are interviewed in every major system (education, health, religious, economic, and others). Then, more detailed surveys and intensive interviews are performed to yield information on organizations and the various roles in each organization.

- A comprehensive assessment describes not only the systems of a community but also how power is distributed throughout the system, how decisions are made, and how change occurs.
- Because comprehensive assessment is an expensive, time-consuming process, it is seldom performed. Indeed, in many cases, such a thorough research plan might be a waste of resources and might repeat, in part, other studies. Performing a more focused study based on prior knowledge of needs is often a better strategy.

## 5- Community Assets Assessment:

- The final form of assessment presented here is assets assessment, which focuses on the strengths and capacities of a community rather than its problems.
- The previously mentioned methods are needs oriented and deficit based – in other words, they are "pathology" models, in which the assessment is performed in response to needs, barriers, weaknesses and problems, in the community.

## Community Assets Assessment:

- This may result in a fragmented approach to solutions for the community's problems rather than an approach focused on the community's possibilities, strengths, and assets.
- This approach requires that the assessor look for the positive, or see the glass as "half full".

# Community Assessment Methods

- Community health needs may be assessed by a variety of methods.
- Four important methods are discussed here:

1. Surveys
2. Descriptive epidemiologic studies.
3. Community forums or town meetings.
4. Focus groups.

## 1. Survey:

- A survey is an assessment method in which a series of questions is used to collect data for analysis of a specific group or area.

### 1. Planning Phase:

Select survey method or instrument to be used (eg, interviews, telephone calls, questionnaires).

Determine sampling size (eg, a percentage of the total population in question).

**Survey:**

**2. Data Collection Phase:**

**3. Data Analysis and Presentation Phase:**

- Report results, including implications, recommendations.

## 2. Descriptive Epidemiologic Studies:

- A second assessment method is a descriptive epidemiologic study, which examines the amount and distribution of a disease or health condition in a population by person (who is affected?), by place (Where does the condition occur?), and by time (When do the cases occur?).

## 3. Community Forums or Town Hall Meetings:

- The community or town hall meetings is a qualitative assessment methods designed to obtain community opinions. It takes place in the neighborhood of the people involved, perhaps in a school gymnasium or an auditorium.
- Is used to elicit public opinion on a variety of issues, including health care concerns, political views, and feelings about issues in the public eye.

## 4. Focus Groups:

- This fourth assessment method, focus groups, is similar to the community forum or town hall meeting in that it is designed to obtain grassroots opinion. However, it has some differences.
- First, there is only a small groups of participants, usually 5 to 15 people.

## Focus Groups:

- The members chosen for the group are homogeneous with respect to specific demographic variables. For example, a focus group may consists of female community health nurses, young women in their first pregnancy, or retired businessmen.
- Usually the group meets for 1 to 3 hours, and there be a series of meetings.

# SOURCES OF COMMUNITY DATA

- There are many places the community health nurse can look for data to enhance and complete a community assessment.

- Data sources can be primary or secondary, and they can be from international, national, state, or local sources.

# Primary and Secondary Sources

- Community health nurses make use of many sources in data collection.
- Community members, including formal leaders, informal leaders, and community inhabitants, can frequently offer the most accurate insights and comprehensive information.
- Information gathered by talking to people provides primary data, because the data are obtained directly from the community.

# Primary and Secondary Sources

- Secondary sources of data include people who know the community well and the records such people create in the performance of their jobs.
- Specific examples are health team members, client records, community health statistics, Census Bureau data, reference books, research reports, and community health nurses.

## Primary and Secondary Sources

- Because secondary data may not totally describe the community and do not necessarily reflect community self-perceptions, they may need augmentation or further validation.

# International Sources

- International data are collected by several agencies, including the World Health Organization (WHO) and its six regional offices and health organizations, such as the Pan-American Health Organization.
- WHO publishes an annual report of their activity.

# International Sources

- In addition, the United Nations and global specialty organizations that focus on certain populations or health problems, such as the United Nations Children's Fund, are major sources of international health-related data.
- International statistics for diseases and illness trends can be found on the Internet.

# National Sources

- There are official and nonofficial sources of national data that community health nurses can access if needed.
- Official sources develop documents based on data compiled by the government.
- Examples for official sources:

✓ U.S. Department of Health and Human Services, (USDHHS)
✓ National Center for Health Statistics (NCHS)
✓ U.S. Bureau of the Census
✓ National Institutes of Health (NIH)

# National Sources

- Nonofficial agencies have data sources generated from research they conduct that focuses on the population, disease, or condition they were developed to serve.

# National Sources

- Examples of Nonofficial agencies are:

The American Cancer Society (ACS), the American Association of Retired Persons (AARP), Mothers Against Drunk Drivers (MADD), and Students Against Drunk Drivers (SADD).

Information from such national sources allows community health assessment teams to compare local data with national and state statistics and trends—a very valuable function.

# What is a Healthy Community?

The following descriptors serve as a guide for assessing a healthy community:

1. Is one in which members have a high degree of awareness that "we are a community".
2. Uses natural resources wisely while taking steps to conserve them for future generations.

# What is a Healthy Community?

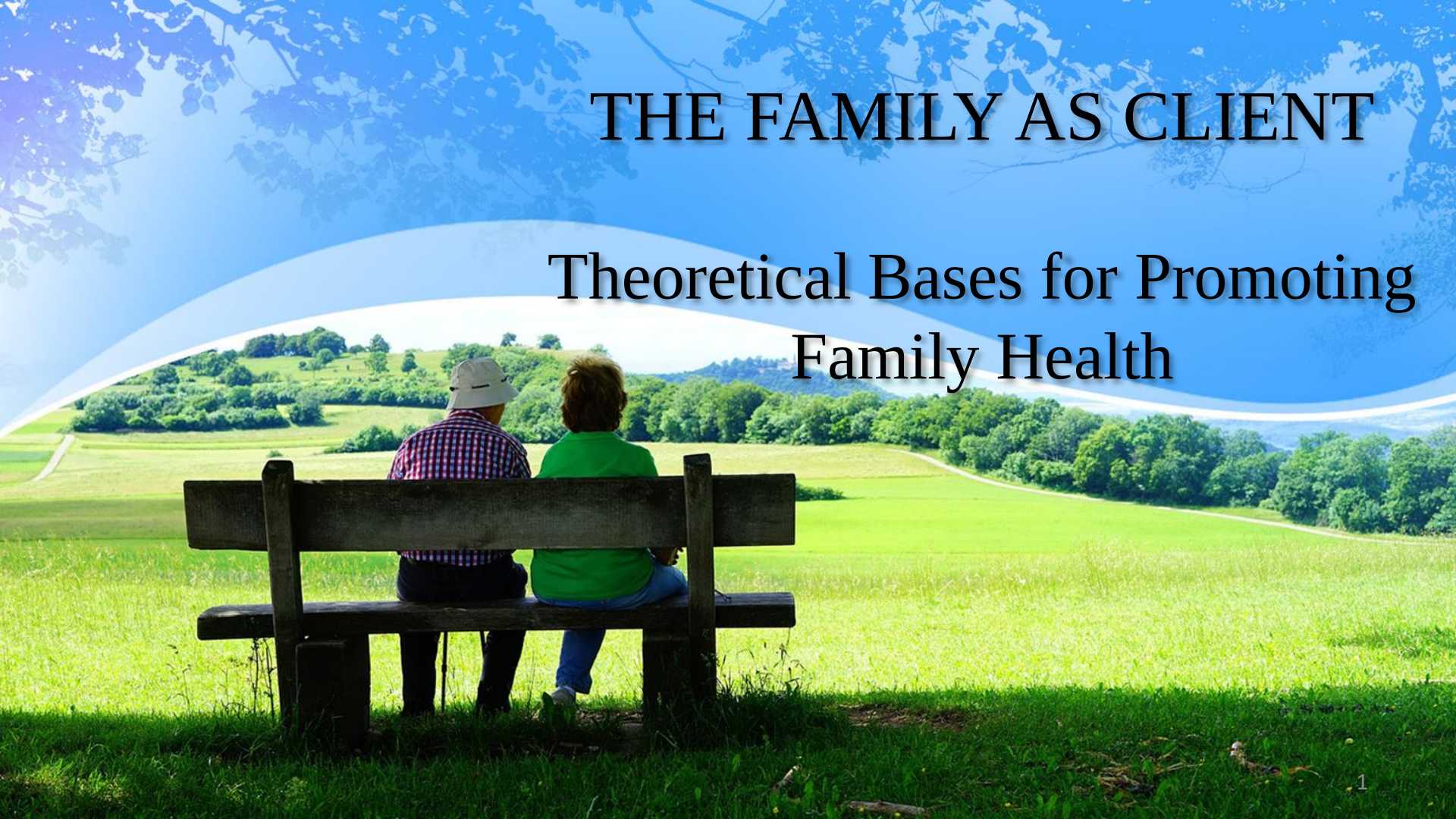
3. Openly recognizes the existence of subgroups and welcomes their participation in community affairs.
4. Is prepared to meet crises.
5. Is a problem-solving community; it identifies, analyzes, and organizes to meet its own needs.
6. Seeks to make each of its systems' resources available to all members of the community.

# What is a Healthy Community?

7. Has legitimate and effective ways to settle disputes that arise within the community.
8. Encourages maximum citizen participation in decision-making.
9. Promotes a high level of wellness among all its members.

# THE FAMILY AS CLIENT

## Theoretical Bases for Promoting Family Health



When you hear the word *family,* what do you think of?
How would you define your own family?
Is your grandfather a member of your family?





Although many different definitions exist, most family theorists agree that:

"***a family*** *consists of two or more individuals who share a residence or live near one another; possess some common emotional bond; engage in interrelated social positions, roles, and tasks; and share a sense of affection and belonging.*"

- The family is a separate entity with its own structure, functions, and needs.
- In every society throughout history, the family is the most basic unit; so, too, in community health.
- It is the family, more than any other societal institution, that nurtures and shapes a society's members.





- The effectiveness of the community health nurse depends on knowing how to work with a family as a unit of care.
- This chapter examines the nature of families, family functioning, and family health.
- This information will increase the effectiveness of interventions with families at the primary, secondary, and tertiary levels of prevention.

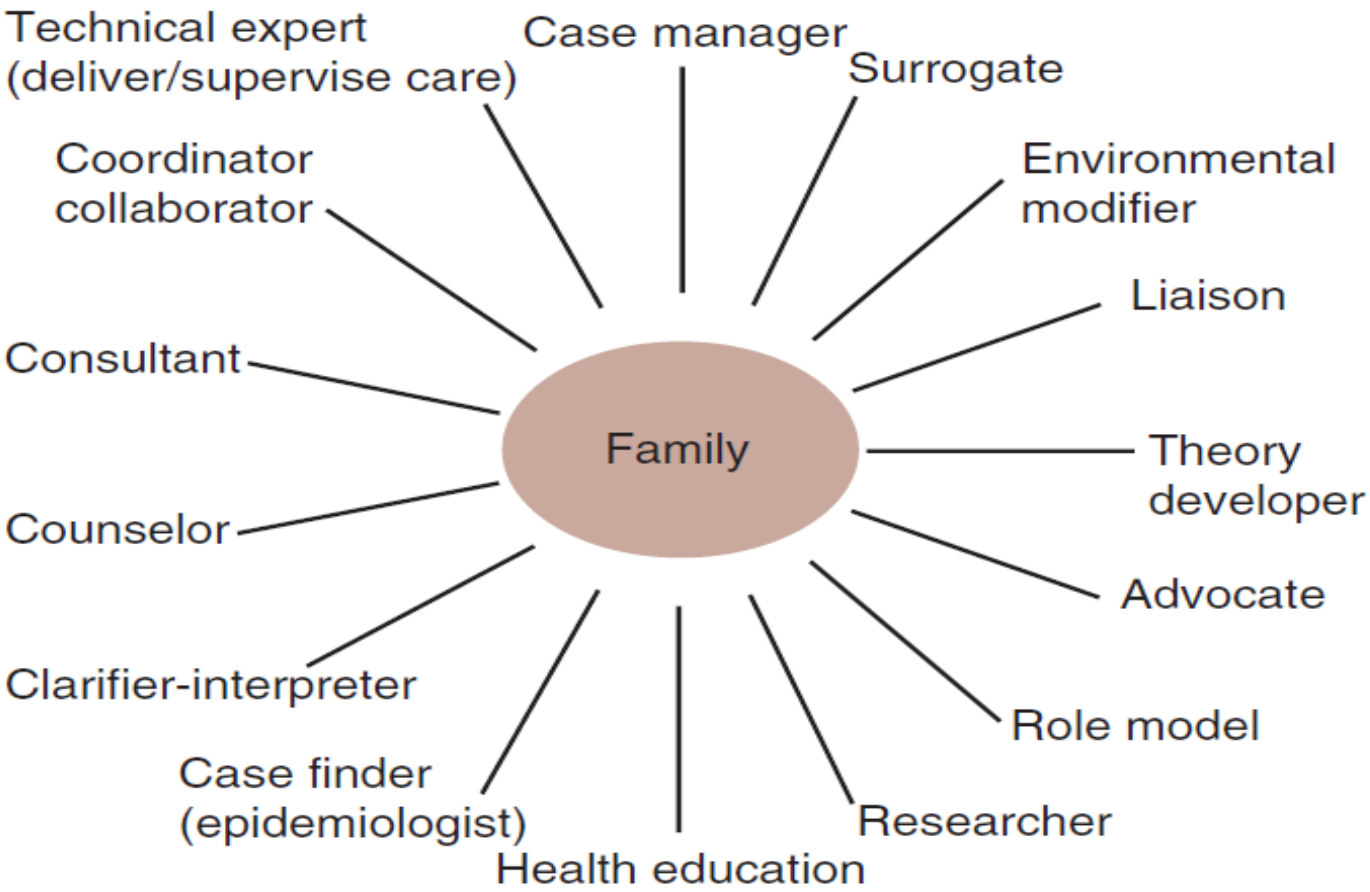



**FIGURE 1-9** Family Nursing Roles

## ☺ Family functioning is defined as:

“Those behaviors or activities by family members that maintain the family and meet family needs, individual member needs, and society's view of family”.





## Family health

is concerned with how well the family functions together as a unit. It involves not only the health of  the members and how they relate to other members, but also how well they relate to and cope with the community outside the family.

- In fact, family health, like individual health, ranges along a continuum from wellness to illness.
- A family may be at one point on that continuum now and at a much different point 6 months from now.
- Family health refers to the health status of a given family at a given point in time.





- Each family is unique. Each have their own distinct problems and strengths.
- Every family shares some universal characteristics with every other family.
- **Five of the most important family universal characteristics for community health nursing are:**

1. Every family is a small social system.
2. Every family has its own cultural values and rules.
3. Every family has structure.
4. Every family has certain basic functions.
5. Every family moves through stages in its life cycle.

These five universal of family life, are based on systems theory, sociologic theories, and theories of family development.





# Attributes of Families as Social Systems

- Many viewing families merely as collections of individuals.
- Systems theory offers some insights about how families operate as social systems.

There are five attributes of open systems that help explain how families function:

1. Families are interdependent.
2. Families maintain boundaries.
3. Families exchange energy with their environments.
4. Families are adaptive.
5. Families are goal-oriented.





# 1-Interdependence Among Members

- All the members of a family are interdependent; each member's actions affect the other members.
- The way parents relate to each other, for instance, influences the quality of their parenting. When the interactions between them are frequent, honest, and nurturing, they have more to offer their children. Marital, parent–child, and sibling relationships all significantly influence family functioning.
- They determine how well the family as a system handles conflict, provides a support system for its members, copes with crises, solves daily problems, and capitalizes on its own resources.

# 2-Family Boundaries

- Families, as systems, set and maintain boundaries: ego-boundaries, generation boundaries, and family – community boundaries (Barker, 1998).
- These boundaries, which result from shared experiences and expectations, link family members together in a bond that excludes the rest of the world.





# 3-Energy Exchange

- Family boundaries are semipermeable; although they provide protection and preservation of the family unit, they also allow selective linkage with the outside world.
- As open systems, in order to function adequately, families exchange materials or information with their environment.
- This process is called energy exchange.

# 4-Adaptive Behavior

- Families are adaptive, equilibrium-seeking systems. In accordance with their nature, families never stay the same.
- They shift and change in response to internal and external forces.





# 5-Goal-Directed Behavior

- Families as social systems are goal directed. Families exist for a purpose - to establish, maintain and promote the development of their members.
- To fulfill this purpose, a family must perform basic functions such as providing love, security, identity, a sense of belonging; assisting with preparation for adult roles in society; and maintaining order and control.

# Family Culture

**Family culture:** is the acquired knowledge that family members use to interpret their experiences and to generate behaviors that influence family structure and function.





## Three aspects of family culture deserve special consideration:

1. Family members share certain values that affect family behavior.
2. Certain roles are prescribed and defined for family members.
3. A family's culture determines its distribution and use of power.

# 1.Shared Values and Their Effect on Behavior

- ☺ Although families share many broad cultural values drawn from the larger society in which they live, they also develop unique variants.
- ☺ Every family has its own set of values and rules for operation that can be considered as family culture (Barker, 1998). Such values. " Don’t' tell anyone about our problems".





- A family that values free expression for every member engages comfortably in loud, noisy debates.
- Another family uses birth control based on belief's about human life and parental responsibility; another family chooses not to use birth control because the members hold a different set of values.

# 2. Prescribed Roles

- Families distribute among their members all the responsibilities and tasks necessary to conduct family living.
- The responsibilities of breadwinner and homemaker, with their accompanying tasks, may belong to husband and wife, respectively, or may be shared if both husband and wife hold jobs outside the home.





- The intra-role functioning can be exceptionally taxing. A woman, may play the role of wife to her husband, daughter to her mother who lives with her, and mother to each of her children.
- The mother role may involve taking on several additional roles and responsibilities and varies with each child's needs.

# 3. Power Distribution

- Power -the possession of control, authority, or influence over others- assumes different patterns in each family.
- In some families, power is concentrated primarily in one member, in others, it is distributed on a more egalitarian basis.





# Family Structures

Families come in many shapes and sizes. The varying family structures or compositions comprise the collective characteristics of individuals who make up a family unit (age, gender, and number).

# Traditional Families

Traditional family structures are those that are most familiar to us and that are most readily accepted by society; they include:

- Nuclear family – husband, wife, and children living together in the same household.





- Nuclear – dyad "twoness" family consists of a husband and wife living together who have no children or who have grown children living outside the home.
- Single-parent families include one adult (either father or mother) caring for a child or children as a result of divorce, or the death of a spouse.
- Extended families: multiple generations.

# Implications for the Community Health Nurse

- First, community health nurses: they must be prepared to work with all types of families and accept them.
- Second, the structure of an individual's family may change several times over a lifetime.



**Table 4-1** Family Structures

| Structure | Participants |
|---|---|
| Nuclear family | Married couple with children<br>Unmarried, heterosexual, or same-sex couples with children |
| Nuclear dyad | Couple, married or unmarried; heterosexual or same sex |
| Single-parent family | One adult with children (separated, divorced, widowed, or never married) |
| Single adult | One adult |
| Multigenerational family | Any combination of the first four family structures |
| Kin network | Two or more reciprocal households (related by birth or marriage) |



# Family Functions

Families in every culture throughout history have engaged in similar functions: families always have produced children, physically maintained their members, protected their health, encouraged their education or training, given emotional support and acceptance, and provided supportive and nurturing care during illness.

**Six functions** are typically of families today and are essential for maintenance and promoting of family health:

1. Providing affection.
2. Providing security
3. Instilling identity.
4. Promoting affiliation.
5. Providing socialization.
6. Establishing controls (Duvall & Miller, 1985).





- These tasks help promote the growth and development of family members.
- Understanding these functions and how well individual families provide them enables the community health nurse to work effectively with each family at its level of functioning.

# Family Life Cycle

- Many of the characteristics and defined developmental stages of individual growth also apply to families.
- For example, it is known that families, while maintaining themselves as entities, change continuously.





- Families inevitably grow and develop as the individuals within them mature and adapt to the demands of successive life changes.
- A family's composition, structures, set of roles, and network of interpersonal relationships change with passage of time.

- Community nurses who are knowledgeable about this cycle can provide anticipatory guidance to families.
- For instance, while teaching prenatal care to a pregnant women, the nurse can help the soon-to-be mother anticipate the responsibility and costs of raising her child by helping her calculate child care needs that must be met.





# Stages of the Family Life Cycle

There are two broad stages in the family life cycle; one of expansion as new members are added and roles and relationships are increased, and one of contraction as family members leave to start lives of their own or age and die.

**Table 2-5 Traditional Family Life Cycle Stages and Developmental Tasks**

| Stages of Family Life Cycle | Family Developmental Tasks |
|---|---|
| Married couple | Establishing relationship as a married couple. |
| | Blending of individual needs; developing conflict-and-resolution approaches, communication patterns, and intimacy patterns. |
| Childbearing families with infants | Adjusting to pregnancy and then infant. |
| | Adjusting to new roles as mother and father. |
| | Maintaining couple bond and intimacy. |
| Families with preschool children | Understanding normal growth and development. |
| | If more than one child in family, adjusting to different temperaments and styles of children. |
| | Coping with energy depletion. |
| | Maintaining couple bond and intimacy. |
| Families with school-age children | Working out authority and socialization roles with school. |
| | Supporting child in outside interests and needs. |
| | Determining disciplinary actions and family rules and roles. |
| Families with adolescents | Allowing adolescents to establish their own identities but still be part of the family. |
| | Thinking about the future, education, jobs, and working. |
| | Increasing roles of adolescents in family, cooking, repairs, and power base. |
| Families with young adults: launching | After member moves out, reallocating roles, space, power, and communication. |
| | Maintaining supportive home base. |
| | Maintaining parental couple intimacy and relationship. |
| Middle-aged parents | Refocusing on marriage relationship. |
| | Ensuring security after retirement. |
| | Maintaining kinship ties. |
| Aging families | Adjusting to retirement, grandparent roles, death of spouse, and living alone. |

# Community Health Nursing

## Nursing Process and Families

# Family Health Practice guidelines

- **Family Nursing:**

✓ Is a kind of nursing practice in which the family is the unit of services.

✓ It is a holistic approach.

# Characteristics of Healthy Family

1. A facilitative process of interaction exists among family members.
2. Individual member development is enhanced.
3. Role relationships are structured effectively.
4. Active attempts are made to cope with problems.
5. There is a healthy home environment and lifestyle.
6. Regular links with the broader community are established.

# Family Health Practice guidelines

- **Five principle guide and enhance family nursing practice:**

1. Work with the family collectively.
2. Start where the family is.
3. Adapt nursing intervention to the family stage of development.
4. Recognize the validity of family structural variations.
5. Emphasize family strength.

# Application of nursing process on promoting Family health

**Family health Assessment-**

- provides information on the measuring a current health situation of family member and emotional support that can be expected to be offered to an individual from the family.

# Guide lines for family health assessment:

1. Focus on the family as total unit.
2. Ask goal-directed questions.
3. Collect data over time.
4. Combine quantitative and qualitative.
5. Exercise professional judgment.

# Family Health Assessment:

- To assess a family's level in a systematic three tools are needed:

1. A conceptual framework on which to base the assessment.
2. A clearly defined set of assessment categories for data collection.
3. A method for measuring family's level of functioning.

# Family Health Assessment:

1. **Conceptual Framework:** is a set of concepts integrated into a meaning explanation that helps one interpret human behavior or situation.

- Three framework that are particularly useful in community health nursing:

A. The interactional framework.
B. The structural-functional framework.
C. The developmental framework.

# 1. Conceptual Framework

**A. The Interactional framework:** describes the family as a unit of interacting personalities and emphasize communication, roles, conflict, coping pattern, and decision making process.

- This framework focuses on the internal relationships but neglects the family interaction with the external environment.

# 1. Conceptual Framework

**B. The structural framework:** describes the family as a social system relating to other social system in the external environment, such as school, work, and the health care system.

- This framework examines the interacting functions of society and the family (External), considers family structure and analyzes how a family's structure affects its function.

# 1. Conceptual Framework

C. **The developmental framework:** studies families from a lifecycle perspective by examining members' changing roles and tasks in each progressive life-cycle stage.

- This framework incorporates elements from interactional and structural-functional approaches so that family structure function, and interaction are viewed in the context of the environment at each stage of family development.

# Developmental Frameworks
# (Family Developmental Tasks)

## 1. Beginning family

- Establishing a mutually satisfying marriage.
- Planning to have or not have children.

## 2. Childbearing family

- Having and adjusting to infant.
- Support needs of all three members.

## 3. Family with pre- school children

- Adapting to needs of pre-school children to stimulate growth and development.

## 4. Family with school age children

- Adjusting to the activity of growing children.
- Promoting joint decision making between children and parents.
- Encouraging and supporting children's educational achievements.

## 5. Family with teenagers and young adults

- Maintaining open communication among members.
- Supporting ethical and moral values within the family.
- Balancing freedom with responsibility of teenagers.

## 6. Post- parental family

- Preparing for retirement.
- Maintaining ties with younger and older generations.

## 7. Aging family

- Adjusting to retirement.
- Adjusting to loss of spouse.

**TABLE 18.2 Selected Stage-Critical Family Developmental Tasks**

| Stage of Family Life Cycle | Family Position | Stage-Critical Family Developmental Tasks | Role of the Community Health Nurse |
|---|---|---|---|
| Forming a partnership | Female partner<br>Male partner | Establishing a mutually satisfying relationship | Interact with family where they are at |
| Childbearing | Partner-mother<br>Partner-father<br>Infant child(ren) | Adjusting to pregnancy and the promise of parenthood<br>Fitting into the kin network<br>Having and adjusting to infants, and encouraging their development<br>Establishing a satisfying home for both parents and infant(s) | Assist them in developing strong relationships |
| Preschool-age | Partner-mother<br>Partner-father<br>Child, siblings | Adapting to the critical needs and interests of preschool children in stimulating, growth-promoting ways<br>Coping with energy depletion and lack of privacy as parents | Assist in preparing for family expansion through education and anticipatory guidance |
| School-age | Partner-mother<br>Partner-father<br>Child, siblings | Fitting into the community of school-age families in constructive ways<br>Encouraging children's educational achievement | Encourage time for each other as adults in a relationship separate from parenting role |
| Teenage | Partner-mother<br>Partner-father<br>Child, siblings | Balancing freedom with responsibility as teenagers mature and emancipate themselves<br>Establishing outside interests and careers as growing parents | Provide anticipatory guidance for the school-age children as they grow into adulthood |
| Launching center | Partner-mother-grandmother<br>Partner-father-grandfather<br>Child, sibling, aunt or uncle | Releasing young adults into work, military service, college, marriage, etc., with appropriate rituals and assistance<br>Maintaining a supportive home base | Provide anticipatory guidance for the contracting family as children leave home |
| Middle-aged parents | Partner-mother-grandmother<br>Partner-father-grandfather | Rebuilding the relationship<br>Maintaining kin ties with older and younger generations | Prepare adults for grandparenting role |
| Aging family members | Widow or widower<br>Partner-mother-grandmother<br>Partner-father-grandfather | Adjusting to retirement<br>Coping with bereavement and living alone<br>Closing the family home or adapting it to aging | Assist aging adults with emotional and financial security as they approach retirement<br>Prepare the aging adults with ways to cope with the losses of old age, including changes in space, work, health, status, and loss of friends and family members |

# 2. Data collection categories:

- **There are 12 categories:**

1. Family demography.
2. Physical environment.
3. Psychological and spiritual environment.
4. Family structure and roles.
5. Family function.
6. Family values and beliefs.

# 2. Data collection categories:

7. Family communication pattern.
8. Family decision-making patterns.
9. Family problem solving patterns.
10. Family coping pattern.
11. Family health behavior.
12. Family social and cultural patterns.

# 3. Assessment methods

- Many different methods are used to assess families.

- These methods serve to generate information about selected aspects of family structure and function; the methods must match the purpose for assessment.

# 3. Assessment methods

## 1. Ecomap

Is a diagram of the connection between a family and the other system in its ecological environment, or it is a picture of the family's patterns.

**Nurses can use an ecomap to identify:**

a. Family resources that are present
b. Family needs
c. Conflicts
d. Connections that are present or absent
e. The balance or lack of balance between a family's needs and the resources available to the family

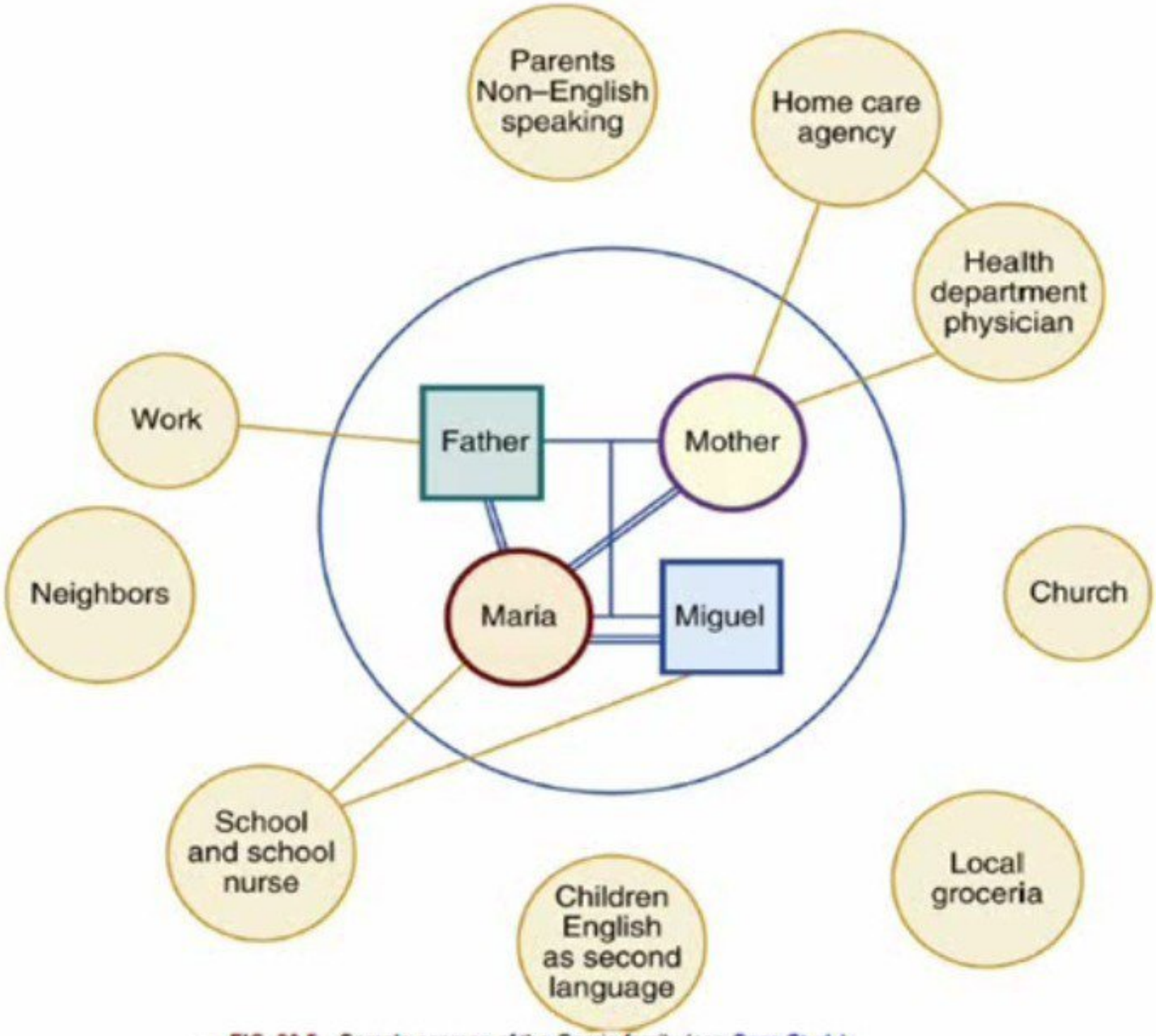
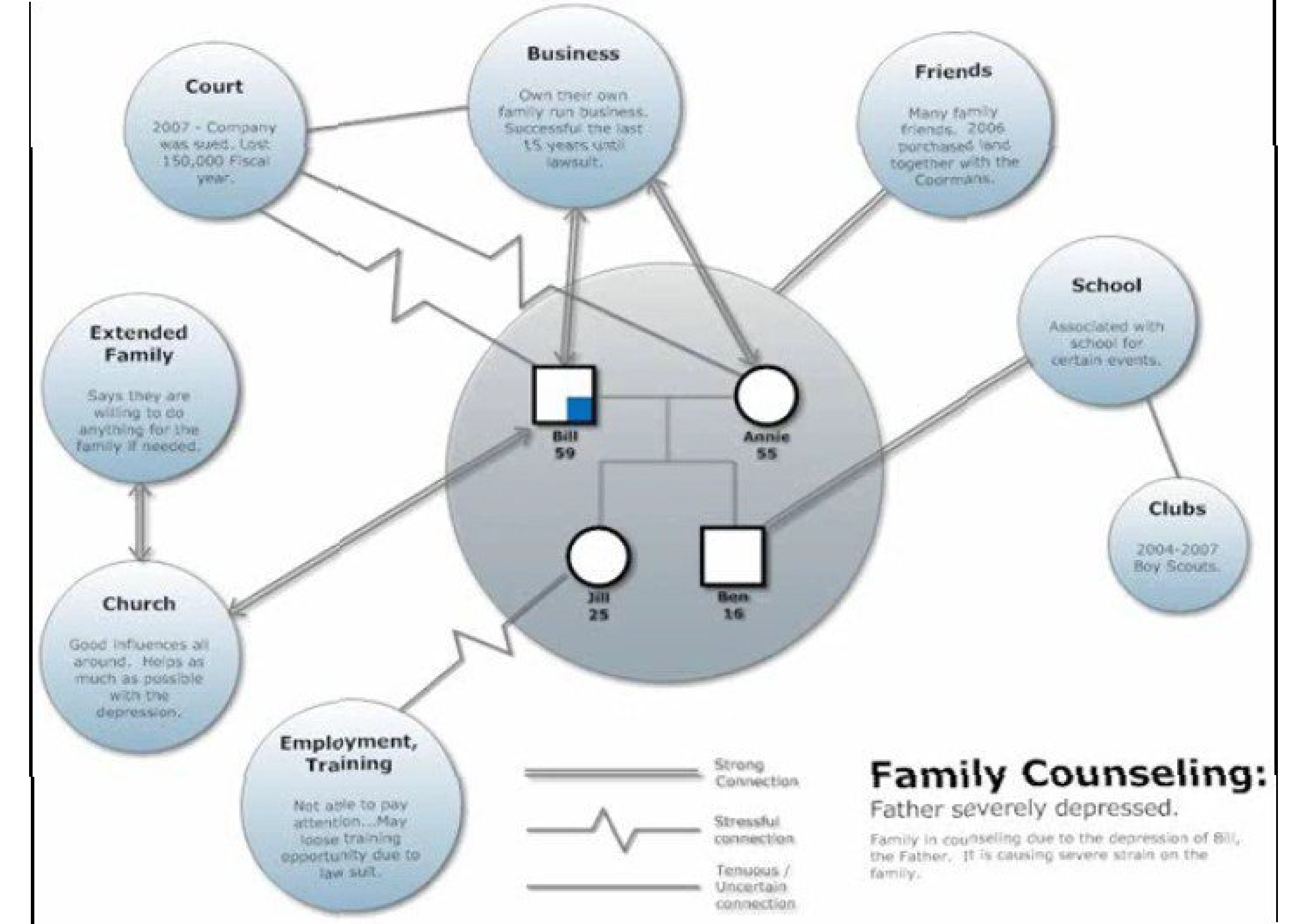



FIG. 20.5 Sample ecomap of the Garcia family (see Case Study).


Court
2007 - Company was sued. Lost 150,000 Fiscal year.
Business
Own their own family run business. Successful the last 15 years until lawsuit.
Friends
Many family friends. 2006 purchased land together with the Coormans.
School
Associated with school for certain events.
Clubs
2004-2007 Boy Scouts.
Extended Family
Says they are willing to do anything for the family if needed.
Church
Good influences all around. Helps as much as possible with the depression.
Employment, Training
Not able to pay attention...May loose training opportunity due to law suit.
Bill
59
Annie
55
Jill
25
Ben
16
Strong Connection
Stressful connection
Tenuous / Uncertain connection
Family Counseling:
Father severely depressed.
Family in counseling due to the depression of Bill, the Father. It is causing severe strain on the family.

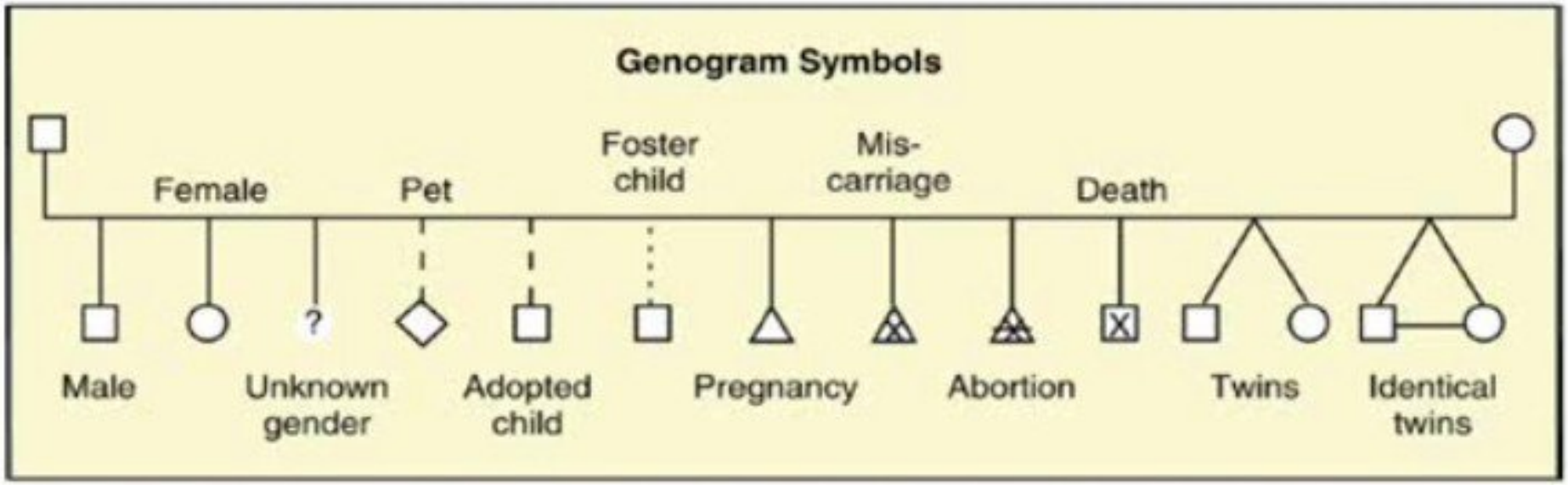
# 3. Assessment methods

**2. Genogram**

displays family information graphically in a way that provides a quick view of complex family pattern, usually used over three or more generations.

**The genogram maps such information as:**

a. Relationships among family members
b. Important life events
c. Place of residence
d. Characteristics such as race, culture and religious affiliations




**Sample Genogram: Woman with Multiple Husbands**

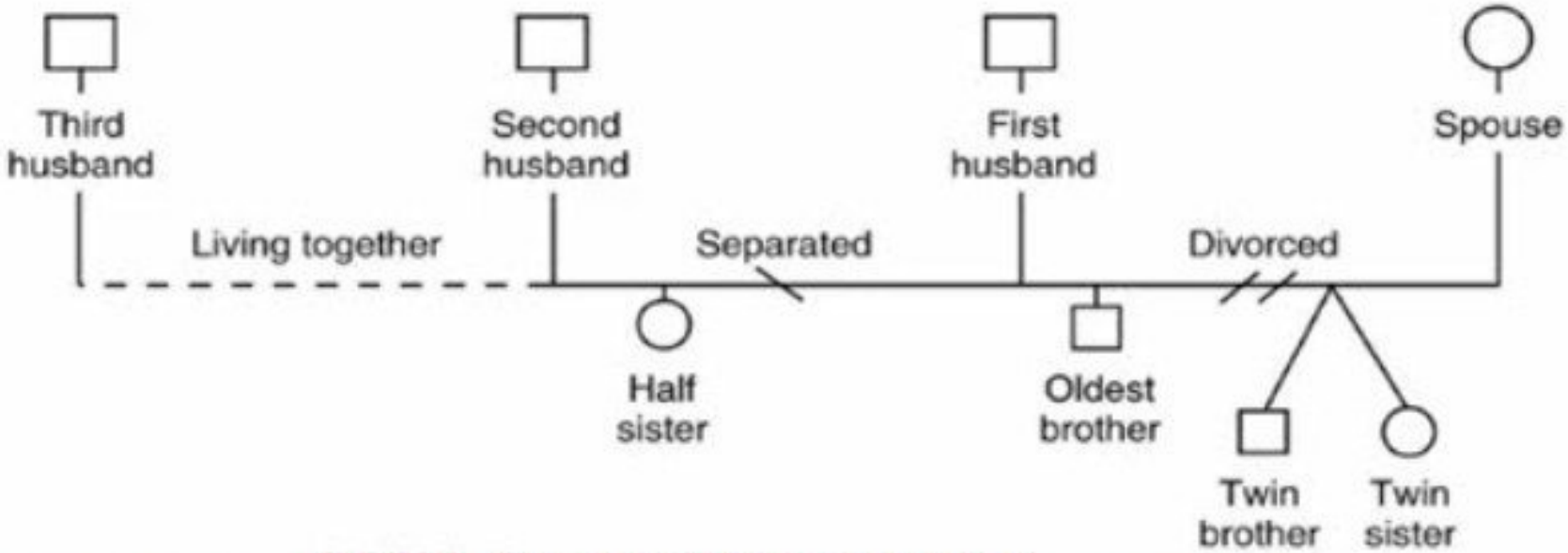
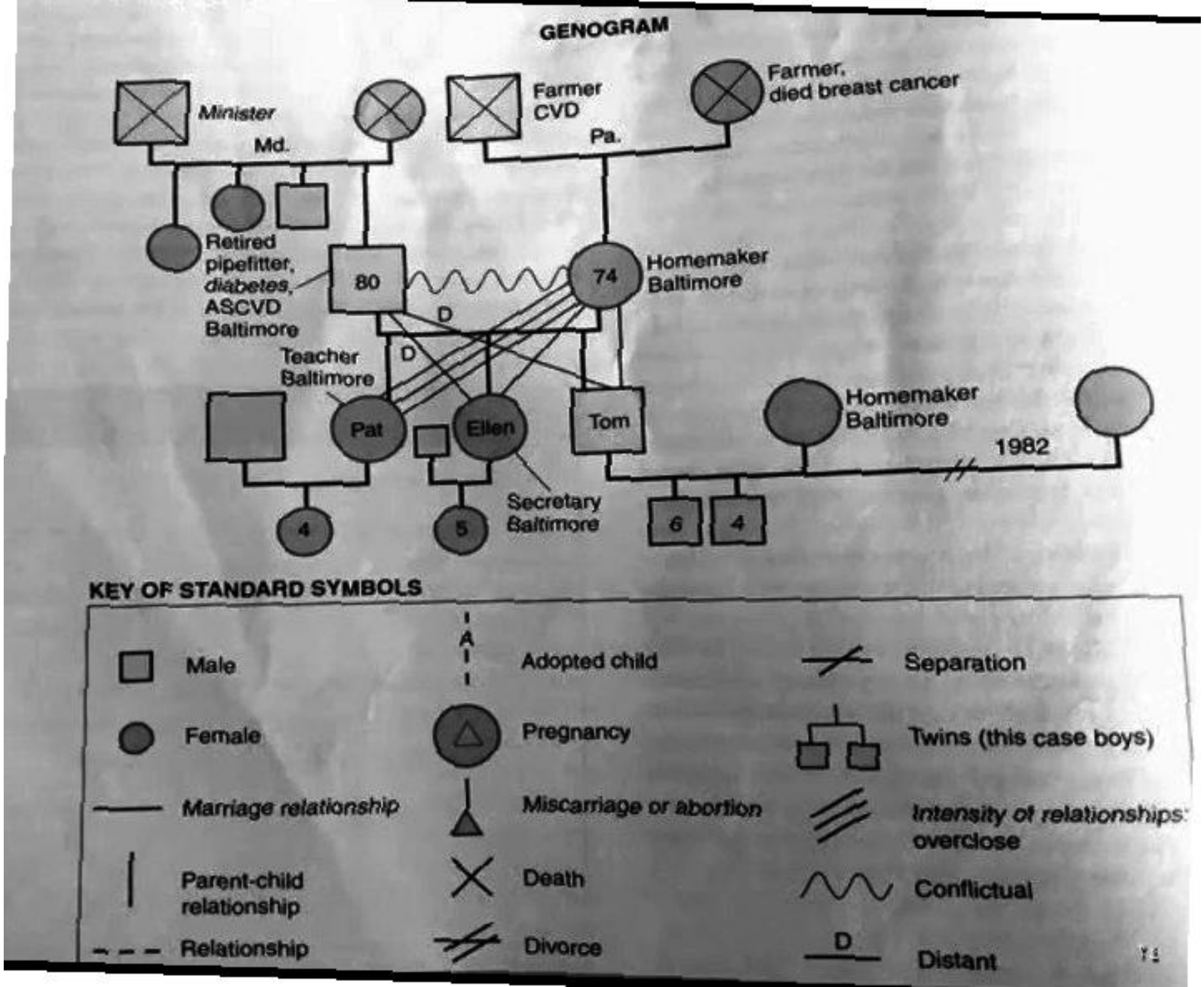



FIG. 20.2 Commonly used genogram symbols.
Redrawn from Genopro Software: *Symbols used in genograms, 2016*. www.genopro.com.


GENOGRAM
Minister
Md.
Farmer
CVD
Pa.
Farmer,
died breast cancer
Retired
pipefitter,
diabetes,
ASCVD
Baltimore
80
74
Homemaker
Baltimore
D
D
Teacher
Baltimore
Pat
Ellen
Tom
Homemaker
Baltimore
1982
4
5
Secretary
Baltimore
6
4
KEY OF STANDARD SYMBOLS
Male
Female
Marriage relationship
Parent-child
relationship
Relationship
A
Adopted child
Pregnancy
Miscarriage or abortion
Death
Divorce
Separation
Twins (this case boys)
Intensity of relationships:
overclose
Conflictual
D
Distant

# 3. Assessment methods

## 3. Social Network Support Map or Grid

Gives details about the quality and quantity of social connections.

- Strengths within the system can be elaborated with words, or numbers, or a combination of these.
- The nurse uses this tool to help the family understand its sources of support and relationships and to form a basis for nursing care planning and intervention.

# 3. Assessment methods

**4. Family Health Tree**

A record of diseases that occur in a family.

**It can be used to track:**

a. Diseases that have genetic bases
b. Environmental diseases
c. Mental health disorders

# Family nursing diagnosis:

**There are two systems of family nursing diagnosis:**

**A. NANDA system**

- Uses nursing diagnosis labels

**B. Omaha system**

- Develops for community health nurses, consists of:

1. Problem classification
2. Intervention
3. Problem rating scale for outcomes

# Family nursing diagnosis:

**Examples for family nursing diagnosis:**

- Family Processes, Dysfunctional: Alcoholism (substance abuse).
- Family Processes, Interrupted.
- Family Processes, Readiness for Enhanced.
- Family coping, ineffective.
- Altered family process related to emergency hospital admission of child.
- Altered family process related to unplanned pregnancy.

# Family nursing diagnosis:

**Examples for family nursing diagnosis:**

- Family coping, potential for growth.
- Potential for role conflict related to prolonged separation.
- Parent/infant/child attachment, altered, risk for.
- Parenting, altered.

# Family nursing diagnosis:

**Examples for family nursing diagnosis:**

- Parenting, altered, risk for.
- Role performance, altered.
- Social interaction, impaired.
- Social isolation.

# HOME VISITING

# Introduction

- The home visit is a family-nurse contact which allows the health worker to assess the home and family situations in order to provide the necessary nursing care and health related activities.

- In performing this activity, it is essential to prepare a plan of visit to meet the needs of the client and achieve the best results of desired outcomes.

# Introduction

- Is a professional contact between the community health nurse and the family.
- The services provided is an extension of the health services agency (Health center).
- The community health nurse doing home visits usually works as part of a home health care team that can includes social workers, rehabilitation specialists, and home health nurses or aides. Cooperation and communication with other care providers are essential.

# Objectives of the home visit:

- Assessment
- Nursing care
- Treatment
- Health education
- Referral (if care fails)

# Purpose of home visit

- To give care to the sick, to a postpartum mother and her newborn with the view teach a responsible family member to give the subsequent care.
- To assess the living condition of the patient and his family and their health practices in order to provide the appropriate health teaching.
- To give health teachings regarding the prevention and control of diseases.
- To establish close relationship between the health agencies and the public for the promotion of health.
- To make use of the inter-referral system and to promote the utilization of community services.

**The goals of primary health nursing are often met through providing health care to families in their home. It can be provided by:**

a. Visiting nurse association

b. Hospice

c. Public health departments

d. Home health agencies

e. School districts

# Principles involved in preparing for a home visit

- A home visit must have a purpose or objective.
- Planning for a home visit should make use of all available information about the patient and his family through family records.
- In planning for a home visit, we should consider and give priority to the essential needs of the individual and his family.
- Planning and delivery of care should involve the individual and family.
- The plan should be flexible.

# Guidelines to consider regarding the frequency of home visits

1. The physical needs, psychological needs, and educational needs of the individual and family.
2. The acceptance of the family for the services to be rendered, their interest and the willingness to cooperate.
3. Take into account other health agencies and the number of health personnel already involved in the care of a specific family.
4. Careful evaluation of past services given to the family and how the family avails of the nursing services.
5. The ability of the patient and his family to recognize their own needs, their knowledge of available resources and their ability to make use of their resources for their benefits.

## Phases and Activities of a Home Visit

| Phase | Activity |
|---|---|
| I. Initiation phase | Clarify source of referral for visit<br>Clarify purpose for home visit<br>Share information on reason and purpose of home visit with family |
| II. Previsit phase | Initiate contact with family<br>Establish shared perception of purpose with family<br>Determine family's willingness for home visit<br>Schedule home visit<br>Review referral and family record |
| III. In-home phase | Introduction of self and professional identity<br>Social interaction to establish rapport<br>Establish nurse-client relationship<br>Implement nursing process |
| IV. Termination phase | Review visit with family<br>Plan for future visits |
| V. Postvisit phase | Record visit<br>Plan for next visit |

# Steps in conducting home visits

- Greet the patient and introduce yourself.
- State the purpose of the visit
- Observe the patient and determine the health needs.
- Put the bag in a convenient place.
- Perform the nursing care needed and give health teachings.
- Record all important date, observation and care rendered.
- Make appointment for a return visit.

# Advantages of home visits:

A. These visits cost less than hospital care, with better outcomes, especially when chronic health issues are involved.
B. Clients have greater control over their health and lives.
C. The community health nurse gains access to families to provide health education and other prevention strategies.
D. The nurse can observe family and environment factors that influence health.
E. Home visits allow for primary intervention, to prevent disease or injury from occurring.
F. Home visits facilitate family participation and promote family focus.

# Disadvantages of home visits:

A. The nurses skills, personality, or physical ability may not be compatible with providing home visit.
B. Home visits are time consuming; travel time is required to get to the persons needing care.
C. There is no easy access to emergency equipment or consultation with other health professionals if needed.
D. Home visits may present issues regarding the nurse personal safety in some community or family settings.
E. The nurse has less control over the care setting (for example, cleanliness, noise, privacy, or distractions).

# Community health nurse bag

- Requirements include equipment for basic assessment, medical asepsis, and waste disposal.
- These precautions must be followed to avoid contamination from blood, body secretions, excretions, or contaminated items.

✓ Wash hands
✓ Use gloves
✓ Wear eye and face protection
✓ Wear gown
✓ Handle client care equipment carefully
✓ Clean environmental surfaces
✓ Use proper sharps disposal container

# The three most common intervention in home health care include:

a. Helping families deal with stress created by health problems.

b. Making referrals for community services.

c. Teaching and educating clients, with the focus on strengths rather than weakness.

# Promoting Nurse's Safety

1. **Clarifying the nurse's self – responsibility.**

- Why beginning home visit, reason of visit, family express anxiety and fear.

2. **Promoting Safe Travel:**

- Obtain the family permission to work with them by telephone. Ask for direction to their road.
- Don’t carry purses or jewelry.

## Promoting Safe Travel Cont. ……

- Wear appropriate dress or uniform (as determined by the agency).
- Carry card for telephone and small amount of money.
- Carry an identification badge.
- Avoid secluded and alleys area, empty building.
- Avoid areas where persons are loitering or obtain escort.

## Promoting Safe Travel Cont. ……

- Use discretion about visiting a family; if you feel unsafe don't visit.
- Consider whether an escort is needed to avoid visit alone male if you are female.
- Request a nurse partner or escort to home visit if needed.
- Avoid entering a home in which fight or drug use.
- Always report back to agency in person or by telephone at the end of clinical day.
- Visit only during schedule work hours.

## 3. Handling threats during home visit:

"family members have altercation, fight, under influence of drugs, display a weapon" so the community health nurse ask the family whether another time to visit would be better, notify your supervisor.

## 4. Protecting the safety of family members:

From violence, injury, neglect, abuse. The CHN needs contact supervisor and follow agency's policies and procedures, communication with family, crisis intervention, family's self-expression.

**Evaluation:** is the ongoing process that continually assesses clients progress toward expected outcome.

**Termination:** of home visits occurs when both client and nurse are satisfied that goals have been met or that appropriate referrals have made.

**Telemedicine:** which use phone and computer technologies to monitor clients and provide care without the nurse making a home visit.

## DISPLAY 19.1 GUIDELINES FOR MAKING HOME VISITS: 30 STEPS TO SUCCESS

The following guidelines can be followed to evaluate yourself after making a home visit; or it can be a tool used when you are evaluated by another nurse (peer or instructor). Rate yourself using the following scale: 0 = does not apply, 1 = unsatisfactory, 2 = satisfactory.

| Rating | Assessment |
|---|---|
| _____ | 1. Studies referral, record, or other available data about the family. |
| _____ | 2. Gathers community resource information potentially appropriate to the family. |
| _____ | 3. Obtains appropriate supplies or educational material in anticipation of family needs. |
| | **Planning** |
| _____ | 4. Contacts family to set up an appropriate time for the home visit. |
| _____ | 5. Ascertains correct address and directions to the family for the home visit. |
| _____ | 6. Formulates a written plan for nursing intervention with each family member. |
| _____ | 7. Organizes a chart with forms and charting tools based on the focus of the visit. |
| _____ | 8. Plans a route to the family's home that is the most direct, being resource efficient. |
| | **Implementation** |
| _____ | 9. Travels the community with safety, locating the family home with ease. |
| _____ | 10. Knock on the door loudly enough to be heard and in a friendly manner. |
| _____ | 11. Introduces self to family members in an appropriate manner. |
| _____ | 12. Clearly states the reason for the visit. |
| _____ | 13. Allows a few moments of socialization before beginning the visit. |
| _____ | 14. Smiles, speaks in a pleasant, friendly tone of voice, and maintains eye contact. |
| _____ | 15. Uses aseptic technique when providing nursing care. |
| _____ | 16. Respects the dignity, privacy, safety, and comfort of family members. |
| _____ | 17. Listens attentively to ascertain what family members are saying or implying. |
| _____ | 18. Converses with family members during the home visit. |
| _____ | 19. Communicates accurate and meaningful information to family members. |
| _____ | 20. Responds to family members in a way that encourages them to continue talking. |
| _____ | 21. Uses appropriate words of explanation for family member understanding. |
| _____ | 22. Utilizes opportunities for incidental teaching. |
| _____ | 23. Commends progress made by individual family members. |
| _____ | 24. Explains nursing measures before, during, and after each procedure. |
| _____ | 25. Shares the results of nursing measures with family members when indicated. |
| _____ | 26. Closes the home visit by summarizing the main points of the visit. |
| _____ | 27. Makes plans for the next visit, considering family member wishes. |
| | **Evaluation** |
| _____ | 28. Utilizes information gathered on the home visit to plan care for next visit. |
| _____ | 29. Documents home visit in an appropriate and timely manner. |
| _____ | 30. Completes a self-evaluation of the home visit. |

# Expanded Program on Immunization (EPI)

## Immunity

Is the ability of the body to resist becoming infected upon exposure to a microorganisms or parasite, it is classified as:

1. **Nonspecific:** (Skin, Mucosa Membrane, Phagocytosis)
2. **Specific:** (Active, Passive).

## Types of Specific Immunity

### A. Active immunity

- Long-term resistance, permanent – sometime lifelong (acquired immunity naturally or artificially).
- Is stimulation of the immune system to produce antigen – specific humeral (antibody) and cellular immunity.
- Protection produced by a person's own immune system.

  ***Examples***:

  **Artificial active immunity**: can be induced by a vaccine (a substance that contains antigen). A vaccine stimulates a primary response against the antigen without causing symptoms of the disease. e.g. *vaccination*.

  **Naturally active immunity:** Naturally acquired active immunity occurs when a person is exposed to a live pathogen, and develops a primary immune response, which leads to immunological memory. This type of immunity is "natural" because it is not induced by deliberate exposure. e. g. *Measles*.

### B. Passive immunity

- Short-term resistance (acquired naturally or artificially)
- Usually provide effective immunity but this immunity disappears by time.

***Examples:***

**Artificial passive immunity**: Achieved by the transfer of antibodies, which can be administered in several forms; as human or animal blood plasma or serum, as pooled human immunoglobulin for intravenous (IVIG) or intramuscular and as monoclonal antibodies. It is also used in the treatment of several types of acute infections and to treat poisoning.

**Naturally passive immunity:** refers to antibody-mediated immunity conveyed to a fetus by its mother during pregnancy. Maternal antibodies (MatAb) are passed through the placenta to the fetus.

**Antigen & antibody definition**

- **Antigen:** a live or inactive substance capable of producing an immune response, e.g. (protein, polysaccharide).
- **Antibody:** protein molecules (immunoglobulin) produced by B-lymphocytes to help elimination of antigen.

**The expanded program on immunization is program adopted by the WHO in 1974, it included:**

1- *Child immunization*: immunization of children with vaccines to protect them against six-target disease (T.B– Poliomyelitis – Measles– Diphtheria –Tetanus – Pertussis ), then in 1996 WHO added Hepatitis B vaccine.

2- *Vaccination of pregnant woman*: with tetanus toxoid vaccine to protect woman and newborn from neonatal tetanus.

**The EPI objectives:**

- Achieve and sustain high immunization coverage.
- Establish reliable diseases surveillance.
- Implement disease control and elimination.

**Immunization**

- Is one of the most effective examples of the primary prevention methods of controlling communicable diseases in population.
- Vaccines are not presently available for all communicable disease.
- Vaccination programs have been responsible for global eradication of small box and are close to elimination others (poliomyelitis, measles).

**Causes of Inadequate Immunization**

1. Limited access.
2. Vaccination cost.
3. Patient disinterested.

**Prevention and Control of Communicable Diseases:**

- **The goal of prevention and control programs:** is to reduce the prevalence of disease to a level at which it no longer poses major public health problem.
- **The goal of elimination:** is to remove a disease from a large geographical area.
- **Eradication:** is irreversible termination of all transmission of infection by extermination of infectious agent worldwide (smallpox: 1980).

**World Health Organization**

1. Reduce mortality, disability and morbidity from vaccine preventable diseases(VPD).
2. Attain the WHO of elimination and eradication.
3. Reduce undesirable outcomes of immunization.

**Vaccine preventable diseases (VPDs)**

VPDs are those preventable by currently available vaccines. The most effectively way to prevent childhood illness, death or disability is providing effectively immunization.

**Vaccine:**

Is a suspension of live or killed microorganisms, these agents presented to potential host to induce immunity to a specific disease caused by that organism.

**Types of Vaccine:**

1. Live attenuated (Sabin–Measles–Mumps– BCG).
2. Inactivated or killed (Salk–Pertussis–Rabies).
3. Cellular fraction (Pnumoccocal Pneumonia–Meningococcal).
4. Recombined vaccine (Hepatitis B).
5. Toxoids or Antisera (Tetanus, Diphtheria).
6. Immunoglobulin (Hepatoglobulin).
7. Antitoxin (Tetaglobulin).

**General Rules:**

- For live vaccines, the first dose usually provides protection, the additional doses is given to insure seroconversion; example 95% - 98% of recipients will response to a single dose of measles vaccine, the second dose is given to insure that 100% of people are immune.
- Live attenuated vaccines generally produce long- lasting immunity with single dose.
- The minimum interval between the injections of the same live vaccine is 3-4 weeks.
- Decrease the interval between doses of multi-dose vaccine may interfere with antibody response and protection.
- For the inactivated vaccines, the first dose usually does not provide protection, the proactive immune response may not develop until second or third doses.
- The antibody titer for inactivated vaccines may decrease below protective levels after few years (e.g. tetanus, diphtheria), so required multi doses and may require periodic boosting to maintain immunity.

## Tuberculosis vaccine (BCG)

### What is BCG vaccine?

- BCG vaccine protects infants' agonist tuberculosis; the letters B.C.G stand for Bacilli Calmette –Guerin, Bacilli describes the shape of a bacterium, Calmette and Guerin are the names of people who developed the vaccine.
  - ✓ *BCG vaccine should be kept at 2°– 8°C after reconstitution.*
  - ✓ *Must be discarded after six hours or at end of immunization session, which comes first.*

### When BCG should not be given

- BCG vaccine should not be given to people who are immunosuppressed (e.g. people have symptomatic HIV infection or organs transplant).
- Pregnant should not be given
- Patients with past history of TB.
- Clients with acute illness with fever or systemic disorder.
  - ✓ *Vaccine comes in powder, it must be reconstituted with diluents before used.*

### How safe is BCG vaccine and what potential side – effects are:

- Most children do have a reaction at the site of injection, normally, when BCG vaccine is injected a small raised lump appears at the injection site, usually disappears within 30 minutes.

  After about two weeks a red sore forms that is about the size of the end of an unsharpened pencil, the sore remains for another two weeks and then heals and a small scar about 5 mm across remains, this is a sign that the child has been effectively immunized.

***Other reactions include:***

- Swelling or abscesses, sometimes the glands in a child's armpit or near the elbow swell up after injection with BCG vaccine, may develop an abscess, swollen glands or abscesses occur because an unsterile needle or syringe was used, too much vaccine was injected, or most commonly, the vaccine was injected incorrectly under the skin instead of in its top layer.
- There are very few serious reactions following BCG vaccine, generalized infection due to BCG vaccination occurs at a rate of five per million doses of vaccine given, primarily in HIV – infected persons or those with severe immune deficiencies.

**Administration Summary: BCG Vaccine**

| | |
|---|---|
| **Type of vaccine** | Live attenuated bacteria |
| **Number of doses** | One |
| **Schedule** | At or as soon as possible after birth |
| **Booster** | None |
| **Contraindication** | Symptomatic HIV infection |
| **Adverse reactions** | Local abscess, regional lymphadenitis (rare), distant spread to osteomyelitis, disseminated disease |
| **Special precautions** | Correct intradermal administration is essential,<br>a special syringe and needle is used for administration |
| **Dosage** | 0.05 ml |
| **Injection site** | Outer upper left arm or shoulder |
| **Injection type** | Intradermal (I.D.) |
| **Storage** | Store between 2°– 8°C<br>Vaccine may be frozen for long term storage but not the diluent. |

## Hepatitis B (Hep B) vaccine

### What is hepatitis B Vaccine?

- Hepatitis B vaccine is a cloudy liquid that is provided in a single or multi- dose vials or in prefilled auto-disable (AD) device injection. Hep B vaccine have monovalent and vaccine is also available in combination PENTA = (Hib+ HepB+ DPT).
  - ✓ Only monovalent hep B vaccine should be used as a birth dose, the dose given within the first week of life. Combination vaccines should not be used at birth, but may be used in subsequent doses.
  - ✓ If hep B vaccine stands for a long time the vaccine may separate from the liquid, so shake the vial to mix the vaccine and liquid before using the vaccine.
  - ✓ Hep B should never be frozen.

***Some people should not get hepatitis B vaccine or should wait:***

- Anyone with a life-threatening allergy to baker's yeast, or to any other component of the vaccine, should not get hepatitis B vaccine. Tell your provider if you have any severe allergies.
- Anyone who has had a life-threatening allergic reaction to a previous dose of hepatitis B vaccine should not get another dose.
- Anyone who is moderately or severely ill when a dose of vaccine is scheduled should probably wait until they recover before getting the vaccine.
- Not harmful for pregnant woman.
- If mother's HBsAg - Positive, administer hep B vaccine + immunoglobulin (HBIG) for infants weighting ≥ 2000g (no later than 1 week).

**How Safe Is HB Vaccine, What Are Their Potential Side- Effects?**

Hep B vaccine is one of the safest vaccines.

**Mild reaction includes:**

- ✓ Soreness: about 15% of adults and 5% of children have tenderness or mild swelling at the injection site.
- ✓ Fever: about 1% to 6% of those who receive the vaccine develop a mild fever that lasts one or two days after injection of the vaccine.

***Notes****:* Reactions and complications due to the vaccine are rare; allergic reactions (such as rash, difficulty in breathing, and choking) occur about once every 600,000 doses, no fatal allergic reaction has been reported.

## Administration summary: Hepatitis B vaccine

| | |
|---|---|
| **Type of vaccine** | Recombinant DNA or plasma derived or yeast. |
| **Number of doses** | Four doses |
| **Schedule** | In Palestine; monovalent before age 6 weeks, 2,4,6 month of age (PENTA) |
| **Booster** | None |
| **Contraindication** | Anaphylactic reaction to a previous dose |
| **Adverse reactions** | Local soreness and redness, rarely anaphylactic reaction |
| **Special precautions** | Birth dose must be given if there is a risk of perinatal transmission |
| **Dosage** | 0.5ml – 1ml for adult |
| **Injection site** | Outer mid-thigh (infants), outer upper arm (child and adult) |
| **Injection type** | Intramuscular |
| **Storage** | Store between 2°– 8°C |

## Haemophilus influenzae type b (Hib) conjugate vaccine

### What is Hib?

- Haemophilus influenzae type b vaccine prevents meningitis, pneumonia, epiglottis, and other serious infections caused by *Haemophilus influenzae* type b bacteria. The vaccine will not protect against these conditions if they are caused by other agents.
- In 2000, H. influenzae type B (Hib) was estimated to have caused two to three million cases of serious disease, notably pneumonia and meningitis, and 450 000 deaths in young children.
- *Pneumonia* and *meningitis* are the most important diseases caused by Hib bacteria, in developing countries, *pneumonia* is more common than meningitis in children with Hib disease. Hib disease should be suspected in the case of any child with signs and symptoms of meningitis or pneumonia.
- Hib vaccine is available in two forms, liquid or freeze – dried, each is available as monovalent vaccine or in combination with other vaccines, many countries give Hib combined with DTP and Hep B vaccines (DTP +Hep B + Hib) is called PENTA.

**Some people should not get Hib vaccine or should wait**:

1. People who have a life-threatening allergic reaction to a previous dose of Hib vaccine should not get another dose.
2. Children less than 6 weeks of age should not get Hib vaccine.
3. People who are moderately or severely ill at the time the shot is scheduled should usually wait until they recover before getting Hib vaccine.
4. Children over 5 years old usually do not need Hib vaccine.

5. But some older children or adults with special health conditions should get it (include sickle cell disease, HIV/AIDS, removal of the spleen, bone marrow transplant, or cancer treatment with drugs).

**How Safe is Hib Vaccine and What are its Potential Side – Effects?**

Haemophilus influenzae type b vaccine is very safe and there are no known serious reaction to the vaccine.

***Mild reactions include:***

- ✓ Soreness.
- ✓ 5% - 15% of those who receive Hib vaccine develop redness, swelling, mild pain at the site of injection.
- ✓ Fever; a short time after immunization, between 2%-10% of people may develop a mild fever.

## Administration Summary (Hib) Conjugate Vaccine

| | |
|---|---|
| **Type of vaccine** | Conjugate |
| **Number of doses** | 2 or 3 doses (depending on manufacturer) |
| **Schedule** | 6,10,14 weeks of age for three doses schedule (depending on manufacturer).<br>In Palestine; 2,4,6 month of age (PENTA) |
| **Booster** | None |
| **Contraindication** | Hypersensitivity to previous dose |
| **Adverse reaction** | Mild local reaction |
| **Special precaution** | None |
| **Dosage** | 0.5ml |
| **Injection site** | Outer mid-thigh for infants, outer upper arm for older children |
| **Injection type** | Intramuscular (IM) |
| **Storage** | Storage between 2°– 8°C |

## Trivalent Oral Polio Vaccine (TOPV)

- **Two polio vaccines** are used throughout the world to combat polio. The first was developed by Jonas Salk, first tested in 1952, and announced to the world by Salk on April 12, 1955, it consists of an injected dose of inactivated (dead) poliovirus.
  The second was an oral vaccine developed by Albert Sabin using attenuated poliovirus, human trials of Sabin's vaccine began in 1957 and it was licensed in 1962.
- Trivalent Oral Polio Vaccine (TOPV or Sabin) and Inactivated Polio Vaccine (IPV or Salk) protect the child against poliomyelitis.
- Trivalent Oral Polio Vaccine is liquid vaccine that is provided in two types of containers;
  1. Small plastic dropper bottles.
  2. Glass vials with dropper in a separate plastic bag.

**How safe is TOPV and What Are Its Potential Side-Effect?**

- ✓ TOPV causes almost no side –effects.
- ✓ Less than 1% of people who receive the vaccine develop headache, diarrhea or muscle pain.
- ✓ There is a very small risk of vaccine-associated paralytic polio (VAPP), with approximately two to four cases having been reported for every one million.

**What is supplementary immunization with TOPV?**

A key strategy for polio eradication is supplementary immunization with OPV. This is usually conducted in large scale campaigns (National Immunization Days) where two doses of OPV, one month apart, are given to all children under five years of age regardless of how many doses they received in the past.

***Notes:***

1. Keep the vaccines in a cup containing ice during the vaccination session.
2. Throw away the vaccine that remains in the vial after every vaccination session.
3. Vaccine is should not give in cases of other viral infection, to avoid overlap between the viruses.
4. HIV or impaired immunity should not give OPV so can give Salk (IPV).
5. Prevent breast feeding before and after OPV for 2 hours to avoid vomiting and ensures absorption in the digestive system.
6. If child has diarrhea or vomiting or split out when you give OPV, administer an extra dose – that is, a fifth dose – at least four weeks after he has received the last dose in the schedule.

## Administration Summary: TOPV

| | |
|---|---|
| **Type of vaccine** | Live oral polio vaccine (TOPV) |
| **Number of doses** | Five in endemic countries (including birth dose) |
| **Schedule** | At birth, 6,10,14 weeks<br>In Palestine; 2,4,6,18 month and 6 years |
| **Booster** | 18 months and 6 years |
| **Contraindication** | None |
| **Adverse reaction** | Vaccine-associated paralytic polio (VAPP) very rarely (approximately 2 to 4 cases per million children vaccinated). |
| **Special precaution** | Children known to have rare congenital immune deficiency syndromes should receive IPV rather than OPV. |
| **Dosage** | 2 drops, some vials the dose may be 3 drops |
| **Storage** | Storage between 2°– 8°C |

## Salk – Inactivated Polio Vaccine

*WHO does not – as of July 2003 – recommend the adoption of IPV, either alone or in a sequential schedule, in the developing countries for the following reasons:*

1. **Unresolved issues related to immunogenicity** of IPV when administered at birth, six weeks, ten weeks and 14 weeks of age in the EPI vaccination schedule.
2. **High cost of IPV:**
    - The operational complexities of introducing a vaccine, which requires syringes and needles in infants until new combination products, are available, while OPV is given orally.
    - The price of IPV is over 5 times that of OPV.
    - Administering of IPV requires trained health workers.

**These people should not get IPV:**

- Anyone who has ever had a life-threatening allergic reaction to the antibiotics **neomycin**, **streptomycin** or **polymyxin B** should not get the polio shot.
- Anyone who has a severe allergic reaction to a polio shot should not get another one.

**These people should wait:**

- Anyone who is moderately or severely ill at the time the shot is scheduled should usually wait until they recover before getting polio vaccine. People with minor illnesses, such as a cold, ***may*** be vaccinated.
- Safe and can give for pregnant women.

## Administration Summary: IPV

| | |
|---|---|
| **Type of vaccine** | Killed – inactivated polio vaccine (IPV) |
| **Number of doses** | Two doses |
| **Schedule** | In Palestine; 1, 2 month |
| **Booster** | None |
| **Contraindication** | None |
| **Dosage** | 0.5ml |
| **Injection site** | Outer mid-thigh for infants, outer upper arm for older children. |
| **Injection type** | Subcutaneous (S.C.) |
| **Storage** | Storage between 2°– 8°C |

## Pneumococcal Conjugate Vaccine (PCV)

**What is pneumococcal conjugate vaccine**?

- **Pneumococcal conjugate vaccine** (**PCV**) is a pneumococcal vaccine used to protect infants and young children against diseases caused by the bacterium *Streptococcus pneumoniae* (pneumococcus), diseases include; pneumonia, ear infections, meningitis, and blood infections.
- Invasive pneumococcal infections can even lead to death in some children. It is spread through person-to-person contact and is most common during winter and early spring.
- The dose**;** four doses of PCV are recommended, they are given at 2, 4, 6 months, and again between 12 and 15 months.
- There are two types of pneumococcal vaccine — pneumococcal polysaccharide vaccine (**PPSV**) and pneumococcal conjugate vaccine **(PCV)**.
- The first pneumococcal polysaccharide vaccine, containing 14 serotypes, was licensed in the United States in 1977.
  In 1983, an improved pneumococcal polysaccharide vaccine was licensed, containing purified protein from 23 types of pneumococcal bacteria.
- The first pneumococcal conjugate vaccine (PCV7) was licensed in early 2000. Ten years later, in 2010, a new pneumococcal conjugate vaccine product (PCV13) was licensed and replaced PCV7 for use in the routine vaccination of children. PCV13 offers additional protection against the types of pneumococcal bacteria that cause the majority of invasive pneumococcal diseases. PCV13 is recommended for use in preventing pneumococcal diseases in infants and young children, beginning as young as 6 weeks.

**What kind of vaccines?**

- The pneumococcal polysaccharide vaccine (PPSV) contains long chains of polysaccharide (sugar) molecules that make up the surface capsule of the bacteria. Generally speaking, a pure polysaccharide vaccine induces only short-term immunity and doesn't work as well in children younger than 2 years.
- The pneumococcal conjugate vaccine (PCV) includes purified capsular polysaccharides from the bacteria that are "conjugated" (or joined) to a harmless variety of diphtheria toxin. The resultant conjugate vaccine is able to produce an immune response in infants and antibody booster response to multiple doses of vaccine.

**How Safe is PCV And What Are its Potential Side – Effects?**

- About half of children were drowsy after the shot, had a temporary loss of appetite, or had redness or tenderness where the shot was given.
- About 1 out of 3 had swelling where the shot was given.
- About 1 out of 3 had a mild fever, and about 1 in 20 had a higher fever.
- Up to about 8 out of 10 became fussy or irritable.
- Signs of a severe allergic reaction can include; difficulty breathing, hoarseness or wheezing, hives, paleness, weakness, a fast heart beat or dizziness.

## Pneumococcal Conjugate Vaccine

| | |
|---|---|
| **Type of vaccine** | Polysaccharide - Conjugate |
| **Number of doses** | Four – five doses. |
| **Schedule** | In Palestine; three doses 2,4,12 month |
| **Booster** | None |
| **Contraindication** | Severe allergic reaction (e.g., anaphylaxis) after a previous dose (of PCV, or any diphtheria toxoid-containing vaccine) or to a component of a vaccine (PCV or any diphtheria toxoid-containing vaccine)<br><br>Moderately or severely ill should usually wait until they recover before getting the vaccine. |
| **Adverse reaction** | Drowsy, temporary loss of appetite, or had redness or tenderness, sometime irritable |
| **Special precaution** | None |
| **Dosage** | 0.5ml |
| **Injection site** | Outer mid-thigh in infant / outer upper arm if older |
| **Injection type** | Intramuscular |
| **Storage** | Store between 2°– 8°C |

## Diphtheria and Tetanus toxoid and Acellular Pertussis (DTaP) vaccine

**What is DTaP vaccine?**

- Diphtheria – Tetanus – Pertussis vaccine is made from:
  - Diphtheria toxoid (prepared by formalin inactivation of diphtheria toxin);
  - Tetanus toxoid (a preparation of formalin inactivated toxin); and
  - A variable number of purified acellular component antigens of pertussis.
- It is a liquid vaccine, if a vial DTaP vaccine stands for a long time, fine particles may separate from the liquid, they look like fine sand at the bottom of the vial, so before giving the vaccine shake the vial to mix the vaccine and liquid.
- *DTaP vaccine should never be frozen* because it damages Diptheria Toxiod and Tetanus Toxiod, but sunlight damages pertussis.

**Some children should not get DTaP vaccine or should wait**

1. Children with moderately or severely ill should usually wait until they recover before getting DTaP vaccine.
2. Any child who had a life-threatening allergic reaction after a dose of DTaP should not get another dose.
3. Any child who suffered a brain or nervous system disease (Convulsions) within 7 days after a dose of DTaP should not get another dose of pertussis vaccine, but may get a vaccine without pertussis (called **DT**).

**In Older children and adults**

- DTaP is not licensed for adolescents, adults, or children 7 years of age and older, instead they are given a vaccine called **Tdap** which is similar to DTaP, a single dose of Tdap is recommended for people 11 through 64 years of age.

- Another vaccine, called **Td**, protects against tetanus and diphtheria, but not pertussis. It is recommended every 10 years.

**How safe is this vaccine and what is its potential side effects?**

Usually; reactions are mild and the side effects including:

- ✓ **Fever**; up to half of the children who receive DTP vaccine may have a fever in the evening after receiving the injection, this fever should disappear within a day.
  - A fever that begins more than 24 hours after DTP injection is not likely to be a reaction to the vaccine. Administrating paracetamol or any appropriate antipyretic at time and at four and eight hours after immunization decreases the subsequent incidence of febrile and local reaction.
- ✓ **Soreness**; up to half of the children may have pain, redness or swelling at the injection site.
- ✓ **Crying;** for more than three hours (mostly because of pain) can be observed in up to1% of infant.

**More serve reactions:**

- **Convulsions;** usually related to fever (1 case in 12500 doses administrated).
- **Hypotonic** – hyporesponsive episode (1 case in 1750 doses administrated)
- **Anaphylactic** reaction are extremely rare.
- There is no evidence that the vaccine causes any serious neurological disorder such as encephalopathy.

## Administration summary: DTaP Vaccine

| | |
|---|---|
| **Type of vaccine** | Diphtheria, Tetanus as Toxoid, and acellular Pertussis |
| **Number of doses** | Six doses |
| **Schedule** | In Palestine; 2,4,6 (PENTA),18 month |
| **Booster** | 2 doses; DT at 6 years & Td at 15 years |
| **Contraindication** | Anaphylactic reaction to previous dose or to any constituent. |
| **Adverse reaction** | Mild local or systemic reactions are common |
| **Special precaution** | DTaP not usually given over 6 years of age |
| **Dosage** | 0.5 ml |
| **Injection site** | Outer mid-thigh in infant / outer upper arm if older |
| **Injection Site** | Intramuscular |
| **Storage** | Store between 2°– 8°C and should never frozen |

## Tetanus Toxoid (TT) vaccine

**What is TT vaccine?**

- Tetanus Toxoid (TT) vaccine protects against tetanus, it is provided as a liquid in vials and also in prefilled injection devices.
- It is available in a number of different formulations:

1) **TT:** vaccine protects only against tetanus and neonatal tetanus.
2) **DTP:** vaccine protects against Diphtheria, Tetanus, and Pertussis.
3) **DT**: vaccine protects against Diphtheria, Tetanus; because it contains a high level of diphtheria toxoid, it should not be given to children older than six years old or adults.
4) **Td**: Tetanus- diphtheria toxoid for adult dose vaccine, is the same vaccine as DT, but with a lower diphtheria toxoid dose. It is suitable for children older than six years old and adults, including pregnant women. Td has the added advantage of protecting against diphtheria and tetanus.

- When given to women of child bearing age, vaccines that contain tetanus toxoid (TT or Td) not only protect women against tetanus, but also prevent neonatal tetanus in their newborn infant through antibodies that passed from mother to her fetus.
- TT/Td given intramuscular in upper part of left arm (0.5ml).
- When vaccines containing tetanus toxoid stand for a long time, the vaccine separates from the liquid and looks like fine sand at the bottom of the vial. Shake the vial to mix the vaccine and liquid again before giving the vaccine.
- TT/Td/DTP/ DTaP vaccines should never be frozen.

**Some people should not get T.T vaccine or should wait**:

1. Anyone who has a severe allergy to any component of a vaccine should not get that vaccine.
2. Anyone who has had a life-threatening allergic reaction after a dose of DTP, DTaP, DT, or Td should not get another dose.
3. Anyone who had a coma, or long or multiple seizures within 7 days after a dose of DTP should not get another dose.
4. Talk to your provider if the person getting either vaccine:
   - has epilepsy or another nervous system problem.
   - has had Guillain Barré Syndrome (GBS).
   - had severe swelling or severe pain after a previous dose of DTP, DT, Td vaccine.
5. Anyone who has a moderate or severe illness on the day the shot is scheduled should usually wait until they recover before getting Td vaccine.
6. A person with a mild/ illness or low fever can usually be vaccinated.

**How safe are TT, Td and DT vaccines and what are their potential side-effects?**

Vaccines containing tetanus toxoid cause little serious reaction but quite frequent mild reactions.

**Mild reactions to TT, Td and DT vaccines include:**

- ✓ Soreness; about 1:10 people who receive the vaccines have mild pain
- ✓ Redness; warmth and swelling at the injection site for about one to three days after the injection. This mild reaction is likely to be more common after later doses than earlier ones, and may affect between 50% and 85% of people who receive booster doses.
- ✓ Fever; about one in ten people may develop a mild fever after receiving the vaccines.

## Measles – Mumps and Rubella (MMR) Vaccine

**What are the MMR vaccination?**

- MMR vaccine is administered against measles, mumps and rubella. Some countries use combination vaccinations for measles and rubella (MR) or for measles, mumps and rubella (MMR).
- Vaccines are provided in a powder form with diluents and must be reconstituted before they can be used, it is essential that only the diluents supplied with the vaccine be used.
- MR and MMR vaccines should be kept at 2°– 8°C.
- After reconstitution vaccine must be discarded after six hours or at the end of immunization session, whichever comes first.

**Some people should not get MMR vaccine or should wait:**

1. People should not get MMR vaccine who have ever had a life-threatening allergic reaction to gelatin, the antibiotic neomycin, or to a previous dose of MMR vaccine.
2. People who are moderately or severely ill at the time the shot is scheduled should usually wait until they recover before getting MMR vaccine.
3. Pregnant women should wait to get MMR vaccine until after they have given birth. Women should avoid getting pregnant for 4 weeks after getting MMR vaccine.
4. Some people should check with their health care provider about whether they should get MMR vaccine, including anyone who:
    - Has HIV/AIDS, or another disease that affects the immune system.
    - Is being treated with drugs that affect the immune system, such as steroids, for 2 weeks or longer.
    - Has any kind of cancer.
    - Is taking cancer treatment with radiation or drugs.

- Has ever had a low platelet count (a blood disorder).

5. People who recently had a transfusion or were given other blood products should ask their doctor when they may get MMR vaccine.

**Note**:

- The first dose of **MMR** vaccine has been associated with rash and fever; rash has been reported in about 1 person in 20, and fever in about 1 person in 5.
- Seizures caused by a fever are also reported more often after MMR.
- These usually occur 5-12 days after the first dose.

**How safe the MMR vaccine and what are its potential side-effects?**

**Mild reactions to the vaccines include:**

✓ Fever: as with the single – antigen measles vaccine, about 5% to 10% of children develop a mild fever within 5 to 12 days of receiving the vaccine.

✓ Rash: again as with the measles vaccine, about one in 20 children develop a mild rash about 5 to 12 days of receiving the vaccine.

**Severe reactions:**

Are rare and similar to that experienced after receipt of the measles vaccine.

✓ Although an association between MMR and autism has been suggested, there was absolutely no evidence of a link between the MMR vaccine and autism according to a study in 2004, UK.

✓ In addition rubella – containing vaccines may result in temporary arthritis from one to three weeks after vaccination in about one in four post pubertal females, these reactions are very rare in young children.

✓ Mumps – containing vaccines may result in rare cases of parotitis and some cases of septic meningitis, children recover without sequels, although some may need to be hospitalized.

## Administration Summary: MMR Vaccine

| | |
|---|---|
| **Type of vaccine** | Live attenuated viral |
| **Number of doses** | Two doses |
| **Schedule** | In Palestine; 12-18 month |
| **Booster** | Rubella vaccine alone given for females at 12 years, measles immunization is recommended (routine or campaign) |
| **Contraindication** | Sever reaction to pervious dose, congenital or acquired immune disorders.<br>Pregnancy; "although not recommended to administer the vaccine during pregnancy, there has never been any evidence of damage to the fetus from vaccination the mother during pregnancy". |
| **Adverse reactions** | Same as measles vaccine, plus cases of arthritis in adolescent females for rubella – containing vaccine, and rarely parotitis and aseptic meningitis with mumps – containing vaccine may occur. |
| **Special precaution** | None |
| **Dosage** | 0.5ml |
| **Injection site** | Outer mid-thigh in infant / outer upper arm if older |
| **Injection type** | Subcutaneous (S.C.) |
| **Storage** | Store between 2°– 8°C (vaccine may be frozen for long – term storage but not the diluents) |

## Rotavirus Vaccine

**What are the RV vaccination?**

- Rotavirus is a virus that causes diarrhea, mostly in babies and young children. The *diarrhea* can be severe, and lead to *dehydration*, *vomiting* and *fever* are also common in babies with rotavirus, it is transmitted by direct fecal- oral mode.
- Every year before the vaccine was available; more than 400,000 young children had to see a doctor for illness caused by rotavirus, 20 to 60 young children died.
- **RV vaccine** first dose should be administered starting at 6 weeks of age and before 15 weeks of age. To optimize protection, RV vaccine should be initiated as soon after 6 weeks of age as feasible.
- Orally vaccine, rotavirus vaccines can be given with other routine vaccines at 2 and 4 months of age, or 2, 4 and 6 months of age.

**Some babies should not get this vaccine:**

1. A baby who has had a life-threatening allergic reaction to a dose of rotavirus vaccine should not get another dose.
2. Babies with "severe combined immunodeficiency" (SCID) should not get rotavirus vaccine.
3. Babies who have had a type of bowel blockage called "intussusception" should not get rotavirus vaccine.
4. Babies who are moderate to severe ill should wait until they recover, this includes babies with moderate or severe diarrhea or vomiting.

**How safe is the RV vaccine and what are its potential side-effects?**

**Mild problems following rotavirus vaccine:**

✓ Babies might become irritable, or have mild, temporary diarrhea or vomiting after getting a dose of rotavirus vaccine.

**Serious problems following rotavirus vaccine:**

- ✓ Intussusception is a type of bowel blockage that is treated in a hospital, and could require surgery.

**Administration Summary: Vaccine**

| | |
|---|---|
| **Type of vaccine** | Live attenuated viral |
| **Number of doses** | Two doses |
| **Schedule** | In Palestine; 2,4 month |
| **Booster** | None |
| **Contraindication** | Babies with "severe combined immunodeficiency".<br>Babies who have had "intussusception". |
| **Adverse reaction** | Babies might become irritable, or have mild, temporary diarrhea or vomiting.<br>Intussusception is a type of bowel blockage. |
| **Special precaution** | None |
| **Dosage** | 1.5ml |
| **Vaccine route** | Orally |
| **Storage** | Store between 2°– 8°C |

## Contraindications to Immunization

There are not many contraindications to immunization, all infants should be immunized except in these four rare situations:

**Permanent Contraindication to Vaccination**

1. Severe allergic reaction (e.g. anaphylaxis) after a previous dose or to a vaccine component.
2. Encephalopathy (e.g. coma, decreased level of consciousness, prolonged seizures) not attributable to another identifiable cause within 7 days of administration of previous dose of DTP.
3. Known severe immunodeficiency (e.g. from hematologic and solid tumors, patients recipient to chemotherapy, congenital immunodeficiency, long-term immunosuppressive therapy; or patients with HIV infection (exhibits the sign and symptoms of AIDS) who are severely immunocompromised.
4. If a parent strongly objects to an immunization for a sick infant, do not give it, ask the mother to come back when infant is well.

**Invalid Contraindication to Vaccination**

All these conditions are not contraindication, infant with these conditions should be immunized:

1. **Mild Illness**: such as (low-grade fever, upper respiratory infection (URI), colds, Asthma, otitis media, and mild diarrhea) should be vaccinated on schedule.
2. **Chronic illnesses:** e.g. heart, lung, kidney, or liver.
3. **Antimicrobial Therapy**: Antibiotics do not have an effect on the immune response to most vaccines.
4. **Disease Exposure or Convalescence:** if a person is not moderately or severely ill, he or she should be vaccinated.

5. **Pregnant or Immunosuppressed Person in the Household:**
    - Most vaccines, including live vaccines MMR, OPV, varicella, zoster, rotavirus, LAIV, and yellow fever) can be administered to infants or children who are household contacts of pregnant or immunosuppressed persons, as well as to breastfeeding infants.
    - Vaccinia (smallpox) vaccine should not be administered to household contacts of a pregnant or immunosuppressed person.
6. **Breastfeeding:**
    - Breastfeeding does not decrease the response to routine childhood vaccines and is not a contraindication for any vaccine except smallpox.
    - Yellow fever vaccine should be avoided in breastfeeding women.
7. **Preterm Birth and Premature or low-birth weight:**
    - Vaccines should be started on schedule on the basis of the child's chronological age.
    - Preterm infants have been shown to respond adequately to vaccines used in infancy.
8. **Allergy to Products Not Present in Vaccine:**

    Infants and children with nonspecific allergies, duck or feather allergy, or allergy to penicillin, children who have relatives with allergies, and children taking allergy shots can and should be immunized.
9. **Allergy That is Not Anaphylactic**:
    - Anaphylactic allergy to a vaccine component (such as egg or neomycin) is a true contraindication to vaccination.
    - If an allergy to a vaccine component is not anaphylactic or is not severe, it is not a contraindication to that vaccine.
10. **Family History of Adverse Events**:
    - Family or personal history of seizures is a precaution for the use of MMRV vaccine.

- Varicella-containing vaccine should not be administered to persons who have a family history of congenital or hereditary immunodeficiency in first-degree relatives (e.g., parents and siblings) unless the immunocompetence of the potential vaccine recipient has been clinically substantiated or verified by a laboratory.
- Family history of sudden infant death syndrome (SIDS) is not a contraindication to vaccination.

**11. Tuberculin Skin Test**:

- Infants and children who need a tuberculin skin test (TST) can and should be immunized; all vaccines, including MMR, can be given on the same day as a TST, or any time after a TST is applied.
- For most vaccines, there are no TST timing restrictions.

**12. Multiple Vaccines:**

Administration at the same visit of all vaccines for which a person is eligible is critical to reaching and maintaining high vaccination coverage.

**13. Allergies or asthma:** (except if there a known allergy to a specific component of the vaccine).

**14. Known or suspected HIV infection:**

With no signs and symptom of AIDS or long-term immunosuppressive therapy can give inactive vaccine without live vaccine.

**15. Recent or imminent surgery**

**16. Malnutrition**

**17. History of jaundice at birth**

**18. Stable neurological condition**:

Such as Cerebral Palsy (CP) or Down's Syndrome.

**REMEMBER:**

**The "Rights of Medication Administration" should be applied to each encounter when vaccines are administered.**

**These rights include:**

1. The right patient
2. The right vaccine or diluents
3. The right time
4. The right dosage
5. The right route, needle length, and technique
6. The right site
7. The right documentation

**Documentation:**

*All vaccines administered should be fully documented in the patient's permanent medical record. Healthcare providers who administer vaccines covered by the National Childhood Vaccine Injury Act are required to ensure that the permanent medical record of the recipient indicates:*

1) Date of administration.

2) Vaccine manufacturer.

3) Vaccine lot number.

4) Name and title of the person who administered the vaccine and the address of the clinic or facility where the permanent record will reside

5) Vaccine information statement (VIS)

- date printed on the VIS
- date VIS given to patient or parent/guardian

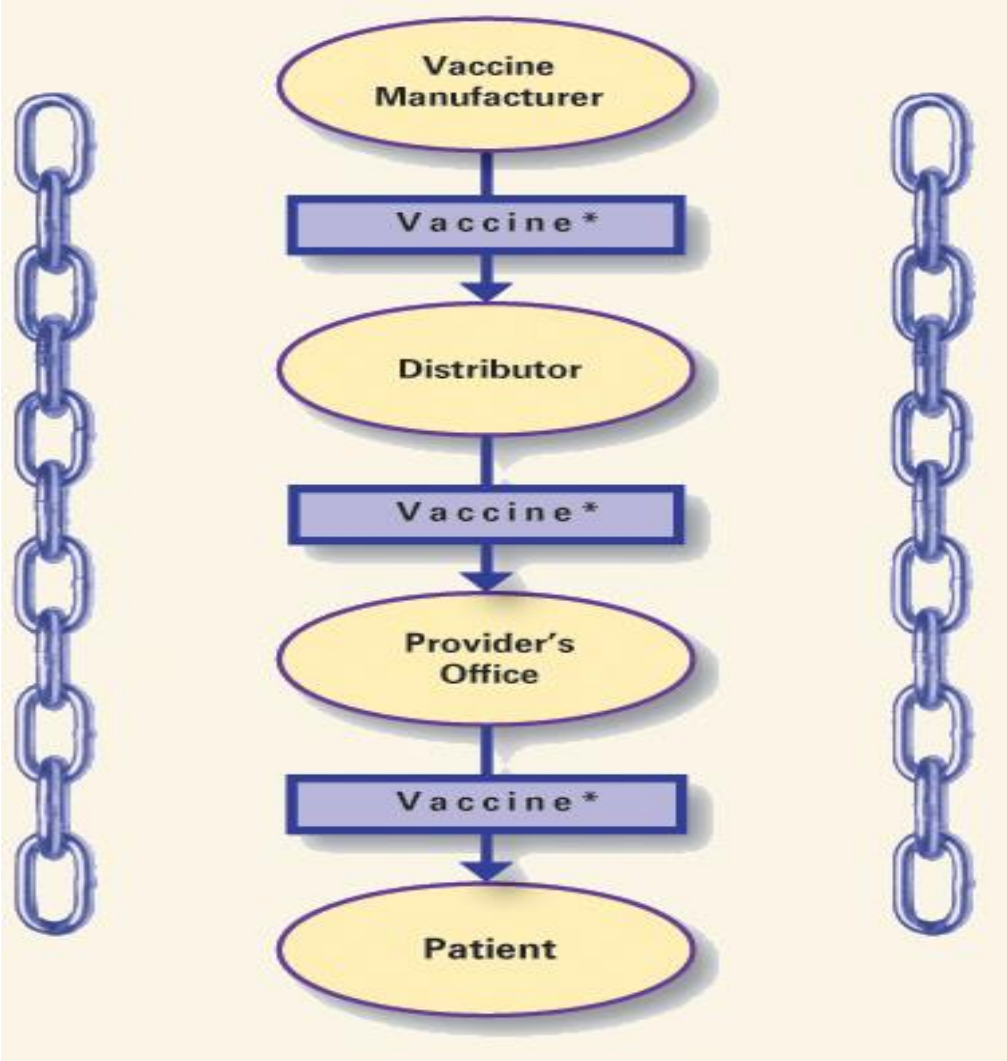
**What Damage Vaccines:**

1. Heat and sunlight can damage vaccines especially the live attenuated ones (BCG, Polio and Measles).
2. Freezing damages the killed and toxoid (DTaP, DPT, DT, TT and hepatitis B).
3. Freezing does not damage live attenuated vaccines (BCG, oral polio and measles vaccines).
4. Disinfectant or antiseptics can damage vaccine (such as detergents) and antibiotics such as streptomycin on BCG.
5. All vaccines lose their potency after a certain time you know that time from the expiry date printed on the vaccine.

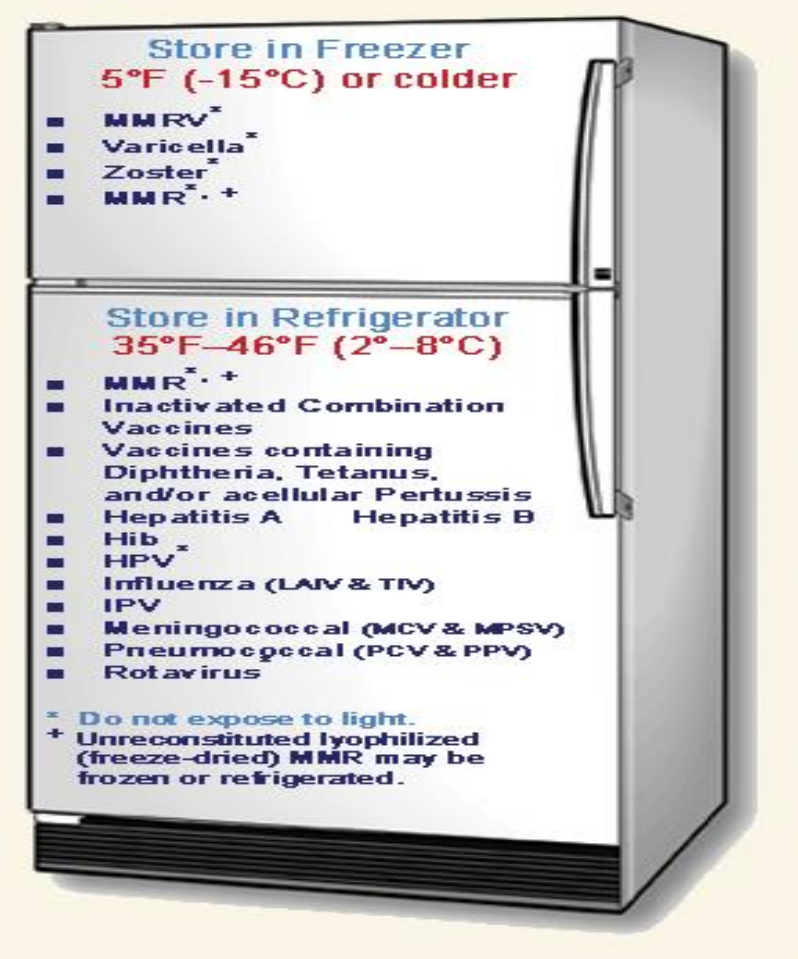
**The Cold Chain**

- Vaccines must be stored properly from the time they are manufactured until the time they are administered. The system used to keep and distribute vaccines in good condition is called the cold chain.
- **The cold chain has three main components:**
  - Transport and storage equipment
  - Trained personnel
  - Efficient management procedures
- All three elements must combine to ensure a safe vaccine transport and storage.
- The cold chain begins with the cold storage unit at the vaccine manufacturing plant, extends through the transfer of vaccine to the distributor and then to the provider's office, and ends with the administration of the vaccine to the patient.

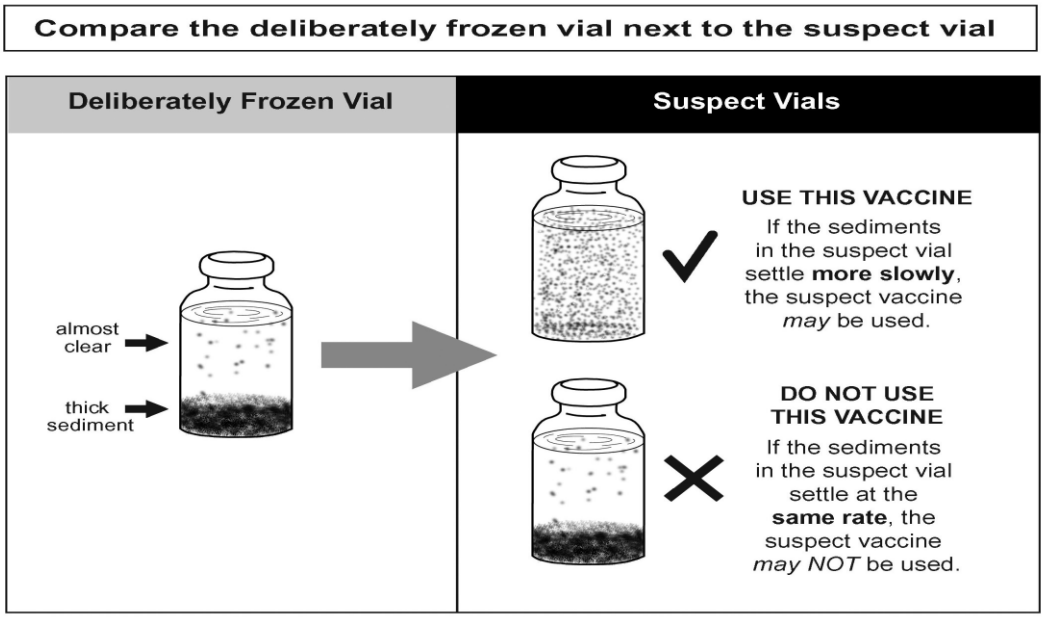
# The Cold Chain



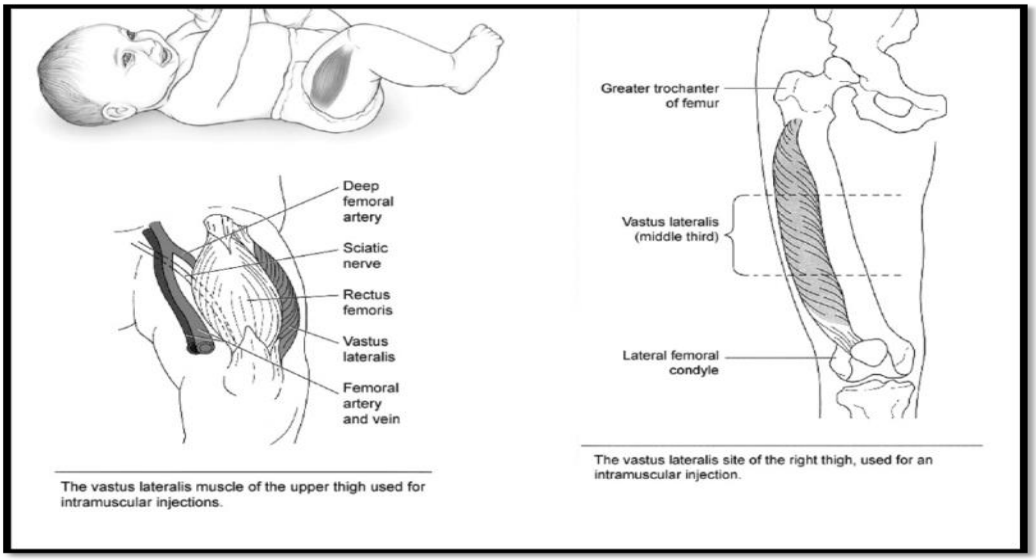

**(Figure 1)**Vaccine is transported in a refrigerated or frozen state, as appropriate (refrigerator [2°– 8°C]; freezer or colder), using an insulated container or a refrigerated truck.



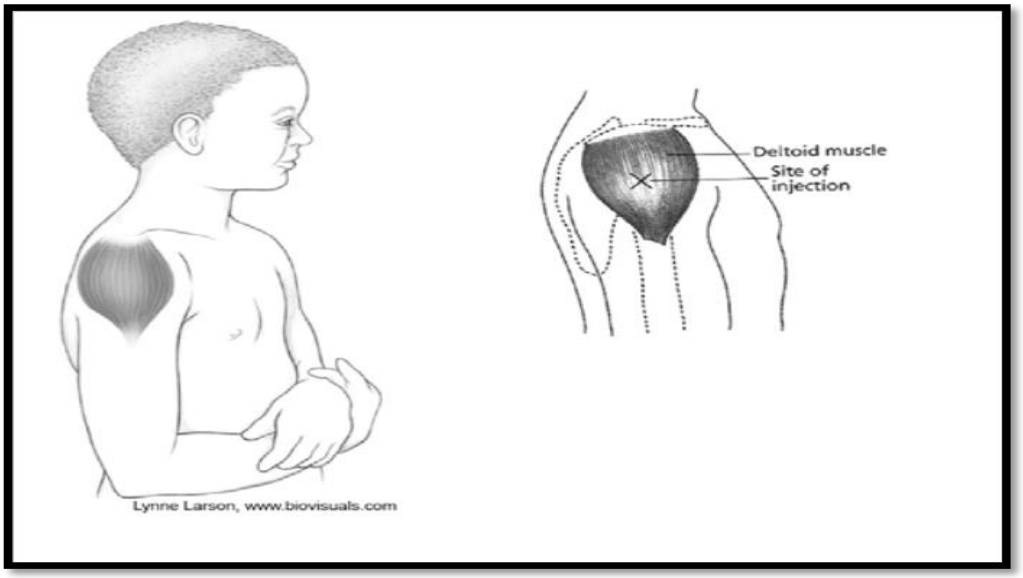

**(Figuer 2) A health center refrigerator showing vaccine vaccine stored correctly**




**(Figure 3) Shake test**




**(Figure 4 A) Site of injection**




**(Figure 4 B) Site of injection**

# Vaccination program

| Age | Vaccine | Dose | Route |
|---|---|---|---|
| **3-7 Day** | BCG | 0.05 ML | ID |
| | HEP B | 0.5 ML | IM |
| **1 Month** | IPV1 | 0. 5 ML | SC |
| **2 Month** | IPV2 | 0.5 ML | SC |
| | PENTA 1 | 0.5 ML | IM |
| | OPV1 | 2 Drops | Orally |
| | PCV1 | 0.5 ML | IM |
| | ROTA 1 | 1.5 ML | Orally |
| **4 Month** | PENTA 2 | 0.5 ML | IM |
| | OPV2 | 2 Drops | Orally |
| | PCV2 | 0.5 ML | IM |
| | ROTA 2 | 1.5 ML | Orally |
| **6 Month** | PENTA3 | 0.5 ML | IM |
| | OPV3 | 2 Drops | Orally |
| **9 Month** | Vitamin A | 1 Capsule | Orally |
| **12 Month** | MMR1 | 0.5 ML | SC |
| | PCV3 | 0.5 ML | IM |
| **15 Month** | Vitamin A | 1 Capsule | Orally |
| **18 Month** | MMR2 | 0.5 ML | SC |
| | DPT Booster | 0.5 ML | IM |
| | OPV Booster | 2 Drops | Orally |
| **6 Years** | DT | 0.5 ML | IM |
| | OPV | 2 Drops | Orally |
| **12 Years** for female | Rubella | 0.5 ML | SC |
| **15 Years** | Td | 0.5 ML | IM |

Pentavalent vaccine is a combined vaccine with five individual vaccines conjugated into one, intended to actively protect people from 5 potentially deadly diseases.
*PENTA = Hib+ HepB+ DPT*